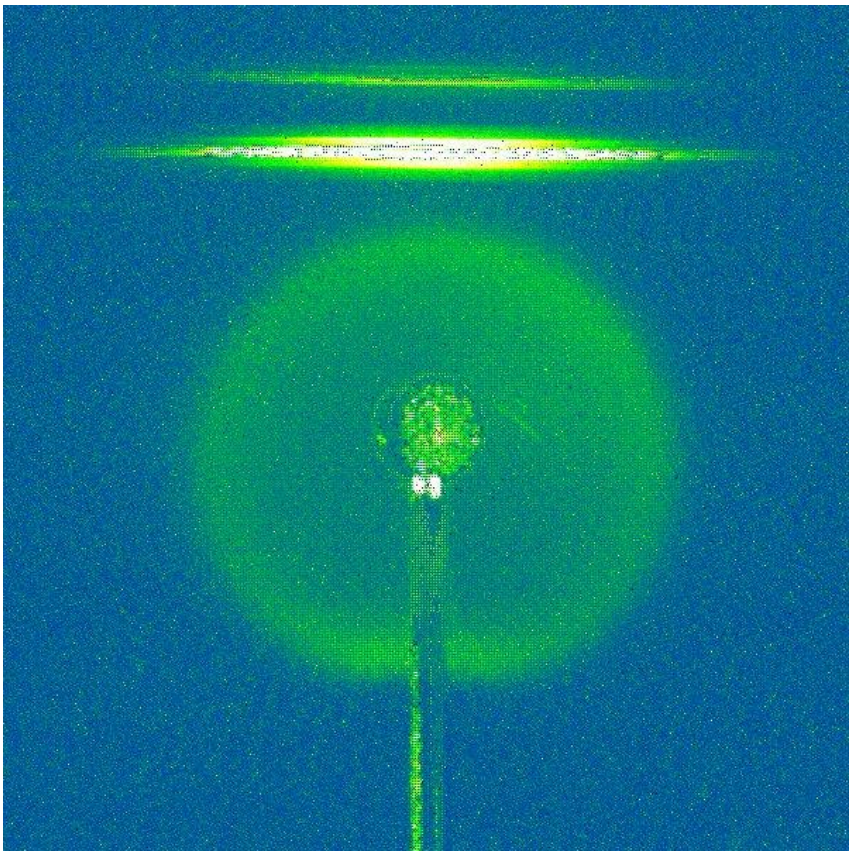
Advanced Combustion via Microgravity Experiments (ACME)
Structure and Response of Spherical Diffusion Flames (s-Flame)

ACME/s-Flame Final Report (2021*)

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A preliminary false-colored image of a hydrogen/helium flame in an oxygen/helium atmosphere, where the porous spherical burner and its supply tube can be seen within the flame. The bright glow of two 14-micron fibers used for Thin Filament Pyrometry (TFP) is also visible.

**Data analysis and the publication of results is still underway, so it is expected that this report will be updated in the future with at least a revision of the list of papers and presentations.*

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1. Overview

The Structure and Response of Spherical Diffusion Flames (s-Flame) experiment is one of six independent experiments that make up the Advanced Combustion via Microgravity Experiments (ACME) project for which tests are conducted within the Combustion Integrated Rack (CIR) on the International Space Station (ISS).

The experiment's Principal Investigator (PI) is Prof. Chung K. Law of Princeton University. The experiment has two Co-Investigators (Co-Is), namely Prof. Stephen D. Tse of Rutgers University and Dr. Kurt R. Sacksteder of the NASA Glenn Research Center. While Sacksteder was actively engaged in defining the experiment, he had transitioned to other NASA work long before the spaceflight operations limiting his overall contributions.

Meanwhile, Dennis P. Stocker of the NASA Glenn Research Center is the Project Scientist (PS) for ACME including s-Flame and the rest of the ACME investigations. In preparatory drop testing, etc., Dr. Fumiaki Takahashi of the Case Western Reserve University (CWRU) served as the Deputy Project Scientist. For ACME's spaceflight operations, the NASA science team includes Drs. Vedha Nayagam and Uday G. Hegde who are also with CWRU.

The s-Flames operations were conducted in two rounds of testing during Increments 60-63, where there was no testing during Increment 62 due to limited crew availability. More specifically, the first round of testing was conducted from Sept. 2019 to Jan. 2020, while the second-round testing – featuring helium rather than nitrogen dilution - occurred from May to July 2020.

2. Science Overview

A. Background

This background is the Executive Summary from the experiment's Science Requirements Document (SRD) of Jan. 2010.

Recognizing that most combustion engines and devices operate on the basis of diffusion flames, the proposed program aims to characterize and gain predictive capability of the structure and dynamics of diffusion flames in simple, well-defined flow fields such that the phenomena of interest can be studied without being unduly complicated and compromised by complex and sometimes non-quantifiable flow field effects. The basic flame configuration adopted for the investigation is the spherically symmetric diffusion flame generated by discharging a fuel gas from a porous spherical burner into a quiescent, oxidizing ambience in microgravity. The one-dimensional nature of the system facilitates data acquisition, data reduction, and computational simulation.

The primary goals of the program are two-fold and strongly interdependent. The first objective is to interrogate the fidelity of the chemical kinetic mechanisms and transport submodels used in the simulation of aerothermochemical phenomena, and consequently identify possible modifications. The second objective is to characterize and gain predictive capability on spherical diffusion flames.

The space flight program consists of two main parts: (1) Mapping and characterization of the movement of spherical diffusion flames subsequent to ignition from a non-steady-state condition and their extinction boundaries due to direct finite-rate chemistry and indirect radiative heat loss. The emphases are on the relative importance of radiative emission and re-absorption, the examination of the adequacy of the detailed chemical kinetics mechanism and transport sub-models used, and the development of flame front instabilities that could modify the extinction boundaries. Results of the study are of relevance to the development of lean-burn engines for improved efficiency and reduced pollutant emissions. (2) Studies of the response of characteristically sooty flames to a non-steady-state initial condition with respect to the aspects given in (1). The studies will include characterizing soot onset as a function of residence time for expanding flames, as well as soot radiation and transport contributions in spreading, extinction, and instability.

The project involves microgravity experimentation, computational simulation, and, where appropriate, theoretical analysis. Interpretation of all phenomena will emphasize the simultaneous and coupled influence of chemistry and transport.

B. Rationale for Microgravity

This is the 'Justification for Extended Duration Microgravity Environment' section (2.5) of the experiment's SRD from Jan. 2010. See the SRD itself for the references, etc. which are not included in this report.

2.5.1 Limitations of Terrestrial (1g laboratory) Testing

We seek to make observations of truly one-dimensional flames whose structure can be studied without complications due to uncertainties in the flow pattern caused by buoyancy. Such fundamental study of flame structure and dynamics is best served via the elimination of buoyancy through the reduction of gravity.

The minimum sizes of these flames imply that buoyancy would be significant if the experiments were conducted in normal gravity. Previous experimentation on earth has amply demonstrated this influence. Although studies of spherical diffusion flames were conducted in normal gravity using the low-pressure, inverse-flame arrangement (see §1), the conditions are limited for small flames, low pressures, low molecular weight fuels. Indeed, only the hydrogen flame can be considered to be nearly buoyancy free. With such constraints, normal-pressure, micro-gravity, hydrocarbon-based experiments are requisite.

One-dimensional spherical flames allow for the unambiguous study of such fundamental flame phenomena as the dynamics and extinction of diffusion flames (including the mechanisms of flame front unsteadiness), soot suppression/formation, and flame front instabilities, without gravity-induced baroclinicity and buoyancy, on the flame responses.

2.5.2 Limitations of Drop Towers and Aircraft

Extensive experimentation has been conducted in the 2.2-s drop tower, 5-s drop tower, and parabolic aircraft, as reviewed previously in §1.3. Although experimental data of good quality were obtained on the transient response of spherical diffusion flames in the 2.2-s drop tower facility, the duration was insufficient to observe radiative extinction. The numerical simulations indicate that, for the characteristic flames under study, more than 2.5s of microgravity time is needed. Subsequent parabolic aircraft experimentation afforded the additional low-gravity duration (up to 10s by free-floating the apparatus), as well as an experimental package incorporating rainbow schlieren (which cannot be contained within a package for the 2.2-s drop tower), although radiative extinction was still not observed. The unavoidable g-jitter associated with the facility, however, renders the observation less definitive. Note that even though the package was free-floated, umbilical restrictions could not be avoided. This g-jitter effect was substantiated by accelerometer data. 5-s drop tower experiments for a mixture composition of 20%H2 / 25%CH4 / 55%N2 did show events with local extinction toward the end of the drop (see Fig. 1.11). While these new results are extremely encouraging and suggest that radiative extinction can indeed be experimentally observed, the 5-s duration is still insufficient. To obtain spherical symmetry for small burners (in order to minimize heat loss to the burner), as well as to ensure quasi-steady flame evolution up to radiative extinction (for the flame sizes to be studied), low mass flow rates are needed, which increase the characteristic time needed. Moreover, for other mixture compositions (such as 50%H2/10%CH4/40%N2 45%H2/55%N2), more than 5s of microgravity duration has been computationally predicted to be needed to achieve radiative extinction (see Fig. 1.12). Additional time is also needed to employ multi-staged flames (where the mass flow rate is changed) so that such phenomena as kinetic extinction can be studied (for example, by establishing a quasi-steady flame at a given flow rate and then reducing it so that the flame front moves inwardly and quasi-steadily until kinetic extinction). Finally, given the characteristic time for diffusion flame oscillation (see Fig. 1.14), which does not occur until the flame nears the quasi-steady extinction limits, extended microgravity duration becomes essential in order to study the phenomenon.

2.5.3 Justification for Space-Flight Facility

As explained above, extended microgravity duration, of high quality, is needed to make all necessary flame structure measurements, allowing for the study of transient phenomena (such as flame oscillations and fuel vapor accumulation) and extinction processes. Moreover, at present, the advanced diagnostics found on the space-flight facility cannot be utilized in the drop tower facilities. Numerical simulations indicate that about 25s of microgravity is needed for non-extinguishing flames to reach 80% of their steady-state radius, for the fuel mixtures and flow rates examined. It is expected that within this duration, the aforementioned phenomenon can be duly investigated. The total experimental duration for a single case would be 25s. This duration is within the 25s bound imposed by constant volume combustion within the specified CIR dimensions (see §Appendix 5.1) regarding oxygen depletion for a “penetration depth” of 20cm.

C. Science Objectives

The proposed objectives text presented here is the ‘Experimental Objectives’ section (2.1) of the experiment’s SRD from Jan. 2010.

Current experimental and theoretical capabilities make detailed, quantitative descriptions of laminar flames situated in well-defined flow fields conceivable provided that comprehensive

chemical-kinetic and transport mechanisms are available. The comprehensiveness of these mechanisms has been extensively tested for premixed but not diffusion flames, especially in simple, well-defined flow fields not complicated by buoyant flows. A burner-generated spherical diffusion flame provides a simple, one-dimensional flame geometry that permits a rigorous comparison between detailed theory and experimental verification. Existing theory and limited low-gravity experimental results suggest that quasi-steady spherical diffusion flames can exist with some combinations of fuel mixture supply rates and quiescent ambient oxidizers. With respect to the steady flame conditions, altering fuel supply rates to limit flame residence time induces a convective-kinetic extinction limit. Alternatively, very large residence times allow radiative heat losses and diffusive transport mechanisms to quench the flame. Near these dual flammability boundaries of quasi-steady flames, flame-front instabilities may appear before the flame is quenched.

We seek to investigate and improve the fidelity and comprehensiveness of the chemical kinetic mechanisms and transport submodels used in the simulation of aerothermochemical phenomena, and consequently identify possible modifications, by characterizing important unit combustion processes, including: (1) the dynamics and extinction of diffusion flames, with emphases on fuel vapor accumulation, radiation heat transfer, flame pulsation, and flame kinetics; and (2) the structure and response of characteristically sooty flames, with emphases on flame front stability and soot formation. Comprehensive predictive capability can be achieved by employing the simple flames of Process (1) to interrogate and suggest modifications of the kinetic and transport data bases, and then validate their adequacy in the chemically and aerodynamically more complex situation of Process (2).

A series of experiments is proposed with the following specific objectives, in both soot-free and sooty flames, to compare with detailed computational simulation to assess transport properties and detailed chemical kinetics (including chemiluminescence):

Objective A. Measure and characterize the transient structure (such as temperature and species distribution) of diffusion flames established at initially non-steady, spherical flame structures for mixtures of different H2/CH4/diluent ratios (soot-free) and different C2H4/diluent ratios (sooty), for different flow rates.

Objective B. Determine the quasi-steady convective/chemical-kinetic extinction limits (low system Damköhler numbers) and radiative/chemical-kinetic extinction limits (high system Damköhler numbers) for mixtures of different H2/CH4/diluent ratios (soot-free) and different C2H4/diluent ratios (sooty).

Objective C. Determine the existence (for different Zeldovich number, Ze), onset (as a function Da), and nature (i.e. mode, frequency, amplitude) of pulsating instabilities theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures that are above a critical Lewis number.

D. Science Success Criteria

The success criteria is from the s-Flame' portion of the 'Success Criteria' section of the ACME Integrated Science Requirements Document (ISRD, ACME-SRD-001.B) from Sept. 2013, which is revision B of the version prepared for the Requirements Definition Review (RDR).

6.4.1. Minimum Success

Minimum success is defined to mean sufficient scientific data return from the experiment to perform a direct comparison with the numerical flame simulation and publish a single archival journal article. This minimal level of success may be achieved by obtaining (for example):

1. Observations of transient flame phenomena leading toward steady state flames for 1 CH4 case and 1 C2H4 case. Such observations must include, at a minimum, the flow and boundary condition measurements, flame temperature data and image sequences from at least two orthogonal imaging systems; or
2. Observations of flame extinction at both low and high system Damköhler number for 1 case. Such observations must include, at a minimum, the flow and boundary condition measurements, flame temperature data and image sequences from at least two orthogonal imaging systems; or
3. Observations of spherical soot formation in 1 C2H4 case. Such observations must include, at a minimum, the flow and boundary condition measurements, flame temperature data and image sequences from at least two orthogonal imaging systems.

6.4.2. High Success

High success is defined to mean sufficient scientific data return from the experiment to perform direct comparison with the numerical flame simulation resulting in multiple archival journal publications, but less return that defined for complete success. This high level of success may be achieved by obtaining (for example):

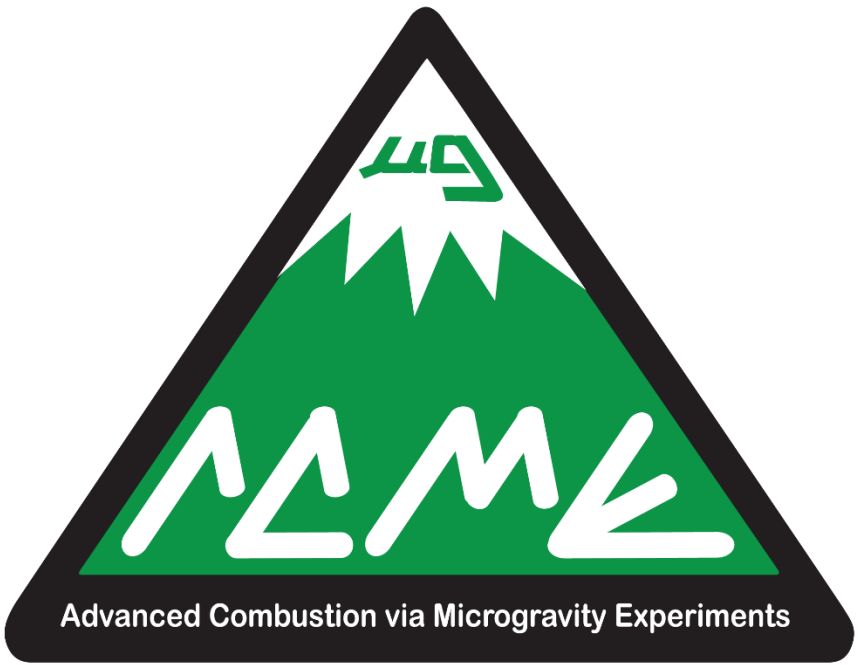
1. Observations of at least two instances of flame front instabilities including at least one hydrocarbon fuel. Such observations must include, at a minimum, the flow and boundary condition measurements, and data quantifying the flame front oscillations from image sequences from at least two of the three imaging systems and from the radiometer instrument; or
2. Combinations or extensions (in terms of number of mixture ratios for a single hydrocarbon fuel or number of hydrocarbon fuels) of at least two of items 1 through 4 defined for minimum success.

6.4.3. Complete Success

Complete Success is defined as meeting all of the experiment objectives,
including as a minimum:

1. Observations of transient flame phenomena leading toward steady state flames for at least three different mixture ratios of each of two hydrocarbon fuels and two diluents. Such observations must include, at a minimum, the flow and boundary condition measurements, flame temperature data, image sequences from two orthogonal imaging systems, and radiometer data; and
2. Observations of flame extinction at both low and high system Damköhler number for at least three different mixture ratios of each of two hydrocarbon fuels and two diluents. Such observations must include, at a minimum, the flow and boundary condition measurements, flame temperature data, image sequences from two orthogonal imaging systems, and radiometer data; and
3. Observations of at least two instances of flame front instabilities each including at least one hydrocarbon fuel. Such observations must include, at a minimum, the flow and boundary

condition measurements, and data quantifying the flame front oscillations from image sequences from at least two orthogonal imaging systems and from the radiometer instrument; and
4. Observations of soot formation. Such observations must include the flow and boundary condition measurements, flame temperature data, image sequences from all imaging systems, temperature data, and radiometer data.



3. Hardware Overview

A. General Description

This section is copied from section 2.A of the April 2021 version of 'ACME Engineering Implementation' which should be consulted for additional details.

This section provides a general description a general description of the ACME engineering from the '3.0 Experimental' section of the following 2014 conference paper:

> *Dennis P. Stocker, Fumiaki Takahashi, J. Mark Hickman, and Andrew C. Suttles, "Gaseous Non-Premixed Flame Research Planned for the International Space Station," Spring Technical Meeting of the Central States Section of the Combustion Institute, Tulsa, OK, 16–18 March 2014.*

IMPORTANT: *The 2104 paper from which this section was extracted proceeded completion of the spaceflight hardware which was mostly launched in 2017. As a result, it does not accurately describe the final engineering in some regards. Therefore updates, clarifications, additional details, shortcomings, references, etc. have been included in boxed notes like this following the original description. As a minor note, the sizes of figures were adjusted in this report, but none were cropped.*

As previously discussed, the ACME experiments will be conducted within the Combustion Integrated Rack (CIR) using a chamber insert, depicted in Figure 2, which was designed to accommodate gas delivery to a variety of exchangeable burners. Besides the chamber insert, other ACME-unique hardware includes a high-definition color camera and an avionics package, which is for data acquisition and control.

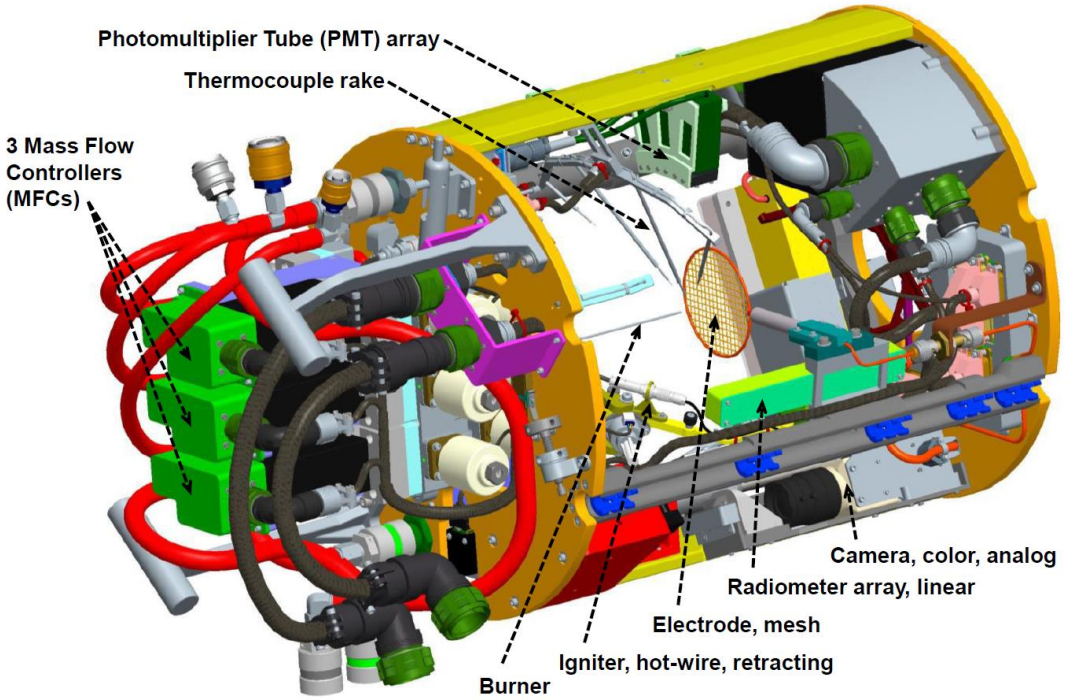


Figure 2: Design of the Advanced Combustion via Microgravity Experiments (ACME) chamber insert that will be used in the Combustion Integrated Rack (CIR) on ISS.

ACME's chamber insert, which is ~0.4 m in diameter, provides common mounting for a variety of burners on the chamber axis (but only one at a time) and includes a retracting resistive igniter. As depicted in Figure 3, there will initially be gas-jet burners (with inner diameters ranging from 0.4 to 3.2 mm), a small coflow burner (where the two concentric tubes have inner diameters of 2.1 and 25 mm), porous spherical burners (with diameters ranging from 6.35 to 12.7 mm), and two flat porous burners for the BRE experiment (with inner diameters of 25 and 50 mm). Most burners will be equipped with a surface thermocouple, and each BRE burner will have two heat flux transducers. The internal pressure of the porous burners will be measured to allow monitoring of the pressure differential across the outlet surface. The igniter will have a crew-exchangeable tip and arm (to accommodate the various burners) and its stepper-motor drive will allow for precise positioning.



Figure 3: ACME burner types, including from left to right, gas-jet burners, a co-flow burner, porous spherical burners, and flat porous burners for the BRE experiment.

Ultimately, the spherical burners were limited to a single diameter of 6.35 mm, and the fabricated gas-jet burners were limited to two sizes with nominal inner diameters of 1.3 and 2.1 mm. However, only the larger gas-jet burner was used during the spaceflight tests. The burners are addressed in detail in section 3.C.

Unfortunately, the spherical burners have not always yielded spherically symmetric flames, presumably as a result of (1) greater burner flow through the joint between the porous sphere and supply tube, and (2) heat loss to the support tube (encompassing the supply tube) causing local flame quenching.

Two types of igniter tips have been used during ACME testing. Most tests have been conducted using a loop igniter tip like those which had been successfully used in the droplet combustion experiments which preceded ACME research in the CIR. However, some tests were conducted with a coil igniter tip, where that included the second round testing for both the Burning Rate Emulator (BRE) and Structure and Response of Spherical Diffusion Flames (s-Flame) experiments, with the exception of the latter's first day testing. The CFI-G investigation will also use the coil igniter tip. For more information about the igniter tips, see section 3.D.

The chamber insert includes three mass flow controllers (visible on the left of Fig. 2) for the delivery of fuel, oxidizer or inert (e.g., helium or carbon dioxide), and nitrogen to the burner. The maximum fuel flow is 2 slpm (on a nitrogen basis), whereas much higher flows, e.g., 10 slpm (on a nitrogen basis), are possible for the oxidizer and nitrogen lines. The crew can exchange the mass flow controllers to provide appropriate ranges that maximize the flow accuracy. The gas delivery system enables on-orbit dilution of the fuel or oxidizer (but not both simultaneously), coflow, inverse flames (e.g., oxidizer delivery to the center of a coflow burner), and premixed flames,

although the latter are not planned for any of the current ACME experiments. Gases can also be delivered premixed to the space station where that is necessary in some circumstances, such as the blending of different fuel gases. CIR's standard bottle sizes are 1, 2.25, and 3.8 liters but ACME will sometimes use smaller bottles as a hazard control to limit the deficient reactant (e.g., fuel) that can be released into the combustion chamber.

Thus far, ACME tests were not conducted with carbon dioxide as a diluent. Furthermore, the fuel diluted with helium was premixed prior to launch rather than on orbit. Ultimately, oxidizer flows of up to 20 slpm (on a nitrogen basis) were possible, while the nitrogen flow was limited to a maximum of 5 slpm. ACME has not used gas bottles smaller than 1 liter as indicated above; the bottle fill pressure was instead limited as an additional hazard control.

ACME's primary flame diagnostics are a suite of imaging systems including five video cameras and two illumination systems, of which three cameras are part of the CIR facility and are already in use on orbit. All but one of the cameras is mounted outside of the chamber. Collectively, the imaging systems will enable color, CH*, and OH* imaging; pyrometry; measurement of the soot volume fraction; and operational support. However, it is unlikely that all functions would be exercised at any one time as will become apparent in the subsequent discussion.

Color imaging is provided by two cameras: a high-definition digital camera mounted outside of the chamber and a standard analog camera mounted on the chamber insert (with a turning mirror). Both cameras are part of the ACME hardware that is now in development. The analog camera signal will be downlinked in near real time to support the experiment operations. Light-emitting diodes within the chamber can be used to illuminate the igniter and burner to both support the operations and provide reference images. The 12-bit digital camera records at up to 30 full frames (1360x1024) per second, and is equipped with a motorized lens with independent control of the zoom, iris, and focus. The digital camera will be equipped with a filter assembly with four options, where selection of a clear "filter" allows for color imaging. Color imaging can also be carried out with a blue-green BG7 Schott glass filter, which balances the color planes to avoid saturation in the red while maintaining sensitivity in the blue and green. The natural flame color can be approximately reconstructed, but the primary purpose of the filter is pyrometry and measurement of the soot volume fraction, again via the techniques of [23-24]. Alternately, CH* imaging can be conducted with this camera because the filter assembly includes a 430-nm bandpass filter with a 10-nm full width at half-maximum transmission (FWHM). A 450-nm bandpass filter (again with a 10-nm FWHM) can be used with the CH* filter to effectively subtract interfering broadband soot emission. However, the filter assembly rotates slowly so such subtraction effectively requires a separate test with each filter and of course reproducible flame conditions.

The video from the analog operations (i.e., ops) camera, also referred to as the internal camera, is not archived in space. The video is instead recorded on the ground in the operations center. Its gain was sometimes adjusted for improved imaging of dim flames, e.g., approaching extinction.

The sensitivity of the color digital camera, often referenced as the ACME DataCam (i.e., Data Camera), to the dim microgravity flames was such that it was often used with long exposure times and thus low framing rates, for example on the order of 1 frame per second. The clear "filter" is of fused silica. Unfortunately, the thickness of the blue-green BG7 Schott glass filter and the added

antireflective (AR) coating caused too great a reduction in the red, so it does not balance the intensity of the three color planes as intended and thereby can limit the accuracy of temperature determination. Meanwhile, rather than the 450-nm bandpass filter described above, the high definition color camera was equipped with a bandpass filter for imaging formaldehyde emissions. For more information, see the Data Camera Characterization Report (ACME-RPT-010 Rev. A).

Imaging of the dim hydrogen flames in the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment was particularly challenging with these, if not all, cameras.

Pyrometry can alternately be accomplished with CIR's monochrome High-Bit Depth Multispectral (HiBMs) camera, equipped with a Liquid-Crystal Tunable Filter (LCTF), which allows for imaging at multiple spectra, but requires a quasi-steady flame for measurement at each desired band. This camera has a telecentric optical system and is recorded at 512x512 and 30 frames per second (fps). Although it has a 1024x1024 pixel array, binning has been necessary because of signal loss through the LCTF. Both soot pyrometry and Thin Filament Pyrometry (TFP) are possible, where ACME is equipped with a crew-replaceable array of 5 silicon carbide (SiC) fibers. The fibers are orthogonal to the flame axis and the array can be translated along the flame axis to the desired position downstream of the burner, e.g., based on the downlink of the analog video. During set up, the crew can manually orient the array so that it is orthogonal to either ACME's color digital camera or the HiBMs camera with the LCTF. An alternate fiber array including a platinum wire will be used for inflight characterization, which in combination with preflight calibration will enable determination of the soot volume fraction [24].

In contrast to the preceding paragraph, ACME did not use a HiBMs camera for pyrometry, but instead relied on use of ACME's color DataCam for such temperature measurements. For an example of how this can be done, see the techniques described in the following papers:

P.B. Kuhn, B. Ma, B.C. Connelly, M.D. Smooke, and M.B. Long, "Soot and Thin-filament Pyrometry Using a Color Digital Camera," Proceedings of the Combustion Institute, 33 (2011) 743-750.

B. Ma, M. B. Long, "Absolute light calibration using S-type thermocouples," Proceedings of the Combustion Institute, 34 (2) (2013) 3531-3539.

Thus far, TFP tests have been conducted with 14-micron fibers spaced apart by 5 mm for a total span of 20 mm (in the case of 5 intact ceramic fibers), neglecting optional translation of the array. Tests have been conducted using arrays with and without the platinum wire in the furthest downstream position. Tests have also been conducted with some of the fibers broken and subsequently removed, leaving only three fibers in the final testing for the BRE and Coflow Laminar Diffusion Flame (CLD Flame) experiments and at least the initial testing for the Cool Flames Investigation with Gases (CFI-G).

CIR has a second HiBMs camera, without a filter, that will be used in combination with CIR's illumination package for alternate measurements of the soot volume fraction. This will be accomplished via an extinction technique previously employed in microgravity [3,6,26], where the camera and the collimated light source (with its two fiber-coupled laser diodes) are positioned at opposite windows. The images are recorded at 1024x1024 and 30 fps. Thus far, the primary purpose of this combined system has been the measurement of the droplet diameter for determination of the droplet burning rate.

A HiBMs camera without the Liquid Crystal Tunable Filter (LCTF) has been used throughout much, but not all, of ACME's operations. In the first round of the CLD Flame experiment, it was used without a filter for flame imaging, but this proved to be largely unsuccessful because the camera is very insensitive to blue. Afterward, it was equipped with a red filter corresponding to the CIR illumination package for the rest of the ACME testing. Unfortunately, that filter is sufficiently broad that significant incandescence can pass through the filter limiting its utility for determinations of soot volume fraction via light extinction, e.g., through the technique described in the following papers:

D.L. Urban, Z.G. Yuan, P.B. Sunderland, K.C., Lin, Z. Dai, and G.M. Faeth, "Smoke-Point Properties of Nonbuoyant Round Laminar Jet Diffusion Flames," Proceedings of the Twenty-Eighth Symposium (International) on Combustion 28 (2000) 1965-1972.

F.J. Diez, C. Aalburg, P.B. Sunderland, D.L. Urban, Z.-G. Yuan, G.M. Faeth, "Soot properties of laminar jet diffusion flames in microgravity," Combustion and Flame, 156 (2009) 1514-1524.

P.S. Greenberg and J.C. Ku, "Soot Volume Fraction Imaging," Applied Optics, 36 (22) (1997) 5514-5522.

Nonetheless, soot volume fraction can also be determined from the color DataCam images.

During ACME operations, the HiBMs camera is typically archived at 15 fps because of data limits for transfer and downlink operations. And because operations are constrained by the data limits, image data from this camera are sometimes deleted without downlinking to enable additional testing. And in some testing, HiBMs images aren't even acquired given their lesser value.

Unfortunately, the HiBMs backlight illumination is significantly obstructed by the bracket for the electrode mesh further limiting its utility for the E-FIELD Flames experiment. This was a design trade-off necessitated primarily by ACME's large number of cameras and other optical instruments.

CIR's Low-Light-Level Ultraviolet (LLL-UV) package is a monochrome camera with an intensifier (for 280-700 nm) and a 310-nm bandpass filter (with a 10-nm FWHM) for OH* imaging. Again as a result of the signal loss through the filter, binning is employed such that the OH* images are recorded at 512x512 and 30 fps.

During ACME operations, the LLL-UV camera has only been used for the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment.

While not discussed in the preceding section, another intensified camera was used throughout ACME testing, although its image data was sometimes deleted or not even acquired to allow the performance of additional tests. This was the Cool Flames Investigation (CFI) camera which was generally configured with its CH filter, where an exception is some s-Flame testing with particularly dim flames. Due to design trade-offs, its Field of View (FOV) was partially obstructed by the arms of the Thin Filament Pyrometry (TFP) fiber array when it was moved inwards to the flame. More information about this camera can be found in its characterization report, CIR-RPT-4600.*

The ACME chamber insert will also be equipped with non-imaging optical diagnostics including both thermopile radiometers and Photomultiplier Tubes (PMTs). There is a crew-exchangeable linear array of radiometers of which there are two versions supporting the initial set of

experiments. The array has a fixed position which is parallel to the chamber axis and centered (along the chamber length) relative to the chamber windows and thus most of the other instrumentation. One array is equipped with a single wide-view radiometer with two narrow-view radiometers that are downstream and two more that are upstream, where all views are orthogonal to the chamber axis, as depicted in Figure 4. The narrow-view radiometers are intended to support studies of gas-jet flames, where ACME can accommodate shorter burners, making the "lower" detectors useful. The other array, also seen in Fig. 4, provides for three wide-view measurements of the flame, where one view is masked to nominally block radiant emission from the burner. It was specifically designed for the BRE experiment but will be a resource available to other ACME experiments. Given that the radiometer array can be exchanged by the crew, alternate arrays could be developed.

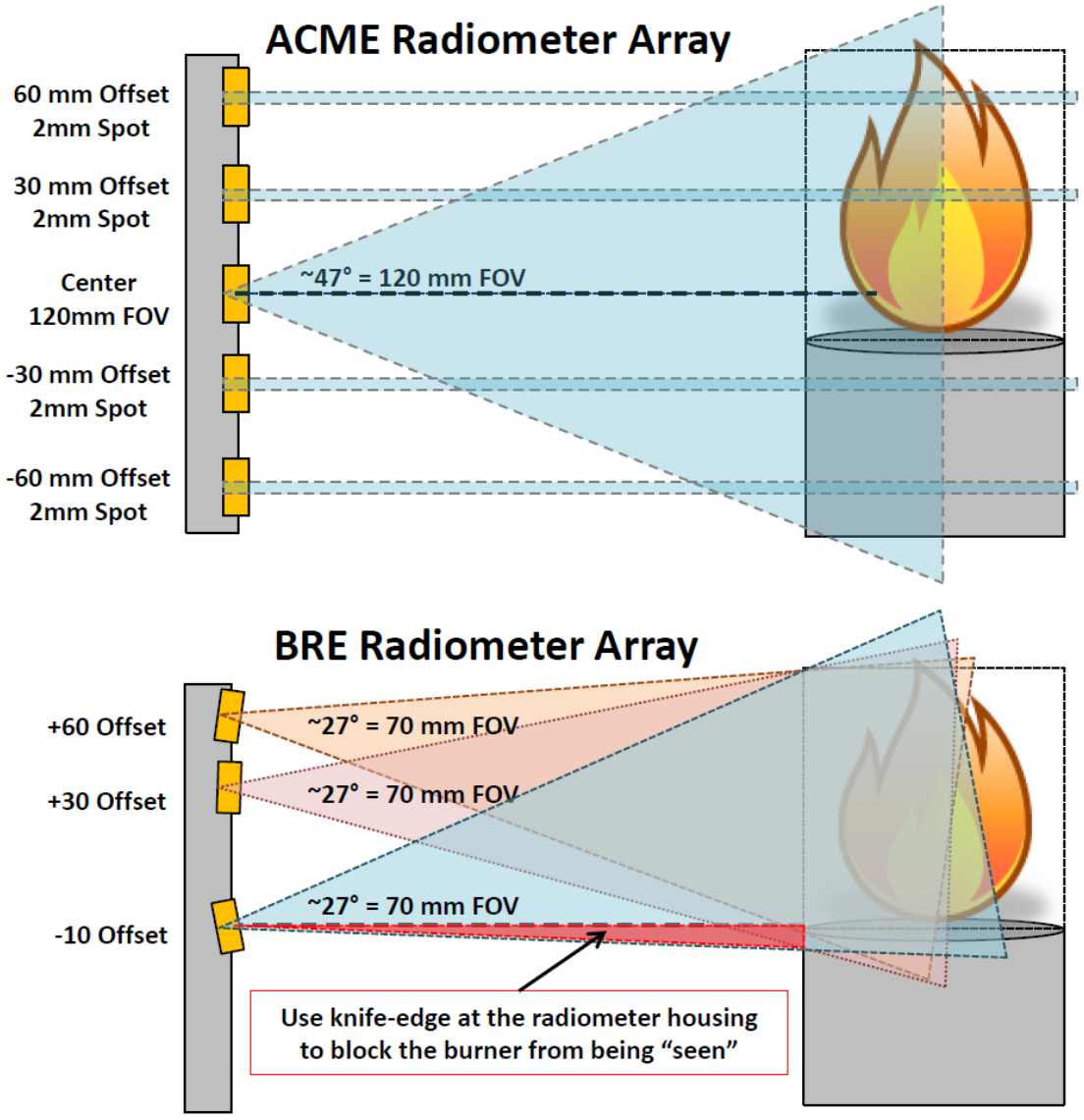


Figure 4: Crew-exchangeable radiometer arrays on the ACME chamber insert: (top) standard (ACME) array of 5 radiometers, and (bottom) array with three radiometers, designed for the BRE experiment, where the lowest detector is masked to avoid view of the burner.

A third radiometer array was developed for the CFI-G experiment, where it is important to understand that only a single radiometer array can be used at a time.

The narrow-view radiometers in the standard array are only fully sensitive in small spots, but they are each sensitive in a much greater Field of View (FOV) than represented in the preceding figure. They do not have lenses; their FOV is a result of an aperture where most of their FOV is within the penumbra which is only visible to a portion of the sensor. Figures representing the measured FOVs of the as-built standard and BRE radiometer arrays are found in section 3.F of this document.

Unfortunately, the radiometers in the BRE array typically saturated in the first round of BRE testing, so the standard array was used in its second round. However, those radiometers similarly saturated in many of round 2 tests. In this regard, each of ACME's BRE and standard radiometers have a fixed gain. Meanwhile, each sensor in the radiometer array developed for the CFI-G investigation has two fixed gains, i.e., a high gain and low gain.

For more information about the standard radiometer array used for most ACME testing, see the Radiometer Characterization Report (ACME-RPT-011). Meanwhile, you can learn about the other radiometer arrays from the BRE Radiometer Characterization Report (ACME-RPT-012) and the Cool Flames Radiometer Characterization Report (CIR-RPT-4600).

An additional wide-view radiometer, that is fixed and not crew exchangeable, is positioned downstream of the burner and flame at 30° from the chamber axis. This detector is primarily intended to support the spherical flame experiments and will have a quicker response than the more sensitive detectors that will used in the linear arrays. Like the radiometers on the two linear arrays, it will have a spectral range of nominally 0.2-11 micron and its field of view will be established by an appropriately selected aperture.

The FOV of the off-set radiometer is also shown in section 3.F. Learn more about this sensor from the Radiometer Characterization Report (ACME-RPT-011).

To detect the quenching of dim flames, the chamber insert will be equipped with a set of three PMTs. Each PMT will have a wide-angle view, a sight line that is orthogonal to the chamber axis, and will be centered (along the chamber length) relative to the chamber windows so that it is aligned with most of the other instrumentation. The PMTs' spectral range will be 230-700 nm, i.e., visible and near ultraviolet, but one will have a 310-nm bandpass filter and another will have a 430-nm bandpass filter (where both are with a 10-nm FWHM). The pair of detectors will enable measurement of OH* and CH* emission, respectively. The third PMT will be equipped with a neutral density filter to prevent saturation. Like many other components, the PMT array can be removed from the chamber insert by the crew.

Unlike the radiometers, each PMT had a readily adjustable gain. However, the PMT signals sometimes saturated when the gains were set too high. See the PMT Characterization Report (ACME-RPT-013) to learn more about the three Photomultiplier Tube (PMTs).

The chamber insert is also equipped with six far-field thermocouples to measure the spread of the thermal field in the spherical flame experiments. They are of limited utility for other experiments because they are positioned upstream of the burner tip. Therefore, they will

generally not be installed by the crew to reduce the chance of accidental damage. There will be two versions of a four-thermocouple rake to allow a choice of near and far measurements. Other arrays could be designed to support future experiments, but they cannot extend downstream of the burner tip without interfering with other (e.g., optical) measurements.

The two lone thermocouples, which are both at nominally 200 mm from the center of the spherical burner, have been installed throughout ACME testing. The tip of one was seen to be slightly bent early in testing, as discussed and depicted in section 3.G of this report.

Only the short (or far) version of the radiometer rake has been used thus far, where factors include the need for crew time to switch between rakes and the heat loss that could result from thermocouples in the long (close) rake. The short (far) rake's thermocouple tips should nominally be at 45, 60, 100, and 100 mm from the spherical burner center. For thermocouples at the same distance from the burner center, the angle between the rays from the burner center to the two thermocouples should be 90 degrees. The bead size of the thermocouples was not measured, but it is undoubtedly at least twice the size of the 0.2 mm diameter wire.

The two thermocouples closest to the burner outlets on each four-thermocouple rake are type B, while the others are type K.

Experiment-unique hardware is also mounted on the chamber insert. For example, to accommodate the E-FIELD Flames experiment, the insert is equipped with a high-voltage supply and a crew-removable electrode mesh which can be set between +10 and -10 kV. The circular mesh is orthogonal to the chamber axis and is positioned (during set up) at either ~35 or ~50 mm downstream of the insulated burner. The ion current between the burner and electrode mesh will be determined via measurement of the voltage drop across a known resistor and application of Ohm's Law.

Additional images of the chamber insert can be seen in subsequent sections of this report. The hardware specific to E-FIELD Flames is discussed further in section 2.B.

Additional supporting measurements will be made, where the chamber pressure is a simple example. The ACME experiments will use the CIR's gas chromatograph for pre- and post-test analysis of the chamber atmosphere, e.g., for oxygen, nitrogen, carbon dioxide, carbon monoxide, and fuel gases. Acceleration measurements will be provided by the Space Acceleration Measurement System (SAMS) [27], which is the primary instrument for such measurements on ISS and previously served that role for the space shuttle.

The CIR's Gas Chromatograph Instrumentation Package (GCIP) was not used for the CLD Flame or E-FIELD Flames investigations and thus far was only sometimes used during BRE, Flame Design, and s-Flame testing. Its use requires significant time and as such effectively reduces the number of tests that can be conducted. However, estimates of the chamber contents were computed assuming that the fuel in a normal flame test (for example) was either fully or not at all oxidized, e.g., for unsustained ignition in the latter case.

The CIR facility was equipped with a 3-axis SAMS sensor head and the acceleration measurements, along with relevant documentation, are accessible via the home page for the Principal Investigator

Microgravity Services (PIMS), https://gipoc.grc.nasa.gov/wp/pims/home/. To be clear, the acceleration measurements are not archived with the ACME experiment data in the Physical Sciences Informatics (PSI) website, www.nasa.gov/PSI.

All ACME experiments will be constant volume (i.e., rather than constant pressure) studies, where the chamber's free volume will be about 90 liters. Burner flows will generally be constrained to limit the pressure rise, and tests will typically be 0.5-2 minutes in duration.

The chamber's free volume is approximately 83 liters depending on the installed hardware.

The ACME experiments will be configured, but not conducted, by the ISS crew. Instead, the experiments will be commanded from the ground, specifically from the Telescience Support Center (TSC) at the NASA Glenn Research Center in Cleveland, Ohio. They can be conducted in a nominally automated mode using pre-programmed scripts. However, it will also be possible to effectively operate the experiments in a "manual" mode, where changes can be made in response to the analog video downlink and the numerical data available in the telemetry stream. For example, a flow rate could be adjusted to approach a stability or soot limit. Step and ramp changes can be made in the flow rate(s) or electric field potential. The TFP fiber array can be translated uniformly through a soot-free flame to get a continuum measurement of the temperature field. In summary, ACME's experiment control will allow for broad flexibility.

The TSC is now also referred to as the Glenn ISS Payload Operations Center (GIPOC). Thus far, the ACME tests have always been conducted in a nominally automated mode.

B. Hardware Shortcomings

Key s-Flame shortcomings include image acquisition, limited imaging sensitivity (especially to the hydrogen flames), and the flow asymmetry from the burner (especially at high flow rates). Other shortcomings are discussed in this section and Appendices B to D, but the reader may want to refer to the *ACME Engineering Implementation* document for additional information.

At high flow rates, the porous spherical burner used for s-Flame did not produce spherically symmetric flames. While they were nominally spherical at moderate flow rates, the flame's center became increasing offset from the burner center when testing with higher flow rates. More specifically, the flame shifted toward the supply tube indicating that the sintered attachment of the porous sphere to the supply tube is more 'open' than the pores of the sintered sphere. In other words, the flow from the burner is not spherically symmetric. At sufficiently high flow rates, the flame becomes obviously non-spherical, where it can appear to be ellipsoidal or even parabolic in shape. In the latter case, the flame locally quenches near the supply tube.

When s-Flame's first fuel bottle was installed on 30 July 2019 (19211), it was at the expected pressure, but the fuel pressure was observed to drop with time, indicating a leak. Troubleshooting, including testing with other fuel bottles, was carried out and it was concluded that the leak was related to the bottle's quick connection and potentially caused by foreign debris. The bottle was returned to Earth unused and tested in the ground copy of the CIR, but a leak was not found in that ground testing. As a result of this anomaly, testing with a planned hydrogen/methane/nitrogen blend was not accomplished.

When a fuel or diluent Mass Flow Controller (MFC) was used at a small fraction of its maximum flow rate, the actual flow rate sometimes differed from that desired. More importantly, small flow perturbations were observed in some s-Flame tests, where the flow irregularities were most apparent through their effect on Photomultiplier Tube (PMT) measurements, e.g., of OH* and CH* chemiluminescence. For example, a momentary increase in fuel flow, naturally leads to increased light from the flame. Steps were made to mitigate such behavior, e.g., by zero-ing the MFCs prior to each test, but they were not completely successful.

Ignition of the flames in the second round, with helium rather than nitrogen dilution, presented a challenge. The properties of the two gases differ dramatically; for example, helium's thermal conductivity is several times greater than that of nitrogen. A cautious approach was taken in increasing the igniter power on the first test day with helium (11 May 2020, i.e., 20132) because of concerns about potential failure of the igniter tip and the stalled operations that would result. The number of tests accomplished with that day's conditions were limited as a result.

The standard igniter tip was subsequently replaced with a replaced with a coil, rather than loop, design which was believed to be more robust because of the thicker wire. The coil igniter was used successfully in subsequent testing, but still with caution where flames were not always ignited successfully. As a result, some tests with helium dilution were conducted in atmospheres with ~30% oxygen (by volume) instead of the planned ~21%, which as one would expect facilitated ignition. As a result, tests were conducted with an inconsistent mix of oxygen concentrations, which may yield valuable insights, but limits the planned test-to-test comparisons.

The CIR's Image Processing and Storage Unit for Gigabit Ethernet cameras (IPSU-G), which is used with the ACME Data Camera (DataCam) and Cool Flames Investigation (CFI) intensified camera, didn't properly archive all of the image data on 13 Sept. 2019 (19256), where a large fraction of the image data was lost while flames were burning. After troubleshooting attempts, the IPSU-G was removed and replaced by astronaut Jessica Meir on 24 October 24 (19297) where subsequent check-out on Oct. 28 (19301) confirmed normal archival of the image data.

The DataCam images were mis-formatted on 18 Dec. 2019 (19352), where the cause is unknown but there were repeated operational issues with the CIR's Image Processing and Storage Units (IPSUs) on that test day. Fortunately, the mis-formatting was not seen on subsequent test days and the problematic test day was nominally repeated on 24 Jan. 2020 (20024) to make up for the lost science.

The ACME DataCam, generally the primary camera for ACME testing, is unfortunately not as sensitive as had been hoped to the dim microgravity flames. A critical challenge for s-Flame was in imaging the pure hydrogen and diluted hydrogen flames, which were not apparent or hardly visible to the DataCam and analog operations camera. This unfortunately inhibits assessment of three-dimensional flame symmetry because the CFI intensified camera and Low Light Level Ultraviolet (LLL-UV) cameras were opposite of each other.

While infrequent, tests also sometimes aborted automatically, typically due to communication issues with an IPSU but sometimes because test parameters were accidentally assigned out-of-range values. Fortunately, such aborts occurred before the start of gas flow so little was lost other than the time in which the aborted tests were conducted.

In contrast, a combination of factors led to testing being cut short on 3 June 2020 (20155). A high pressure reading (for a carrier gas) in the CIR's Gas Chromatograph Instrumentation Package (GCIP) put the Fuel Oxidizer Management Assembly (FOMA) Control Unit, i.e. the FCU, in an off-nominal state. Rebooting the FCU inadvertently set the valve safety timers to zero, and they cannot be reset without venting and refilling the combustion chamber with the intended atmosphere. Those issues combined with an extended loss of commanding prevented further science operations on that day. However, a test day was added on 8 July 2020 (20190) to make up for the lost science.

4. Preliminary Results

A. Changed Test Plans

To limit experiment scope, ACME investigators were asked to develop test matrices including nominally 50 ignitions. The s-Flame investigators accordingly planned within that target limit whereas each of the other ACME experiments exceeded it, significantly so in most cases. To improve the balance between the ACME investigations, the decision was made to treat s-Flame's highly desired tests, featuring diluted ethylene, as required. This was proposed by Dennis Stocker in 2015 (if not earlier) and adopted by 2016, where s-Flame was conducted on the ISS in 2019-2020.

Long in advance of the launch of the ACME hardware, Stocker proposed that each experiment be conducted in two separate rounds to allow the investigators to dig into the results from past tests so that they could make the best use of the remaining testing and thereby maximize the scientific return. Although this plan was adopted, the limited crew availability in 2020 led to the decision to conduct s-Flame's two rounds of testing sequentially. Despite this decision, the second round ISS testing with flames didn't begin until 3.5 months after it was completed for the first round. As a result of this delay, the investigators did get a significant break between the test rounds even though ACME operations didn't shift to a different experiment.

In response to challenges with ignition during ISS tests for the preceding Coflow Laminar Diffusion Flame (CLD Flame) experiment in late 2017, the decision was made to develop a coil igniter as an alternate to the standard loop type. Although it was later determined that the igniter had not been properly aligned, the decision was made to continue with the development of the coil igniter. When ignition was found to be a challenge in s-Flame's second round testing featuring helium dilution, the decision was made to use the coil igniter for the remainder of those tests. The two igniter designs are discussed and depicted in the *ACME Engineering Implementation* document.

Given the ignition challenges in the second round, the decision was made to conduct some tests at elevated oxygen concentrations, whereas the original plan was to conduct all s-Flame tests with an oxygen concentration of nominally 21% (on a volume basis). This decision is discussed in the preceding section, 3.B Hardware Shortcomings.

A few minor changes were made to the script used to run the ISS tests:

2020-01-14	Added ignition detection prior to igniter arm retraction
2020-03-11	burnerOverLimitSec variable changed to global scope
2020-03-13	Fixed ramp-holds not ramping to desired flow delta

The first change was implemented near the end of the first round of testing, while the other two changes were made between the two rounds. The final version of the script is found in the *ACME/s-Flame Experimental Procedure* document.

B. Overview of Experiment Test Operations

This section is copied from the introduction (section 1) of the April 2021 version of 'ACME/s-Flame Experimental Procedure' which should be consulted for additional details.

The narrative description of the experimental procedure is an excerpt from the 'Methods / Experimental' section of the following conference paper.

Y.-C. Chien, J. Tinajero, D. Stocker, U. Hegde, and D. Dunn-Rankin, "Electric Field Effects on Flames in Microgravity on the International Space Station," 2018 Spring Technical Meeting of the Central States Section of The Combustion Institute, Minneapolis, MN, 20-22 May 2018.

While the paper was written for the Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment, much of the discussion applies to ACME in general, including the s-Flame investigation. The figure referenced in this excerpt is not included herein. Clarifications and additional information, not in the conference paper, are included in boxed notes like this.

The CIR experiments are configured, but not conducted, by the ISS crew. Thus far, ACME's activity has included the majority of the crew members, regardless of their nationalities. Most often the astronauts and cosmonauts exchange gas bottles, mass flow controllers, and igniter tips. In the latter case, the igniter tip exchange can occur after failure or proactively before failure when the crew already has changes to make changes in the interior of the insert. While these activities may seem mundane, the crew play a critical role in setting up the hardware for each experiment. As an extreme example, there were crew activities on 8 separate days in Sept.-Oct. 2017 to configure the CIR and ACME hardware for the first ACME experiment, namely the Coflow Laminar Diffusion Flame (CLD Flame) experiment. In contrast, only two crew days were required to reconfigure the hardware for the first E-FIELD Flames tests because of the initial use of the same coflow burner. Planning is critical because crew activity is scheduled three weeks in advance, for which detailed procedures must be prepared and uplinked to the ISS.

Similarly, the s-Flame experiment was conducted using the same burner as the Flame Design investigation which preceded it. However, there was hardware reconfiguration for s-Flame to enable use of an additional intensified camera for imaging OH chemiluminescence, namely the Low Light Level Ultraviolet (LLL-UV) camera.*

All CIR tests are remotely commanded from the ground from NASA's Glenn ISS Payload Operations Center (GIPOC) in Cleveland, Ohio. Typically, four to six people staff four consoles during the testing. The CIR hardware is controlled from the Rack Officer (RO) and Data Management Officer (DMO) consoles, and the ACME hardware is controlled from the Payload Developer (PD) console. Two shifts of personnel are scheduled to staff these three consoles on test days because they typically last 14 hours with the many pre- and post-testing activities. Given that the tests are remotely commanded, the operations are carried out in the day and evening in Cleveland, e.g., from 7 AM to 9 PM. A day's tests, i.e., with flames, are normally conducted in a period of 10 hours of less. Tests are normally conducted on three weekdays per two weeks. On intervening weekdays, the image data is downlinked and the crew often swap one or both gas bottles.

Transferring and downlinking data from a day's tests also requires a 2-shift day for the RO and DMO consoles, where it is always necessary to verify that data was downlinked without errors and that data is not missing. Those two positions are also staffed to guide and oversee the crew activity. As a result, the CIR consoles are normally actively staffed on four weekdays per week. A team of four carry out the detailed planning for the CIR and ACME console staff, and the daily operational plans typically exceed 50 pages on days including testing.

The science console allows for review of the analog video and non-imaging telemetry (at 1 Hz) in near real time but doesn't allow for commanding. In the case of E-FIELD Flames, arrangements were made so that the analog video is streamed to a secure computer at UC Irvine to allow the research team to participate in test operations and to recommend adjustments to the day's planned parameters. That team does not see the telemetry during the testing, but is in communication via phone and e-mail with their NASA research partners at the GRC ISS Payload Operations Center (GIPOC). While there are many preparatory steps, the tests are ultimately conducted using an uploaded software script and parameter files. In advance of the test day, the UC Irvine science team plans the nearly 60 parameters in the voltage sweep script and specifies other parameters that are configured outside of the script, such as camera settings. It is also possible to effectively conduct tests in a 'manual' mode, where parameters are changed in response to the analog video downlink or telemetry. For example, the electric field potential can be adjusted to approach a stability or soot limit.

Provisions have been made to enable the investigators from all ACME experiments to remotely view the analog video while they communicated verbally with NASA science personnel in the operations center. Thus far, all ACME flame tests, including those for s-Flame and E-FIELD Flames, have been conducted in an automated manner using software scripts and extensive parameter files.

Typically, ten E-FIELD Flames tests are conducted per test day. Despite the short test durations, the low rate results from several unavoidable factors. After a test is conducted, the 125-Hz non-imaging data is downlinked for immediate review by the science team. A small sample of images are often downlinked to ensure that the camera exposures are appropriate, where the times for those images are typically selected based on non-imaging data to provide extreme examples. The review helps to verify parameters or plan for adjustments in subsequent tests, e.g., to select the voltage range after determining where the ion current saturates for the given flame conditions. After a test is conducted, a fan circulates the chamber atmosphere after which the chamber is vented (to space) back to atmospheric pressure, if needed. Midway through a day's E-FIELD Flames tests, the atmosphere is circulated for 15 minutes through a filter cartridge filled with molecular sieves to capture any soot and adsorb at least a portion of the combustion products, e.g., carbon dioxide, water vapor, and incomplete combustion products. The tests are only conducted when both Ku and S band communication is available through NASA's Tracking and Data Relay Satellite (TDRS) System. Depending on the various orbits, there are periodic communication outages with a loss of signal (LOS) that can be a short handover (from one satellite to another) or can last as long as 30 minutes.

In contrast to E-FIELD Flames, an average of 15 flames were ignited per test day for s-Flame, but they were of even shorter durations. The pressure rise with an s-Flame test was small and the chamber was rarely vented after a test unless the chamber atmosphere was being adjusted for

other purposes. In this regard, the chamber atmosphere was sometimes scrubbed for the s-Flame investigation by recirculating it through the adsorber cartridge, but this was not a regular routine as in the case of E-FIELD Flames. The chamber was also sometimes partially or completely vented and refilled during s-Flame operations, for example to adjust the atmosphere's oxygen concentration.

Unlike E-FIELD Flames, the s-Flame experiment also made use of the CIR's Gas Chromatograph Instrumentational Package (GCIP) to measure species concentrations of the atmosphere in the combustion chamber. There is no sampling probe and measurements are only made before, between, or after tests when the chamber atmosphere was 'stirred' through use of a fan. For s-Flame, the GC measurements were made on most but not all test days, but they were limited in scope because of the significant time required. The GC measurements were sometimes only made at the end of the test day but were occasionally made earlier in the day, for example prior to an adjustment of the chamber atmosphere.

A variety of constraints limit the number of tests that can be performed, both overall and on a daily basis. The next combustion experiments in the CIR will feature solid fuels and focus on flame spread and material flammability. That follow-on plan limits the duration of ACME operations, where each ACME experiment is allotted four months for testing. Although it requires additional crew time, these allotments are split into two rounds so that the investigators have a break from testing for careful review of the early results so that they can maximize the scientific outcome of their second round. This paper reports on early results from the first round of E-FIELD Flames.

In contrast to earlier plans, the two rounds of the s-Flame experiment occurred sequentially because of limited crew availability. Due to delays in commercial crew transport, the space station had three rather six crew members through much of 2020. As a result of the reduced crew size, the s-Flame investigators did get a 3-month break (in Feb. to April) between their two rounds of flame tests allowing them to study early results and prepare for the second round. Hardware issues also stretched out s-Flame operations in the latter half of 2019.

While nitrogen is available from the space station, the combustion testing is limited by the available gas bottles that were and are launched to support the tests. For E-FIELD Flames and most of ACME's experiments, the test matrices used to plan those gas resources were established in 2010. At that time, NASA limited the scope of the ACME tests matrices with the intent of focused experiments with limited objectives rather than comprehensive experiments. As such, the general target was 50 tests per experiment. For most ACME experiments, the limiting gas constraint is the oxygen/nitrogen bottles used to fill the chamber. This resource constraint is exacerbated by the CIR because it is equipped with a pair of redundant safety timers to limit the fuel that can flow into the combustion chamber (where the timers can be set for the oxidizer flow in the case of inverse flame testing). The timers can only be reset by venting the combustion chamber. Therefore, the timer limit combined with other constraints effectively corresponds to the day's maximum flow time for the fuel (or more generally, the limiting reactant). For E-FIELD Flames, the safety timers are set to the maximum value of 999 seconds, which corresponds to the 10 tests typically completed per day.

The second round of the s-Flame experiment featured helium dilution of both the fuel and the oxidizer, so its testing was also limited by the oxygen/helium bottles launched for that purpose.

The 999-second limit on the valve safety timers has thus far been common for all ACME testing. However, the timers can be reset to 999 by nearly completely venting the chamber. Sufficient time and gas resources are needed to vent and refill the chamber and still conduct tests afterward and that was only occasionally done for s-Flame.

Another constraint is the available time in a day, where the console personnel and ISS resources, such as the video downlink, are planned for a limited number of hours. When necessary for the research, NASA does have continuous 24-hour/day operations for some non-CIR experiments. But as discussed, the ACME tests are typically two minutes or less in duration, so the testing is almost always limited to the planned durations.

Data limits are yet another constraint, where the on-orbit hardware can store a limited amount of data. More importantly, 127,000 MB is the most that can be reasonably managed in a two-shift transfer and downlink day. Therefore, tests days are limited to this amount of data, where image data from select cameras is often archived at reduced rates to mitigate this challenge. Nonetheless, the limit sometimes forces decisions to abandon recorded images.

This detailed description of operations and hardware provides the context under which the ACME experiments are conducted. …

Meanwhile, Table 1 is a summary log of the flames ignited, including a limited overview of the test conditions, where it is an excerpt from this report's Appendix D. 30-Day Operational Accomplishments Report for ISS Increment 63. Note that it is comprehensive and includes the earlier testing conducted in the preceding increments.

Table 1. Flames ignited for the s-Flame experiment

Test Day[1]	Year	Month	Date	Atmosphere[2]	Fuel bottle	Flames ignited[3]
First round of tests, with nitrogen-diluted fuel and oxygen						
19254	2019	9	11	~21% O_2 with N_2	50% C_2H_4 with N_2	13
19256			13			16
19304		10	31		45% CH_4 with N_2	9
19308		11	4	~30% O_2 with N_2		21
19352		12	18	~21% O_2, with N_2	100% H_2	10
20006	2020	1	6		44.4% H_2 with CH_4	17
20013			13		66.7% H_2 with CH_4	22
20024			24		100% H_2	16
20029			29		50% C_2H_4 with N_2	21
Second round of tests, with helium-diluted fuel and oxygen						
20132	2020	5	11	~21% O2 with He	25% H2, 20% CH4, with He	7
20136			15		20% H2, 25% CH4, with He	11
20142			21	~30% O2 with He	30% H2, 15% CH4, with He	17
20155		6	3	~30% and ~21% O2 with He	45% H2 with He	9
20160			8		45% CH4 with He	12
20164			12	~30% O2 with He	20% C2H4 with He	11
20170			18	~30% and ~21% O2 with He	25% C2H4 with He	20
20176			24		30% C2H4 with He	23
20190		7	8		45% H2 with He	21

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Atmosphere** is ~1 atm but pressure increases and O2% drops with vitiation.
[3]**Flames ignited** is the number of nominally sustained flames ignited on that day (neglecting the day change on ISS).

C. Summary of Preliminary Results

The s-Flame experiment aims to characterize and gain predictive capability of the structure and dynamics of diffusion flames in simple, well-defined flow fields such that the phenomena of interest can be studied without being complicated by various flow field effects. The basic flame configuration adopted for the investigation is the spherically symmetric diffusion flame generated by discharging a fuel gas from a porous spherical burner into a quiescent, oxidizing atmosphere in microgravity. The theoretically one-dimensional nature of the system facilitates data acquisition, data reduction, and computational simulation. The s-Flame experiment uses the Combustion Integrated Rack (CIR) and the ACME hardware in the Destiny module of the ISS.

Specifically, the s-Flame tests are intended to:

(i) Advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames;

(ii) Obtain experimental data in simple, well-defined flow fields and understand various key elemental diffusion flame processes;

(iii) Identify the extinction limits for both radiative and convective extinction (i.e., at high and low system Damköhler numbers, respectively);

(iv) Determine the existence, onset, and nature of pulsating instabilities that have been theoretically predicted to occur in such flames with fuel/diluent mixtures that are above a critical Lewis number;

(v) Validate and/or improve accuracy of computational simulation including detailed chemistry and transport.

The project has completed two (2) rounds of microgravity experimentation. Currently, detailed analyses of the acquired data, computational simulation, and theoretical analysis are ongoing. Interpretation of all phenomena will emphasize the simultaneous and coupled influence of chemistry and transport.

Table 1, on the preceding page, summarizes the tests conducted for two rounds of tests, which are meant to address the scientific objectives listed above. All experimental data address transient phenomenon, as the flame evolves from localized ignition towards “steady” behavior. The flames then evolve in a quasi-steady state manner (becoming history independent), after a short initial transient. Conventional gaseous fuels including hydrogen, methane, and ethylene are used. Hydrogen is the simplest fuel and its reaction mechanism is also reasonably established. Methane is the simplest hydrocarbon, and extensive studies have been conducted on its combustion chemistry. Ethylene is a typical sooty fuel and is also an important intermediate for the oxidation of higher hydrocarbons. Diluents mixed with fuel include nitrogen and helium to adjust mixture diffusivities and characteristic flame temperature.

The rationale for the matrix of tests is to utilize various mixtures to assess both chemical and transport effects (including radiative) on flame behavior. Pure H_2 and pure CH_4 are utilized to assess kinetic mechanisms associated with each. Additionally, pure H_2 only produces H_2O as its radiative product, thereby isolating its radiative properties. Mixtures composed of both H_2 and CH_4 are used to examine their combined chemical and transport effects. N_2 and He inerts which affect characteristic flame temperature and transport (molecular and radiative), in the same mole

fractions, are utilized for comparison.

As seen in Table 1, the first round of tests examined nitrogen-diluted fuel burning in a nitrogen-diluted oxidizing atmosphere.

Test day 19254 examines nitrogen-diluted C_2H_4 issuing into 21% O_2 / 79% N_2 atmosphere. C_2H_4 allows for the examination of a fuel with nominal unity Le number, setting a baseline standard for flame stretch and thermal diffusive stability effects. Additionally, its characteristic sooting nature (in normal gravity) directly addresses our basic science objectives focusing on spreading of flame sheets, dual extinction states, and flame-front instabilities. For a case of fuel-mixture of 20% C_2H_4 / 80% N_2 at a flow rate of 3 cc/s, a seemingly quasi-steady flame is attained for extended duration. For the same mixture, higher flow rates of 5-10 cc/s result in radiative extinctions, corresponding to high system Damköhler number. Other mixtures of 25% C_2H_4 / 75% N_2 and 30% C_2H_4 / 70% N_2, also show radiative extinctions at higher flow rates. Since characteristic residence times will affect sooting tendencies, various flow rates are investigated for different mixtures. Soot is only observed during ignition, which goes away as the flames establish themselves, as seen for cases where low flow rates (e.g., 3 cc/s) with low ethylene contents (e.g., 25-30% C_2H_4) are ramped to high flow rates (e.g., 5-10 cc/s) with higher ethylene contents (e.g., 40-50% C_2H_4). Soot will also enhance flame radiation, but our non-sooty flames already extinguish as they spread outwardly. Observations of such radiative extinction behavior at high flow rates is an achievement because such phenomenon was not observed conclusively in our ground-based studies, given the limited microgravity durations. Radiative reabsorption extends flame size at extinction. In ISS testing, radiative extinction has been found to be a common event, which was predicted computationally, but not definitively observed previously for our flame conditions.

Test day 19256 uses nitrogen-diluted C_2H_4 issuing into 21% O_2 / 79% N_2 atmosphere to examine kinetic extinction at low system Damköhler numbers. Similar to the radiative extinction cases, sufficient microgravity duration is needed to observe kinetic extinction because a near steady flame needs to be established first, before ramp down in flow rate to find properly the chemical kinetic extinction limit. For mixtures with 20-30% C_2H_4 content, spherical non-sooty flames are established at 3 cc/s; and kinetic extinctions are observed during ramp down before reaching 1 cc/s. For the same mixtures, radiative extinctions are observed when the non-sooty flames are established at 3 cc/s and the flow rates are ramped up before reaching 8 cc/s. Quantitative determination of the states of flame extinction, especially the dual extinction modes at high and low system Damköhler numbers have been achieved experimentally and are being compared to computational simulation.

Test day 19304 investigates 45% CH_4 / 55% N_2 mixtures issuing into 21% O_2 / 79% N_2 atmosphere. This pure-fuel methane case is used to compare with CH_4/H_2 blend mixtures. However, quasi-steady flames could not be established for the given fuel MFC (0.5 slpm). Consequently, the 19308 tests investigate the nitrogen-diluted methane flames in increased oxygen atmosphere, i.e., 30% O_2 / 70% N_2. Methane concentrations from 45% down to 15% are varied. For 45% CH_4 / 55% N_2, relatively long duration flames of ~15 s and ~10 s are established at flow rates of 2 cc/s and 3 cc/s, respectively. At larger flow rates of 5 and 9.4 cc/s, the flames expand to large radii and then extinguish, demonstrating radiative extinction at high system Damköhler number. Reduction of flow rate from 2 cc/s to 1 cc/s results in extinction during ramp down, indicating kinetic extinction at low system Damköhler number. For 30% CH_4 / 70% N_2, flow rates of 3-10 cc/s produce flames that expand to radiative extinction; and reduction of flow rate from 3 to 1 cc/s contracted to

kinetic extinction. Similar results are obtained for mixtures of 25% CH_4 / 75% N_2 and 20% CH_4 / 80% N_2. For 15% CH_4 / 85% N_2, flames are ignited, but extinguish soon after. These tests of 19308 suggest that like ethylene, methane flames can manifest both radiative and kinetic extinction for the conditions specified. We are currently mapping computationally the quasi-steady isola response for these flames to compare with the experimental extinction responses.

Test day 19352 was intended to examine various concentrations of nitrogen-diluted H_2 (45%-70%) issuing into air. However, various technical problems manifested that necessitated re-running of the tests on 20024 (as a make-up test day).

Test day 20006 examines 20% H_2 / 25% CH_4 / 55% N_2 mixtures issuing into air. Mixture of methane and hydrogen will be compared to those of pure fuels of methane and hydrogen to address chemical kinetic and transport aspects. With regard to radiative extinction, mixture compositions of 20% H_2 / 25% CH_4 / 55% N_2, 25% H_2 / 20% CH_4 / 55% N_2, and 30% H_2 / 15% CH_4 / 55% N_2 have been shown in computational simulations to be characterized by different radiative extinction times. These mixtures also avoid sooty ignition, which can otherwise result in asymmetrically trapped soot that deteriorates flame sphericity. For this mixture, the flames extinguish, some with wavy oscillating behavior just prior. However, it seems that even for very small flow rates, e.g., 2 cc/s, extinction is by the radiative mechanism during flame expansion. There was no parameter space in terms of flow rate and burner size to identify conclusively kinetic extinction during ramp down for these mixtures. Radiative extinction (with local flame extinction and re-ignition seems to occur prior to ultimate extinguishment) during flame expansion after ignition is observed.

Test day 20013 mainly investigates 30% H_2 /15% CH_4 / 55% N_2 mixtures issuing into air. Again, mixture of methane and hydrogen are run to compare with pure fuels of methane and hydrogen. Note that 25% H_2/ 20% CH_4 / 55% N_2 mixtures could not be run on the ISS because the mixture bottles had leaked. Radiative extinction is difficult to identify conclusively because flames with larger flow rates, e.g. 15 cc/s, seem to last, although with patchy extinction areas, until time out. Larger flow rates could not be examined either because of limited MFC range or distorted flame shape caused by burner limitations at high flow rates. Also, kinetic extinction during ramp down could not be identified conclusively, as some flames seem to extinguish because they contract into the burner.

Test day 20024 is a makeup day for 19352 – when the 0.1-slpm fuel MFC range was too low to examine radiative extinction and the analog and intensified cameras did not pick up the flame images at all or well – examining nitrogen-diluted H_2. Here, a 0.5-slpm fuel MFC is used and only a fused silica filter is used for the intensified camera. 45%-15% H_2/N_2 balance mixtures are investigated, where flames at high flow rates, e.g., 10 cc/s, did not extinguish, appearing quasi-steady until time out. The hydrogen flames would likely need to reach a diameter much larger than that accommodated by the experiment to extinguish radiatively. The flames seem to extinguish during ramp down, but further analysis is needed to assess kinetic extinction at low Damköhler number.

Test day 20029 uses and repeats and augments conditions of 19256 – when the images were corrupt – for nitrogen-diluted ethylene flames. Again, the results indicate both kinetic and radiative extinction phenomena under corresponding conditions. No sooting flames are observed.

As seen in Table 1, the second round of tests replaces the inert, N_2, with He, to better isolate Le>1 instabilities. However, higher characteristic flame temperatures (given the lower molar heat capacity of monatomic He versus that of diatomic N_2) correspond to lower Zeldovich numbers, which would suppress oscillations based on Sivashinsky's criterion for *premixed* flames. Moreover, lower Ze may also affect extinction processes.

Test day 20132 examines helium-diluted fuel mixture of 25% H_2 / 20% CH_4 / 55% He issuing into an ambient of 21% O_2 / 79% He. The high thermal conductivity of He drastically shifts the ignition parameters from that using N_2 as inert in Round 1. The tests produced flames at 3-10 cc/s that expand outwardly quickly and extinguish. For 5-10 cc/s, the flames seemed to disappear and 're-ignite' with a "flash" before final extinction. We are currently investigating the flash phenomenon.

Test day 20136 investigates the fuel blend 20% H_2 / 25% CH_4 / 55% He issuing into 21% O_2 / 79% He. For 5 cc/s and 7 cc/s, the mixture ignites but immediately extinguishes, with almost no observable flame-front expansion. For 10-17.5 cc/s, the mixtures ignite, expand quickly, and extinguish. The extinction mechanism is believed to be radiative and is being assessed with computational simulations. For the cases with a ramp down in flow rate, the flames seem to extinguish before ramp down. Thus, purely kinetic extinction could not be assessed. Only flow rates above 7 cc/s showed flame-front expansion leading to extinction, with 17.5 cc/s showing a flash or wavering of the flame surface before final extinction.

Test day 20142 investigates the fuel blend 30% H_2 / 15% CH_4 / 55% He issuing into 30% O_2/ 70% He. Elevated oxygen ambient facilitates ignition and results in smaller characteristic flame size. For 5-16.5 cc/s flow rates, flames expand to wavy/oscillatory radiative extinction. Ramp down cases did not allow for assessment of kinetic extinction.

Test day 20155 examines 45% H_2 / 55% He issuing into 30% O_2/ 70% He and 21% O_2 / 79% He. Radiative and kinetic extinction processes are not conclusively identified for these cases because the automated protocol ended the tests as the burner temperature routinely reached 250°C. Presumably, the thermally-conductive He diluent augments heat transfer to the burner. Nevertheless, nice transient flames are produced, such that the transient flame structure (e.g., temperature and species distribution) can be characterized by the ACME diagnostics and compared with computational simulation and theoretical analysis, along with cross comparisons with the N_2 inert cases of Round 1.

Test day 20160 investigates 45% CH_4 / 55% He issuing into 30% O_2 / 70% He and 21% O_2 / 79% He. For the 30% O_2/ 70% He atmosphere, the 5-15 cc/s flow rates are sooty at ignition, but subsequently become non-sooty, expanding to radiative extinction. At 6 and 7 cc/s, flames ignite less sooty, expanding with no soot to radiative extinction. For tests comprising 21% O_2 / 79% He atmosphere, at 5-15 cc/s, flames ignite sooty, but dissipate, and expand quickly to radiative extinction.

Test day 20164 studies ethylene-based flames, where 20% C_2H_4 / 80% He issues into 30% O_2 / 70% He and 21% O_2 / 79% He atmospheres. For 5 cc/s issuing into 30% O_2 / 70% He, the flame expands and remains quasi-steady until the burner limit temperature (250°C) is reached and the experiment shuts down. At 6-15 cc/s, the flames expand to radiative extinction, with the higher flow rates at 10-15 cc/s exhibiting sooty ignition. Fuel flow of 5.5 cc/s seems to correspond to a

condition where the onset of oscillating/wavy extinction is concurrent with the burner reaching the 250°C temperature limit. For the cases with a ramp down in flow rate, the flames seem to extinguish during ramp down, but heat loss to the burner may be the mechanism for extinction rather than kinetic extinction. Ignition is not achieved for tests conducted with 21% O_2 / 79% He atmosphere, for the ignition parameters utilized.

Test day 20170 investigates 25% C_2H_4 / 75% He issuing into 30% O_2 / 70% He and 21% O_2 / 79% He. At 5 and 5.5 cc/s issuing into 30% O_2 / 70% He, blue chemiluminescent flames are ignited, but eventually cut off because of the burner temperature limit being reached. At 6 cc/s, a blue chemiluminescent flame expands slowly to radiative extinction. At 7-15 cc/s, sooty ignition occurs, but the soot goes away as the flames expand toward radiative extinction. By ramping the flow rates up, starting at 5 cc/s, soot at ignition is minimized. Ramp down of the flow rate results in almost immediate extinction. For tests at 21% O_2 / 79% He atmosphere, cases at 5-15 cc/s produce flames that expand quickly to extinction. Less soot is present at ignition compared to those for 30% O_2 / 70% He atmospheres.

Test day 20176 explores 30% C_2H_4 / 70% He issuing into 30% O_2 / 70% He and 21% O_2 / 79% He. For 30% O_2 / 70% He, at 5 cc/s, a long duration flame is established that eventually reaches the burner temperature limit. At 6-15 cc/s, the flames are sooty at ignition, but the soot dissipates; and the flames expand to radiative extinction. Ramping up the flow rates, starting at 5 cc/s, minimizes soot at ignition and results in flames that slowly expand to radiative extinction. For tests at 21% O_2/ 79% He atmosphere, cases at 5-15 cc/s produce flames that expand to radiative extinction. Ramp down of the flow rate did not produce flames that could confirm kinetic extinction.

Test day 20190 is a makeup of 20155, with 45% H_2 / 55% He issuing into 30% O_2 / 70% He and 21% O_2 / 79% He. For both oxygen concentration environments, flow rates resulting in expanding flames lasted for a while but were cut off by automated protocol for burner temperature limit before radiative extinction. As for cases where the flow rates are ramped down, extinction detection is observed for both oxygen concentration environments. It is not clear if extinction is kinetic or from heat loss to the burner. Detailed analyses are needed. Tests with a reduced oxygen concentration atmosphere of 17.5% O_2 / 82.5% He seems to show both radiative and kinetic extinction. Additional analyses are needed.

D. Surprises and New Discoveries

Many discoveries are expected to arise as the data is analyzed and computational simulations are compared. Initial surprises and discoveries include:

(1) Nitrogen-diluted methane and ethylene flames in air seem to demonstrate kinetic extinction at low flow rates (low system *Da*) and radiative extinction at high flow rates (high system *Da*). Such phenomena were not observed in our ground-based research, given the limited microgravity durations. Nevertheless, comparisons with computational simulations are needed to assess better the extinction behavior. Radiative reabsorption extends flame size at extinction, making the chamber size an important parameter. Surprisingly, radiative extinction has been found to be a common event in ISS testing, which was predicted computationally, but not observed previously for our flame conditions. Similarly, sufficient microgravity duration is needed to observe kinetic extinction because a near steady flame

need to be established first, before ramp down in flow rate to find properly the chemical kinetic extinction limit. The nitrogen-diluted methane flames were tested in air and enriched O_2 environments. The enriched O2 environments produce smaller characteristic flames; nonetheless, both extinction limits appear to have been observed.

(2) Ethylene flames which tend to soot under normal gravity conditions did not soot under microgravity conditions. Sooting only occurred during forced ignition from the hotwire, but soon dissipated. Inherent sooting was also not found for enriched oxidizer conditions or for either of the inerts, i.e., N_2 and He, examined.

(3) Nitrogen-diluted hydrogen flames seem to need to be at very large characteristic sizes to self-extinguish. As such, radiative extinction was not observed for the cases run, despite lowering the H_2 concentration. However, kinetic extinction during ramp down seem to be exhibited for hydrogen flames.

(4) Nitrogen-diluted hydrogen/methane mixture flames manifested radiative extinction, as predicted by computational simulations. However, kinetic extinction at low flow rates could not be identified conclusively. The characteristic extinction radius may be within the burner radius, precluding experimental identification. More analyses of the data are needed.

(5) Helium-diluted cases had shifted extinction limits with respect to the nitrogen-diluted cases. In general, kinetic extinction was difficult to observe for the conditions examined. Ignition was also severely affected by using He as diluent. More analyses are needed to assess instability near the flame limits, which was predicted to occur computationally.

(6) The He inert cases are meant to better isolate pulsating instabilities that are theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures that are above a critical Lewis number. However, such instabilities were not conclusively observed in the ISS tests, at least from the video images. Radiometer and PMT data will need to be examined in detail.

(7) The highly thermal conductive nature of He seems to affect various processes, such as ignition and heat transfer to the burner (affecting isolation of kinetic extinction), despite higher characteristic flame temperatures.

(8) Flame luminosity does not necessarily correspond with flame strength (or combustion intensity).

E. Accomplishment of Approved Objectives

The ISS experimental matrix was proposed to address the following specific objectives, in both soot-free and sooty flames. Comparisons with detailed computational simulation will permit assessment of transport properties and detailed chemical kinetics (including chemiluminescence). The accomplishment aspect is given after the *statement of each objective*.

Objective A. *Measure and characterize the transient structure (such as temperature and species distribution) of diffusion flames established at initially non-steady, spherical flame structures for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty), for different flow rates.*

Transient flames, both expanding and contracting, were achieved for all fuel mixtures. Some flames appeared to be quasi-steady in their evolution from a non-steady ignition origin.

Characterization of the flame structures is still being done, e.g., analyzing and processing data from thin-filament pyrometry to determine the temperature distribution. Species distribution is not explicitly measured and would need to be inferred from data such as radiometry. Characteristically sooty flames were not observed. Transient sooting only occurred during forced ignition; however, the ignition process and associated phenomena are being analyzed. Transient oscillatory/wavy response of the flame surface prior to extinction was often observed and is being analyzed.

Objective B. *Determine the quasi-steady convective/chemical-kinetic extinction limits (low system Damköhler numbers) and radiative/chemical-kinetic extinction limits (high system Damköhler numbers) for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty).*

Nitrogen-diluted ethylene and methane evinced both radiative and kinetic extinctions. Nitrogen-diluted hydrogen/methane blends only demonstrated radiative extinction for the parameters tested. Nitrogen-diluted hydrogen cases may have indicated kinetic extinction, but more analyses are needed. Helium-diluted hydrogen/methane blends may have shown radiative extinction, but more analyses are needed. Helium-diluted hydrogen displayed radiative extinction and showed kinetic extinction for oxygen-reduced environments. Helium-diluted ethylene and methane exhibited radiative extinction.

None of the flames were inherently sooty, so the effect of soot on extinction could not be investigated.

Objective C. *Determine the existence (for different Zeldovich number, Ze), onset (as a function Da), and nature (i.e., mode, frequency, amplitude) of pulsating instabilities theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures that are above a critical Lewis number.*

Many flames presented unstable behavior near the burner tube prior to extinction, likely because of lack of fuel mixture and heat loss in that region, rather than inherent pulsating instability. However, additional analyses, especially of PMT and radiometer data, are needed to assess the existent of pulsating instability, especially for helium as diluent.

F. Additional Objectives Achieved

No additional objectives have ***yet*** to be achieved. However, as we analyze and interpret the data, we expect new discoveries and insights into the fundamental mechanisms. Computational simulations will augment some test cases, and comparisons with experimental data will allow for extraction of basic parameters and assessment of chemical kinetic mechanisms. The role of the inerts will be better revealed, as the tests showed that they are far from passive participants in the flame dynamics and responses.

Some additional objectives that may be achieved in the future would involve the following aspects given below.

(A) Some experimental cases involved initial ramp up with subsequent ramp down in the flow rate, which may reveal hysteresis in the flame response.

(B) The acquired data will allow for investigation of flame ignition and transient dynamics in 2D and 3D geometry, along with analysis of the emergence and dynamics of 2D and 3D wave patterns for near-limit spherical diffusion flames, and analysis of their role in flame extinction.

G. Impacts, Benefits, and Products

The impact of this work consists of both practical and fundamental aspects. Practically, with diffusion flame burning as the basis in many combustion engines and devices, understanding the underlying dynamics and chemistry of diffusion flames is important in improving performance in terms of power, efficiency, pollutant emissions, and reliability. Fundamentally, the unique buoyancy-free environment offered by microgravity affords simple aerodynamic flow fields to acquire diffusion flame data of high fidelity, with the prospect of understanding unit thermochemical processes as well as quantifying the associated physico-chemical parameters.

In analyzing and validating the data sets, the experimental data and corresponding computational simulations should yield quantitative and qualitative understanding of various elemental flame phenomena. In studying the transient motion of diffusion flames ignited from a non-steady-state initial condition, the significance of fuel accumulation will be evaluated, which was originally found to be important in droplet burning. Here, with no droplet surface regression nor coupling of fuel flow to heat transfer from the flame to influence flame-front motion, the effect can be better isolated and the far-field unsteadiness assessed. Quantitative predictability of the flame motion permits assessment of the parameters and mechanisms involved in radiation loss and re-absorption as well as other transport parameters. The microgravity environment greatly facilitates the examination of pulsating instabilities, which can be hidden or coupled to other processes in normal gravity. Moreover, involvement of multi-dimensional effects, rather than the nominally 1-D geometry here, would further complicate the manifestation of the phenomenon. Near-limit phenomena, including extinction, is best evaluated and validated using our one-dimensional spherical diffusion flame (as realized in the microgravity environment), which is probably the most desirable flame configuration for study of this nature. Understanding extinction phenomenon is paramount to revealing basic flame processes, along with highlighting implications in fire safety.

The results are of strong interest to other groups working on diffusion flames, including those also doing ISS-based experiments. The main products will be (*i*) insights from fundamental characterization of transient dynamics and responses of such flames, (*ii*) categorization of processes that are either quasi-steady (e.g., expanding flame front motion toward steady state) or fully transient (e.g., pulsating instabilities prior to extinction), (*iii*) quantitative and qualitative mapping of extinction boundaries and unstable regimes, (*iv*) clarification of the role of diluents in various processes, and (*v*) additional understanding of ignition in non-premixed flames. Engines and devices on Earth can be improved in efficiency, pollutant emissions, and dependability from such products.

H. Patents

No patents have been filed from the s-Flame research, yet.

5. Follow-On Experiments

Although two rounds of testing have been completed, with no further testing planned, if a third round becomes available, there are two main types of experiments that we would like to implement.

We would like to examine the same fuel mixtures (plus C_3H_8 as mentioned below) of the first two rounds, but with CO_2 as inert, including that for the ambient. N_2 and He are radiatively transparent gases; while CO_2 is an optically participating gas. Use of CO_2 should result in a characteristically lower flame temperature, increasing the Zeldovich number, *Ze*. At the same time, the highly reabsorptive CO_2 species can minimize net radiative heat loss affecting flame dynamics and extinction. Objectives A, B, and C would be addressed by these experiments.

Additionally, characteristically sooty flames were not observed in the first two rounds for the fuel mixtures tested. As such, we would like to investigate using C_3H_8 as fuel, which was originally specified in the SRD that was approved [in 2002, but which was not part of the s-Flame plan under ACME, which didn't begin until 2006]. Specific knowledge originally sought in the first two rounds with respect to sooty flames, which were not achieved consist of: (a) the transient and steady state behavior and structure of sooting spherical flames, (b) residence time effect on soot formation, with only curvature effects and no strain, and (c) the effects of soot radiation and transport on extinction, at constant mass flow rate.

The results so far will be compared with computational simulations to address deficiencies in fundamental parameters and models. The aforementioned CO_2 experiments are highly desired; however, experimentation with C_3H_8 is of utmost importance if we are to achieve our original objectives with respect to characteristically sooty flames, which were not attained for the fuel mixtures and conditions tested in the first two rounds.

Appendix A. Abbreviations and Acronyms

#####	ACME days are identified by year (2 digits) and the day of the year (3 digits, 001-366)
#####x#	ACME tests are identified by day (5 digits; see above), a letter, and a 1-digit number
0-g	microgravity
1-g	normal gravity (on the surface of the Earth)
ACME	Advanced Combustion via Microgravity Experiments (*CIR project*)
ADP	Acceptance Data Package
ASGSR	American Society for Gravitational and Space Research
AVP	avionics package
BRE	Burning Rate Emulator (*an ACME experiment*)
C_2H_4	ethylene
C_3H_8	propane
CH_2O	formaldehyde
CASIS	Center for the Advancement of Science in Space
CCD	Charge-Coupled Device (*such as a camera sensor*)
CFI	Cool Flames Investigation (*an MDCA experiment preceding ACME in the CIR*)
CFI-G	Cool Flames Investigation with Gases (*an ACME experiment*)
CH_4	methane
CIA	Chamber Insert Assembly (*inserted into the CIR combustion chamber*)
CIR	Combustion Integrated Rack (*ISS facility in which ACME experiments are conducted*)
CLD Flame	Coflow Laminar Diffusion Flame (*an ACME experiments*)
CM	Combustion Module (*space shuttle precursor to the CIR*)
cm	centimeter
CO_2	carn dioxide
Co-I	Co-Investigator
CRS	Commercial Resupply Service (*a designation of ISS cargo missions)*
CSA	Canadian Space Agency
CSV	comma-separated values
CWRU	Case Western Reserve University
Da	Damköhler number
DataCam	Data Camera (*ACME's primary camera which was characterized for pyrometry, etc.)*
E-FIELD Flames	Electric-Field Effects on Laminar Diffusion Flames (*an ACME experiment*)
EMI	electromagnetic interference
EMS	Experiment Mounting Structure
ERD	Engineering Requirements Document
ESA	European Space Agency
Fiji	Fiji Is Just ImageJ (*a version of ImageJ preloaded with many plug-in features*)
Flame Design	Flame Design (*an ACME experiment*)
FOMA	Fuel and Oxidizer Management Assembly (*CIR's gas management subsystem*)
FOV	field of view
fps	frames per second
GC	Gas Chromatograph (*see GCIP*)
GCIP	Gas Chromatograph Instrumentation Package (*for measuring CIR chamber contents*)
GigE	Gigabit Ethernet (*an interface standard for high performance cameras*)
GIPOC	Glenn ISS Payload Operations Center (*also known as the TSC*)
GMT	Greenwich Mean Time (*where most references in this report are the day*)
H_2	hydrogen

HiBMs	High Bit-depth Multi-spectral camera (*developed for the CIR facility*)
HOBJ	Halcon object (*binary uncompressed format; used for the DataCam*)
HV	high voltage
HVPS	High-Voltage Power Supply
ICDERS	International Colloquium on the Dynamics of Explosions and Reactive Systems
ID	inner diameter (*e.g., of burner tube*)
ImageJ	Image Processing and Analysis in Java (*open-source image processing software*)
IPSU	Image Processing and Storage Unit (*CIR device for recording images on ISS)*
IPSU-G	IPSU for GigE cameras (*including the DataCam and CFI intensified camera*)
ISRD	Integrated Science Requirements Document (*for a set of experiments*)
ISS	International Space Station
JAXA	Japan Aerospace Exploration Agency
kPa	kilopascal
kV	kilovolt
LLL-UV	Low Light Level Ultraviolet camera ((*developed for the CIR facility*)
LSP	Laminar Soot Processes (*a shuttle study conducted in the Combustion Module*)
MDCA	Multi-user Droplet Combustion Apparatus (*a CIR chamber insert*)
MFC	Mass Flow Controller
Mist	Water-Mist Fire Suppression Experiment (*a Combustion Module experiment*)
MIUL	Materials Identification/Usage List
mm	millimeter
N_2	nitrogen
NSF	National Science Foundation
NTSC	National Television System Committee
O_2	oxygen
OMA2	Optical Multi-channel Analyzer 2 (*open-source software for image processing)*
ops	operations
PARIS	Passive Rack Isolation System (*with which the CIR is equipped, but not used for ACME*)
PI	Principal Investigator
PIMS	Principal Investigator Microgravity Services
PMT	Photomultiplier Tube (*where ACME is equipped with three PMT modules*)
PS	Project Scientist
PSI	Physical Science Informatics System
psia	pounds per square inch
PT	pressure transducer
QD	quick disconnect
qty	quantity
RSA	Russian Space Agency, i.e., Roscosmos
s	second
SAMS	Space Acceleration Measurement System
SDP	Safety Data Package
s-Flame	Structure and Response of Spherical Diffusion Flames (*an ACME experiment*)
SiC	Silicon Carbide (*material of the TFP fibers*)
slpm	standard liters per minute (*a measure of the mass or molar flow rate of a gas*)
SOFBALL	Structure Of Flame Balls At Low Lewis-number (*a Combustion Module study*)
SoFIE	Solid Fuel Ignition and Extinction (*project to follow ACME in the CIR facility*)
SORD	Science Operations Requirements Document (*an ACME supplement to the ISRD*)
SRD	Science Requirements Document

TDRS	Tracking and Data Relay Satellite (System)
TFP	Thin Filament Pyrometry (*a method for gas-phase temperature measurements*)
TIFF or TIF	Tagged Image File Format (*for the CFI intensified, HiBMs, and LLL-UV cameras*)
TM	thermistor
TS	Transport Stream (*a video format*)
TSC	Telescience Support Center (*also known as the GIPOC*)
UV	ultraviolet
Ze	Zeldovich number
zero-g	zero gravity, i.e., microgravity
ZIN	ZIN Technologies, Inc. (*ACME engineering and operations provider*)
µA	microamp

While this list of abbreviations and acronyms is long, it is not comprehensive. Contact ACME Project Scientist Dennis Stocker (216-433-2166, dennis.p.stocker@nasa.gov) for additional clarification.

Appendix B. 30-Day Operational Accomplishments Report for ISS Increments 59 and 60

16 December 2019

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increments 59 and 60

ACME Advanced Combustion via Microgravity Experiments
s-Flame Structure and Response of Spherical Diffusion Flames

The s-Flame investigation is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The initial days of s-Flame testing with flames were carried out in the final month of Increment 60. As a result, there is nothing to be reported for Increment 59.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for s-Flame and ACME information.

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I. Introduction

Tests for the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment were conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. The purpose of s-Flame is to advance the ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth.

The spherical flames are ignited at non-steady conditions and allowed to transition naturally toward extinction. Tests are conducted with various inert diluents in both the fuel and chamber atmosphere. The fuel gases include hydrogen and methane mixtures for soot-free flames, and ethylene for sooty flames. One experiment objective is to identify the extinction limits for both radiative and chemical kinetic extinction, i.e., where the flame extinguishes from heat loss or is blown out, respectively. Another objective is to determine the existence, onset, and nature of

pulsating instabilities that have been theoretically predicted to occur in flames with certain fuel/inert mixtures.

More specifically, s-Flame has the following experimental objectives:

A. Measure and characterize the transient structure (such as temperature and species distribution) of diffusion flames established at initially non-steady, spherical flame structures for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty), for different flow rates.
B. Determine the quasi-steady convective/chemical-kinetic extinction limits (low system Damköhler numbers) and radiative/chemical-kinetic extinction limits (high system Damköhler numbers) for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty).
A. Determine the existence (for different Zeldovich number, Ze), onset (as a function Da), and nature (i.e. mode, frequency, amplitude) of pulsating instabilities theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures that are above a critical Lewis number.

As suggested by the experiment's name, s-Flame features a spherically symmetric diffusion flame, which is generated by discharging a fuel gas from a porous spherical burner into a still oxygen/inert atmosphere in microgravity. The one-dimensional nature of the system facilitates data acquisition, data reduction, and computational simulation.

The space flight program consists of two main parts: (1) Mapping and characterization of the motion (e.g., expansion) of spherical diffusion flames subsequent to ignition and at extinction because of the finite-rate chemistry and radiative heat loss. The study is focused on the thermal radiation, the adequacy of the detailed chemistry and transport submodels used in computational modeling, and the development of flame front instabilities that could modify the extinction boundaries. (2) Studies of the response of characteristically sooty flames to ignition, the onset of soot in the expanding flames, the soot radiation, and the soot's effects on the flame spread, extinction, and instability.

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the natural flicker of flames, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. For example, chemi-ion species, electronically-excited chemiluminescent species, are at very low concentration in typical flames, making the reduced buoyancy environment valuable for evaluating the minor species chemistry.

With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

The s-Flame investigation and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the testing is typically conducted in a nominally automated mode using pre-programmed scripts. However, it may alternately be conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. But it is not expected that such control, which might be used to approach a stability or soot limit, will be used for s-Flame. Thus far, that has been the case for this experiment, with the possible mundane exception of 'manually' halting the test after the flame has extinguished.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

A total of 29 flames were ignited for the s-Flame experiment in its two days of flame testing during Increment 60. Note that the investigation's science testing didn't begin until September 2019, so no flames were ignited during Increment 59.

The s-Flame experiment does not have living subjects like a life science experiment, but the tests are independently conducted with various gaseous fuels, including hydrogen, methane, hydrogen/methane blends, and ethylene. These are simple hydrocarbon gases for which the chemistry is relatively well understood. The Increment 60 tests were all conducted with ethylene (C_2H_4), which is typically highly sooting because of its carbon double bond, but where the fuel was diluted with nitrogen for these tests.

No samples (e.g., of soot) were collected for s-Flame; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the flames ignited for s-Flame during Increment 60, including a limited overview of the test conditions, is found in Table 1.

Table 1.

s-Flame log of flames ignited in Increment 60 (*where the investigation did not begin until Increment 60*)							
Test Day[1]	**Year**	**Month**	**Date**	**Atmosphere[2]**	**Chamber Pressure**	**Fuel bottle**	**Flames ignited[2]**
First round of tests, with nitrogen-diluted fuel, in Sept. 2019 (Increment 60)							
19254	2019	9	11	~21% O2, with N2	~1 atm	50% C2H4 with N2	13
19256			13				16
[1]Test Day includes the year as the first two digits followed by the 3-digit day of the year. **[2]Flames ignited** is the number ignited on that day (neglecting the day change on ISS).							

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

When s-Flame's first fuel bottle was installed on the CIR on July 30 (19211) it was at the expected pressure, but the fuel pressure was observed to drop with time, indicating a leak. Troubleshooting, including testing with other fuel bottles, was carried out and it was concluded that the issue was related to the first s-Flame fuel bottle, which contained a hydrogen/methane blend. It was concluded that the leak was related to the quick connection, where it might have been caused by foreign debris. The bottle was returned to Earth and tested in the ground copy of the CIR, but surprisingly a leak was not found in that testing.

During the review of the leaked fuel bottle, it was determined that the bottles with the hydrogen/methane blends (H_2/CH_4) were over-filled in that the flammable and explosive limits could be reached if the fuel were to leak into the open volume of the CIR rack. A mistake had been made in the determination of the bottle pressure which would be safe in the event of a leak. The two fuels were considered individually instead of collectively using Le Chatelier's mixing rule. Therefore, before their use, the plan is to vent each H_2/CH_4 bottle to a pressure that will be safe if a bottle leaks into the CIR free volume.

The CIR's IPSU-G Image Processing and Storage Unit for Gigabit Ethernet cameras, which is used with the ACME Data Camera (DataCam) and Cool Flames Investigation (CFI) intensified camera, didn't properly archive all of the image data on Sept. 13

(19256). A large fraction of the image data while the flame was burning was lost. Trouble shooting was subsequently conducted to hopefully resolve the issue before further flame testing was conducted. Ultimately, the IPSU-G was removed and replaced by astronaut Jessica Meir on October 24 (19297) where subsequent check-out on Oct. 28 (19301) confirmed normal archival of the image data. Regular testing resumed on Oct. 31 (19304).

2. Any other difficulties in obtaining scientific data

During Increment 60, one s-Flame test automatically aborted (due to the controlling software script), where that was prior to gas flow and the activation of the igniter – which is typical for other ACME experiments. As a result, there was little lost other than the time in the failed test attempt. ACME aborts normally occur because of communication issues with an IPSU, although in this case it was because of operator error. ACME aborts are infrequent, but there are sometimes days when they occur repeatedly.

Tests are of course also unsuccessful when the flame fails to ignite. Two tests on the first day of testing failed to ignite, where adjustments to the tests parameters associated with the relative timing of the fuel flow and igniter activation thereafter resulted in successful ignition, with the exception of the abort discussed above. For other ACME tests, ignition has sometime been unsuccessful because the settings for the automatic flame detection are such that detection threshold is reached because of the igniter glow rather than a flame. As a result of the detection, the hot-wire igniter is deactivated and retracted, which is of course an issue if the flame has yet to ignite. This is typically avoided by detecting ignition with measurement of the flame's OH* chemiluminescence with a Photomultiplier Tube (PMT) module. The OH* chemiluminescence, in the ultraviolet at 310 nm, is a good indicator of an ignited flame rather than the glowing igniter, which emits primarily in the infrared and visible spectrum. However, the combination of igniter settings and PMT gains can lead to false ignition detection.

At high flow rates, the burner does not produce spherically symmetric flames. While nominally spherical at moderate flow rates, the flame's center becomes increasing offset from the burner center. More specifically, the flame is shifted toward the supply tube indicating that the sintered attachment of the porous sphere to the supply tube is more 'open' than the pores of the spherical burner. In other words, the flow from the burner is not spherically symmetric. At sufficiently high flow rates, the flame becomes obviously non-spherical, where it can appear to be ellipsoidal or even parabolic in shape. In the latter case, the flame locally quenches near the supply tube.

Small flow perturbations have been noticed in some s-Flame tests, where ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel, nitrogen, and oxidizer have unfortunately not always performed satisfactorily. The flow irregularities have been most apparent through their effect on the Photomultiplier Tube (PMT)

measurements, e.g., of OH* and CH* chemiluminescence. A momentary increase in fuel flow, as has occurred in some s-Flame tests, naturally leads to increased light from the flame. The MFCs are tested - without flames - after they are first installed. Adjustments are typically made to (1) the setting that controls when the valves open and (2) the Proportional-Integral-Derivative (PID) settings. This ACME activity has involved significant testing, where there are an assortment of MFCs with different flow ranges for each of the three gas types. While the MFC adjustments have been beneficial, they have not eliminated all gas flow problems. It was seen in prior ACME testing that nitrogen oscillations (if any) tend to damp with time, so a nitrogen pre-flow can be used so that such oscillations might end before the flame is ignited. As another measure, the MFCs are re-zeroed prior to each test, where that was another measure adopted in earlier ACME testing.

C. Completeness/Quality of Data

As a result of the anomalies discussed in III.B.1, only two days of flame tests were conducted for the s-Flame investigation during Increment 60, whereas approximately 16 days of such testing are planned. However, 29 flames were successfully ignited which is significant in comparison to the 58 tests specified as required in the ACME Integrated Science Requirements Document (ISRD). Therefore a wealth of data was obtained for a limited set of test conditions.

However, a large fraction of the key image data from the tests conducted on Sept. 13 (19256) was not archived (again as discussed in III.B.1). As a result, that test day will be repeated.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually become 'hot' (i.e., dead) over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS). In this regard, note that the Combustion Integrated Rack (CIR), in which the experiment is conducted, is equipped with a Passive Rack Isolation System (PaRIS).

D. Status of Data Analysis

Image files are currently being processed. Radial position of maximum CH* chemilumenscence is being plotted against time. Different radii are being considered and plotted together, as some flames are not completely symmetric. MFC and PMT data are

being compared with video images to assess and confirm experimental settings. Radiometer data is being reviewed. Analysis of the fiber pyrometry data has not begun.

Computational simulations with detailed chemistry and transport of the test cases are underway. Time dependent simulations will be compared with the time-dependent experimental data. Steady-state simulations will produce the Isola curves mapping flame response versus system Damkhöler number for given fuel mixtures. The experimental cases will then be pin-pointed on the computational mappings to assess whether the experimental conditions were in quasi-steady state or near extinction limits, either kinetic or radiative. Oscillation behavior will be correlated to possible pulsating oscillations found in the simulations.

E. Preliminary Research Findings

The s-Flame experiment utilizes spherically-symmetric diffusion flames and their dynamics and responses to investigate fundamental processes, such as extinction, instability, and soot formation. ISS testing for 19254 and 19256 examined nitrogen-diluted ethylene flames in air.

The experiments of 19254 revealed that a quasi-steady flame could be attained for extended duration at low flow rate, i.e., 3cc/s for 20% C_2H_4 / 80% N_2. The flame was shut down by the automated system because the burner reached a cut-off temperature of 250°C. Although the flame oscillated near the end of the experiment, it was likely due to influence of the burner tube. Comparisons with computation simulations will help to clarify the involved processes. For higher flow rates, e.g., 5-10cc/s, with the same mixture, radiative extinctions were observed. Generally, the flames showed unstable behavior near the burner tube prior to extinction, likely because of lack of fuel mixture and heat loss in that region. Other mixtures, e.g., 25% C_2H_4 / 75% N_2 and 30% C_2H_4 / 70% N_2, also showed radiative extinctions at higher flow rates. For some cases, the pyrometry fiber interacted with the flame, allowing extraction of temperature in on-going analyses. Soot was observed only during ignition, which went away as the flame established itself. To asses whether these nitrogen-diluted ethylene flames are non-sooty or sooty, cases were run where mixtures were ignited and established at low flow rates (e.g., 3 cc/s) with low ethylene contents (e.g., 25-30% C_2H_4) and then ramped to high flow rates (e.g., 5-10 cc/s) with higher ethylene contents (e.g., 40-50% C_2H_4). The flames all radiatively extinguished, but no sooting was observed. Observations of such radiative extinction behavior at high flow rates is an achievement because such phenomenon was not observed in our ground-based experiments, given the limited microgravity durations. Radiative reabsorption extends flame size at extinction. In ISS testing, radiative extinction has been found to be a common event, which was predicted computationally, but not observed previously for our flame conditions.

The experiments of 19256 were aimed at examining kinetic extinction. Similar to the radiative extinction cases, sufficient microgravity duration is needed to observe kinetic extinction because a near steady flame needs to be established first, before ramp down

in flow rate to find properly the chemical kinetic extinction limit. High flow rate cases (e.g., 13.5 cc/s, 15 cc/s) showed asymmetrical flames which radiatively extinguished. Flow rates were lowered so that quasi-steady flames could be established before ramp down in flow rate. The 15% C_2H_4 mixtures were not strong enough to sustain a quasi-steady flame at 3 cc/s before ramp down. However, for stronger mixtures, i.e., 20-30% C_2H_4, very nice flames were established at 3 cc/s; and kinetic extinctions were observed during ramp down before reaching 1 cc/s. Radiative extinctions for ramp up cases were also examined. Similarly, for stronger mixtures, i.e., 20-30% C_2H_4, very nice flames were established at 3 cc/s; and radiative extinctions were observed during ramp up before reaching 8 cc/s.

Future tests will investigate soot formation processes, which have not been observed for the fuel mixtures and conditions tested thus far.

Early ISS results were presented in the following presentations:

1. Chung K. Law and Stephen Tse, "ACME - s-Flame," ISS Increments 61 & 62 Science Symposium, 17-19 Sept. 2019. (oral presentation)
2. Stephen D. Tse, Jonathan Shi, Delin Zhu, and Chung K. Law, "Dynamics of Burner-Generated Spherical Diffusion Flames In Microgravity," 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, 20-23 Nov. 2019. (poster presentation)

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Video(s) describing the ACME project can be found at https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/.

Videos related to s-Flame and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

None.

Appendix C. 30-Day Operational Accomplishments Report for ISS Increments 61 and 62

11 June 2020

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increments 61 and 62

ACME **Advanced Combustion via Microgravity Experiments**

s-Flame **Structure and Response of Spherical Diffusion Flames**

The s-Flame investigation is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The s-Flame testing began at the end of Increment 60 and continued through Increment 61, where progress was limited by both technical issues and limited crew availability. Because of the latter, no tests were conducted during Increment 62.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for s-Flame and ACME information.

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NASA Project Scientist (PS):	Dennis P. Stocker, NASA Glenn Research Center, Cleveland, OH

I. Introduction

Tests for the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment were conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. The purpose of s-Flame is to advance the ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth.

The spherical flames are ignited at non-steady conditions and allowed to transition naturally towards steady-state or extinction. Tests are conducted with various inert diluents in both the fuel and chamber (oxidizer) atmosphere. The fuel gases include hydrogen and methane mixtures for soot-free flames, and ethylene for sooty flames. One experimental objective is to identify the extinction limits for both radiative and chemical kinetic extinction, i.e., where the flame

extinguishes from radiative heat loss or is "blown out," respectively. Another objective is to determine the existence, onset, and nature of pulsating instabilities that have been theoretically predicted to occur in flames with certain fuel/inert mixtures at near-limit conditions.

More specifically, s-Flame has the following experimental objectives:

A. Measure and characterize the transient structure (such as temperature and species distribution) of diffusion flames established at initially non-steady, spherical flame structures for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty), for different flow rates.
B. Determine the quasi-steady convective/chemical-kinetic extinction limits (low system Damköhler numbers) and radiative/chemical-kinetic extinction limits (high system Damköhler numbers) for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty).
B. Determine the existence (for different Zeldovich number, Ze), onset (as a function Da), and nature (i.e. mode, frequency, amplitude) of pulsating instabilities theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures that are above a critical Lewis number, at near-limit conditions.

As suggested by the experiment's name, s-Flame features a spherically symmetric diffusion flame, which is generated by discharging a fuel gas from a porous spherical burner into a still oxygen/inert atmosphere in microgravity. The one-dimensional nature of the system facilitates data acquisition, data reduction, and computational simulation.

The space flight program consists of two main parts: (1) Mapping and characterization of the motion (e.g., expansion) of spherical diffusion flames subsequent to ignition and at extinction because of the finite-rate chemistry and radiative heat loss. The study is focused on the thermal radiation, the adequacy of the detailed chemistry and transport submodels used in computational modeling, and the development of flame front instabilities that could modify the extinction boundaries. (2) Studies of the response of characteristically sooty flames to ignition, the onset of soot in the expanding flames, the soot radiation, and the soot's effects on the flame spread, extinction, and instability.

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the natural flicker of flames, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Characteristic length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion systems, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. For example, chemi-ion species, electronically-excited chemiluminescent species, are at very low

concentration in typical flames, making the reduced buoyancy environment valuable for evaluating the minor species chemistry. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

The s-Flame investigation and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the testing is typically conducted in a nominally automated mode using pre-programmed scripts. However, it may alternately be conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. But it is not expected that such control, which might be used to approach a stability or soot limit, will be used for s-Flame. Thus far, that has been the case for this experiment, with the possible mundane exception of 'manually' halting the test after the flame has extinguished.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

Thus far, a total of 145 flames were ignited for the s-Flame experiment, with 29 and 116 respectively ignited during Increment 60 and 61. In this regard, science testing began at the end of Increment 60 and the first round of tests were completed during Increment 61. No tests were conducted in Increment 62 due to a lack of crew time.

The s-Flame experiment does not have living subjects like a life science experiment, but the tests are independently conducted with various gaseous fuels, including hydrogen, methane, hydrogen/methane blends, and ethylene. These are simple hydrocarbon gases for which the chemistry is relatively well understood. The Increment 60 tests were all

conducted with ethylene (C_2H_4), which is typically highly sooting because of its carbon double bond (under normal gravity conditions), but where the fuel was diluted with nitrogen for these tests.

No samples (e.g., of soot) were collected for s-Flame; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the flames ignited for s-Flame thus far, including a limited overview of the test conditions, is found in Table 1. The first two test days occurred at the end of Increment 60.

Table 1.

s-Flame log of flames ignited thus far, i.e., in Increments 60-62							
Test Day[1]	**Year**	**Month**	**Date**	**Atmosphere[2]**	**Chamber Pressure**	**Fuel bottle**	**Flames ignited[2]**
First round of tests, with nitrogen-diluted fuel (in Increments 60-61)							
19254	2019	9	11	~21% O_2, with N_2	~1 atm	50% C_2H_4 with N_2	13
19256			13				16
19304		10	31			45% CH_4 with N_2	9
19308		11	4	~30% O_2, with N_2			21
19352		12	18	~21% O_2, with N_2		100% H_2	10
20006	2020	1	6			44.4% H_2 with CH_4	17
20013			13			66.7% H_2 with CH_4	22
20024			24			100% H_2	16
20029			29			50% C_2H_4 with N_2	21
[1]Test Day includes the year as the first two digits followed by the 3-digit day of the year. **[2]Flames ignited** is the number ignited on that day (neglecting the day change on ISS).							

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

The only significant anomaly to occur during this increment pair was the mis-formatting of image data from ACME's primary camera on Dec. 18 (19352). The cause is unknown, but there were repeated operational issues with the CIR's Image Processing and Storage Units (IPSUs) on that test day. Fortunately, the mis-formatting was not seen on subsequent test days and the problematic test day was nominally repeated on Jan. 24 (20024).

However, work was carried out during Increment 61 to address an anomaly that occurred near the end of Increment 60. During testing on Sept. 13 (19256), the CIR's IPSU-G Image Processing and Storage Unit for Gigabit Ethernet cameras, which is used with the ACME Data Camera (DataCam) and Cool Flames Investigation (CFI) intensified camera, didn't properly archive all of the image data on Sept. 13 (19256). A large fraction of the image data was lost. The issue wasn't resolved by remote trouble shooting and the IPSU-G was ultimately removed and replaced by astronaut Jessica Meir on October 24 (19297). Subsequent check-out on Oct. 28 (19301) confirmed normal archival of the image data, where regular testing resumed on Oct. 31 (19304).

2. Any other difficulties in obtaining scientific data

Neglecting the anomalies, the most noteworthy challenge was in imaging the hydrogen and hydrogen-diluted flames, which were not apparent or hardly visible to the operations camera and ACME DataCam, where the latter is the primary camera for ACME testing. The testing with the extremely dim flames was limited to that conducted on Dec. 18 (19352) and Jan. 24 (20024), where the testing on the latter day was to make up for the earlier image mis-formatting discussed in the preceding section. This situation unfortunately inhibits assessment of the three-dimensional flame symmetry as the two intensified cameras are opposite of each other.

During Increment 61, several s-Flame tests automatically aborted (due to the controlling software script), where that was prior to gas flow and the activation of the igniter – which is typical for other ACME experiments. As a result, there was little lost other than the time in the failed test attempt. ACME aborts normally occur because of communication issues with an IPSU, and that happened for tests 19304A5, 19352D3, 19352P1, 20024N1, and 20024J1. A few other tests aborted because flow rates were accidentally specified outside of the range of the installed mass flow controller(s), where that included 19352A1, 19352D2, and 20013C2. ACME aborts are infrequent, but there are sometimes days when they occur repeatedly as can be seen here.

Tests are also unsuccessful when the flame fails to ignite, where that occurred during Increment 61 in tests 19304A4, 19304A6, 19308C1, and 20006A1. That can be corrected by adjusting various test parameters and other settings. In this case, adjustments were subsequently made to the ignition position (e.g., after 19304A4 and 19304A6), the relative timing of the fuel flow and igniter activation (e.g., after 19308C1), and the duration of the igniter activation (e.g., after 20006A1). As another example, failed ignition can result from false ignition detection and deactivation resulting from the combination of settings for ignition and detection, e.g., the gain for the Photomultiplier Tube (PMT) module measuring the OH* chemiluminescence. The latter, in the ultraviolet at 310 nm, is a good indicator of an ignited flame rather than the glowing igniter, which emits primarily in the infrared and visible spectrum.

At high flow rates, the burner does not produce spherically symmetric flames. While nominally spherical at moderate flow rates, the flame's center becomes increasing offset from the burner center. More specifically, the flame is shifted toward the supply tube indicating that the sintered attachment of the porous sphere to the supply tube is more 'open' than the pores of the spherical burner. In other words, the flow from the burner is not spherically symmetric. At sufficiently high flow rates, the flame becomes obviously non-spherical, where it can appear to be ellipsoidal or even parabolic in shape. In the latter case, the flame locally quenches near the supply tube.

When at a small fraction of the maximum flow rate, Mass Flow Controller (MFC) performance has been seen to be poor during s-Flame testing, i.e., where actual flow rate differed from that desired. However, this was not a surprise to those familiar with the hardware.

Flow perturbations have been noticed in some s-Flame tests, where ACME's MFCs used to control the flow of fuel, nitrogen, and oxidizer have unfortunately not always performed satisfactorily. In tests conducted using the 1-slpm fuel MFC (where the flow range is on a nitrogen basis), the flow has been seen to overshoot and undershoot when first opening. Small flow irregularities have been most apparent through their effect on the Photomultiplier Tube (PMT) measurements, e.g., of OH* and CH* chemiluminescence. A momentary increase in fuel flow, as has occurred in some s-Flame tests, naturally leads to increased light from the flame. The MFCs are tested - without flames - after they are first installed. Adjustments are typically made to (1) the setting that controls when the valves open and (2) the Proportional-Integral-Derivative (PID) settings. This ACME activity has involved significant testing, where there are an assortment of MFCs with different flow ranges for each of the three gas types. While the MFC adjustments have been beneficial, they have not eliminated all gas flow problems. It was seen in prior ACME testing that nitrogen oscillations (if any) tend to damp with time, so a nitrogen pre-flow can be used so that such oscillations might end before the flame is ignited. As another measure, the MFCs are re-zeroed prior to each test, where that was another measure adopted in earlier ACME testing.

C. Completeness/Quality of Data

During Increments 60 and 61, nine days of tests were conducted during which 145 flames were ignited for the first round of testing. In contrast, only 29 tests were specified as required or highly desired in the ACME Integrated Science Requirements Document (ISRD) for that round. Therefore, a wealth of data was obtained for this round of testing which featured gaseous fuels diluted by nitrogen.

However, there was a significant loss of data. Most importantly, the first fuel bottle appeared to leak on installation (in Increment 60) and later on-orbit testing (in Increment 61) with other hydrogen/methane blends suggests that its fuel mixture would have been ideal for achieving science success. Furthermore, there was significant image data loss on two different test days, specifically Sept. 13 (19256) and Dec. 18 (19352). As a result,

those test days were nominally repeated on Jan. 29 (20029) and Jan. 24 (20024), respectively (i.e., in a reversed order). For these reasons, a simple comparison of the total number of flames ignited (145) with that specified as required or highly desired (29) for the first round of testing is somewhat misleading.

As discussed in the preceding section, the hydrogen and diluted-hydrogen flames were not apparent or extremely dim in imaging from the operations camera and ACME DataCam. Neglecting that factor, the cameras are sensitive to radiation where pixels gradually become 'hot' (i.e., dead) over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this; and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS).

D. Status of Data Analysis **[provided by Prof. Stephen Tse, Rutgers U.]**

Image files are currently being processed. Radial positions of maximum CH* chemiluminescence is being plotted against time. Different radii are being considered and plotted together, as some flames are not completely symmetric. MFC and PMT data are being compared with video images to assess and confirm experimental settings. Radiometer data is being reviewed. Analyses of the fiber pyrometry data has not begun.

Computational simulations with detailed chemistry and transport of the test cases are underway. Time-dependent simulations will be compared with the time-dependent experimental data. Steady-state simulations will produce the Isola curves mapping flame response versus system Damkhöler number for given fuel mixtures. The experimental cases will then be pin-pointed on the computational mappings to assess whether the experimental conditions are in quasi-steady state or near extinction limits, either kinetic or radiative. Oscillation behavior will be correlated to possible pulsating oscillations found in the simulations.

E. Preliminary Research Findings **[provided by Prof. Tse and lightly edited by Stocker]**

The s-Flame experiment utilizes spherically-symmetric diffusion flames and their dynamics and responses to investigate fundamental processes, such as extinction, instability, and soot formation.

The tests conducted on 19254 (near the end of Increment 60) revealed that a quasi-steady flame could be attained for extended duration at low flow rate, i.e., 3 cc/s for 20% C_2H_4/80% N_2 (on a volume basis). The flame was shut down by the automated system because the burner reached a cut-off temperature of 250°C. Although the flame oscillated near the end of the test, it was likely due to influence of the burner tube. Comparisons with computation simulations will help to clarify the involved processes. For higher flow rates, e.g., 5-10 cc/s, with the same mixture, radiative extinctions were

observed. Generally, the flames showed unstable behavior near the burner tube prior to extinction, likely because of lack of fuel mixture and heat loss in that region. Other mixtures, e.g., 25% C_2H_4/75% N_2 and 30% C_2H_4/70% N_2, also showed radiative extinctions at higher flow rates. For some cases, the pyrometry fiber interacted with the flame, allowing determination of temperature in on-going analyses. Soot was observed only during ignition, which went away as the flame established itself. To assess whether these nitrogen-diluted ethylene flames are non-sooty or sooty, cases were run where mixtures were ignited and established at low flow rates (e.g., 3 cc/s) with low ethylene contents (e.g., 25-30% C_2H_4) and then ramped to high flow rates (e.g., 5-10 cc/s) with higher ethylene contents (e.g., 40-50% C_2H_4). The flames all radiatively extinguished, but no sooting was observed. Observations of such radiative extinction behavior at high flow rates is an achievement because such phenomenon was not observed in our ground-based studies, given the limited microgravity durations. Radiative reabsorption extends flame size at extinction. In ISS testing, radiative extinction has been found to be a common event, which was predicted computationally, but not observed previously for our flame conditions.

The 19256 tests (in Increment 60) were aimed at examining kinetic extinction. Similar to the radiative extinction cases, sufficient microgravity duration is needed to observe kinetic extinction because a near steady flame needs to be established first, before ramp down in flow rate to find properly the chemical kinetic extinction limit. High flow rate cases (e.g., 13.5 cc/s, 15 cc/s) showed asymmetrical flames which radiatively extinguished. Flow rates were lowered so that quasi-steady flames could be established before ramp down in flow rate. The 15% C_2H_4 mixtures were not strong enough to sustain a quasi-steady flame at 3 cc/s before ramp down. However, for stronger mixtures, i.e., 20-30% C_2H_4, very nice flames were established at 3 cc/s; and kinetic extinctions were observed during ramp down before reaching 1 cc/s. Radiative extinctions for ramp up cases were also examined. Similarly, for stronger mixtures, i.e., 20-30% C_2H_4, very nice flames were established at 3 cc/s; and radiative extinctions were observed during ramp up before reaching 8 cc/s.

Testing on 19304 investigated 45% CH_4/55% N_2 mixtures issuing into an air atmosphere. The results indicated that the flow rates specified by the fuel MFC (0.5 slpm) were in the regime of kinetic extinction. Quasi-steady flames could not be established. Furthermore, larger flow rates could not be provided by the MFC, so observations of radiative extinction could not be achieved. Even with a larger capacity MFC, characteristically larger flames would be expected to be asymmetric. As such, nitrogen-diluted methane flames were planned to be reexamined in increased oxygen atmosphere, i.e., 30% O_2/70% N_2.

Consequently, the 19308 tests investigated nitrogen-diluted methane flames in 30% O_2/70% N_2 atmosphere. Various methane concentrations from 45% down to 15% were examined. For 45% CH_4/55% N_2, relatively long duration flames of ~15 s and ~10 s were established at flow rates of 2 cc/s and 3 cc/s, respectively. At larger flow rates of 5 and 9.4 cc/s, the flames expanded to large radii and then extinguished. These flames

appeared to demonstrate radiative extinction. Reduction of flow rate from 2 cc/s to 1 cc/s resulted in extinction during ramp down, indicating kinetic extinction. For 30% CH_4/70% N_2, flow rates of 3, 5, and 10 cc/s produced flames that expanded to extinction (assumed radiatively); while reduction of flow rate from 3 to 1 cc/s resulted in presumably kinetic extinction during ramp down. For 25% CH_4/75% N_2, flow rates of 5 and 10 cc/s yielded flames that expanded to extinction. For 20% CH_4/80% N_2, constant flow rate of 10 cc/s created a flame that expanded until extinction (assumed radiatively); while reduction of flow rate from 10 cc/s to 1 cc/s resulted in wavering extinction during ramp down, suggesting kinetic extinction. For 15% CH_4/85% N_2, flames were ignited, but extinguished very quickly. These tests of 19308 suggest that they could manifest both radiative and kinetic extinction for the conditions specified.

The 19352 tests examined various concentrations of H_2 (45%-70%) diluted with N_2 issuing into air. The installed fuel MFC had a maximum output of 0.1 slpm, which unfortunately provided a less-than-sufficient flow range where its use may have precluded the study of radiative extinction corresponding to larger flame sizes. Imaging the diluted hydrogen flames was a challenge, especially since the hobj image files from the ACME DataCam were mis-formatted. Furthermore, little could be seen in the analog ops imaging, e.g., where it, like the DataCam and Cool Flames Investigation (CFI) intensified camera, cannot image the ultraviolet OH* flame emission (at 310 nm). Analysis of images from the Low Light Level Ultraviolet (LLL-UV) camera and its OH* filter is ongoing. And as one might expect, little could be seen with the CFI intensified camera using its broadband CH_2O filter. Quasi-steady flames seem to be present at low flow rates (e.g., 2.4 cc/s for 70% H_2), as inferred from PMT, pyrometry fiber emission, and burner temperature observations. Kinetic extinction seems to occur as the flow rate is ramped down. There were various problems with the IPSU and some tests aborted. Using the 100% H_2 bottle with a higher flow rate MFC was later performed for tests of 20024 (see below), which was a make-up test day added because of the mis-formatting of the DataCam image files.

The tests conducted 20006, using the 44.4% H_2/55.6% CH_4 bottle, examined 20% H_2/25% CH_4/55% N_2 mixtures issuing into air. Flames extinguished, some with wavy oscillating behavior prior to extinction. However, it seems that even for very small flow rates, e.g., 2 cc/s, extinction was by the radiative mechanism during flame expansion. There was too little parameter room in terms of flow rate and burner size to identify conclusively kinetic extinction during ramp down for these mixtures. In short, radiative extinction (with local flame extinction and re-ignition seemed to occur prior to ultimate extinguishment) during flame expansion after ignition was observed. However, it is unclear whether ramping down the flow rate resulted in kinetic extinction. Detailed analyses of the data are needed. The various cameras seemed to capture the tests well.

The 20013 tests, which used the 66.7% H_2/33.3% CH_4 bottle, mainly investigated 30% H_2/15% CH_4/55% N_2 mixtures issuing into air. Radiative extinction was difficult to identify conclusively because flames with larger flow rates, e.g. 15 cc/s, seemed to last, although with patchy extinction areas, until time out. Larger flow rates could not be examined

either because of limited MFC range or distorted flame shape cause by burner limitations at high flow rates. Also, kinetic extinction during ramp down could not be identified conclusively, as some flames seemed to extinguish because they were "going into the burner."

The 20024 tests using the 100% H_2 bottle repeated and augmented conditions for those of 19352 where the 0.1-slpm fuel MFC range was too low to examine radiative extinction and the analog and intensified cameras did not pick up the flame images at all or well. Here, the 0.5-slpm fuel MFC was used and only a fused silica filter was used for the Cool Flames Investigation (CFI) intensified camera. 45%-15% H_2/N_2 balance mixtures were investigated. The flames at high flow rates, e.g., 10 cc/s, did not extinguish, appearing quasi-steady until time out. The pyrometry fibers glowed, perhaps making good flame-structure measurements, which will be evaluated subsequently. The flames seemed to extinguish during ramp down, but further analysis is needed to assess kinetic extinction.

The tests conducted on 20029 used 50% C_2H_4/50% N_2 and repeated and augmented conditions of 19256, when the images were not good. Again, the results indicated both kinetic and radiative extinction phenomena under corresponding conditions. However, no sooting flames were established, necessitating 100% C_2H_4 and an enriched oxygen atmosphere to facilitate sooting behavior in possible future tests.

In summary, selected preliminary findings from this testing include:

- Cases run for nitrogen-diluted methane and ethylene flames and in air seemed to indicate kinetic extinction at low flow rates and radiative extinction at high flow rates. As previously mentioned above (and in our report for the preceding increment pair), such phenomena were not observed in our ground-based research, given the limited microgravity durations. Nevertheless, comparisons with computational simulations are needed to assess better the extinction behavior.
- Cases run for nitrogen-diluted hydrogen flames seemed to show kinetic extinction at ramp down. However, for the mass flow rates that could be accommodated, radiative extinction was not observed.
- Cases run for nitrogen-diluted hydrogen/methane mixture flames manifested extinction, which are presumed to be radiative. However, kinetic extinction at low flow rates could not be identified. More analyses of the data are needed.

Early ISS results were presented in the following presentations:

1. Chung K. Law and Stephen Tse, “ACME - s-Flame,” ISS Increments 61 & 62 Science Symposium, 17-19 Sept. 2019. (oral presentation)
2. Stephen D. Tse, Jonathan Shi, Delin Zhu, and Chung K. Law, “Dynamics of Burner-Generated Spherical Diffusion Flames In Microgravity,” 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, 20-23 Nov. 2019. (poster presentation)

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Video(s) describing the ACME project can be found at https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/.

Videos related to s-Flame and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

None.

Appendix D. 30-Day Operational Accomplishments Report for ISS Increment 63

16 Dec. 2020

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increment 63

ACME Advanced Combustion via Microgravity Experiments
s-Flame Structure and Response of Spherical Diffusion Flames

The s-Flame investigation is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The first round of s-Flame tests was conducted during Increments 60 and 61, while the second round was conducted fully within Increment 63. There were no tests conducted in Increment 62 because of limited crew availability.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for s-Flame and ACME information.

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I. Introduction

Tests for the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment were conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. The purpose of s-Flame is to advance the ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth.

The spherical flames are ignited at non-steady conditions and allowed to transition naturally towards steady-state or extinction. Tests are conducted with various inert diluents in both the fuel and chamber (oxidizer) atmosphere. The fuel gases include hydrogen and methane mixtures for soot-free flames, and ethylene for sooty flames. One experimental objective is to identify the extinction limits for both radiative and chemical kinetic extinction, i.e., where the flame extinguishes from radiative heat loss or is "blown out," respectively. Another objective is to determine the existence, onset, and nature of pulsating instabilities that have been theoretically predicted to occur in flames with certain fuel/inert mixtures at near-limit conditions.

More specifically, s-Flame has the following experimental objectives:

A. Measure and characterize the transient structure (such as temperature and species distribution) of diffusion flames established at initially non-steady, spherical flame structures for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty), for different flow rates.
B. Determine the quasi-steady convective/chemical-kinetic extinction limits (low system Damköhler numbers) and radiative/chemical-kinetic extinction limits (high system Damköhler numbers) for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty).
C. Determine the existence (for different Zeldovich number, Ze), onset (as a function Da), and nature (i.e. mode, frequency, amplitude) of pulsating instabilities theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures that are above a critical Lewis number, at near-limit conditions.

As suggested by the experiment's name, s-Flame features a spherically symmetric diffusion flame, which is generated by discharging a fuel gas from a porous spherical burner into a still oxygen/inert atmosphere in microgravity. The one-dimensional nature of the system facilitates data acquisition, data reduction, and computational simulation.

The space flight program consists of two main parts: (1) Mapping and characterization of the motion (e.g., expansion) of spherical diffusion flames subsequent to ignition and at extinction because of the finite-rate chemistry and radiative heat loss. The study is focused on the thermal radiation, the adequacy of the detailed chemistry and transport submodels used in computational modeling, and the development of flame front instabilities that could modify the extinction boundaries. (2) Studies of the response of characteristically sooty flames to ignition, the onset of soot in the expanding flames, the soot radiation, and the soot's effects on the flame spread, extinction, and instability.

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the natural flicker of flames, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Characteristic length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion systems, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. For example, chemi-ion species, electronically-excited chemiluminescent species, are at very low concentration in typical flames, making the reduced buoyancy environment valuable for evaluating the minor species chemistry. With these advantages, microgravity testing can lead to

new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

The s-Flame investigation and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the tests were all conducted in a nominally automated mode using pre-programmed scripts. As expected, no s-Flame tests were conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. Thus far, that has been the case for all ACME tests, with the mundane exception of 'manually' halting the test after the flame has extinguished or perhaps if a temperature has reached a critical value.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

A total of 276 flames were ignited for the s-Flame experiment, with 29, 116, and 131 respectively ignited during Increments 60, 61, and 63. These numbers neglect some flames which quenched promptly after ignition. In this regard, science testing began at the end of Increment 60 and the first round of tests were completed during Increment 61. No tests were conducted in Increment 62 due to a lack of crew time, where the second round of tests were fully conducted during Increment 63.

The tests were conducted in two rounds, where the first was conducted in increments 60 and 61 featured 145 flames (nominally) in which the fuel and oxygen (i.e., in the chamber atmosphere) were diluted with nitrogen. The second round conducted in increment 63 featured 131 flames (nominally) in which the fuel and oxygen were instead diluted with helium.

The s-Flame experiment does not have living subjects like a life science experiment, but the tests are independently conducted with various gaseous fuels, including hydrogen, methane, hydrogen/methane blends, and ethylene. These are simple hydrocarbon gases for which the chemistry is relatively well understood.

No samples (e.g., of soot) were collected for s-Flame; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the flames ignited for s-Flame, including a limited overview of the test conditions, is found in Table 1.

Table 1. Flames ignited for the s-Flame experiment

Test Day[1]	Year	Month	Date	Atmosphere[2]	Fuel bottle	Flames ignited[3]
First round of tests, with nitrogen-diluted fuel and oxygen						
19254	2019	9	11	~21% O_2 with N_2	50% C_2H_4 with N_2	13
19256			13			16
19304		10	31		45% CH_4 with N_2	9
19308		11	4	~30% O_2 with N_2		21
19352		12	18	~21% O_2, with N_2	100% H_2	10
20006	2020	1	6		44.4% H_2 with CH_4	17
20013			13		66.7% H_2 with CH_4	22
20024			24		100% H_2	16
20029			29		50% C_2H_4 with N_2	21
Second round of tests, with helium-diluted fuel and oxygen						
20132	2020	5	11	~21% O2 with He	25% H2, 20% CH4, with He	7
20136			15		20% H2, 25% CH4, with He	11
20142			21	~30% O2 with He	30% H2, 15% CH4, with He	17
20155		6	3	~30% and ~21% O2 with He	45% H2 with He	9
20160			8		45% CH4 with He	12
20164			12	~30% O2 with He	20% C2H4 with He	11
20170			18	~30% and ~21% O2 with He	25% C2H4 with He	20
20176			24		30% C2H4 with He	23
20190		7	8		45% H2 with He	21

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Atmosphere** is ~1 atm but pressure increases and O2% drops with vitiation.
[3]**Flames ignited** is the number of nominally sustained flames ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

While there were certainly issues with testing discussed in the next section, there were no significant anomalies during s-Flame's Increment 63 testing. However, anomalies did occur during Increment 60 and 61 which led to corrective actions and/or make-up testing.

2. Any other difficulties in obtaining scientific data

Ignition of the flames presented a significant challenge to the experiment during this increment. All of the flame tests were conducted where the fuel and chamber atmosphere were both diluted with helium rather than the nitrogen used in Increments 60 and 61. The two gases' properties are dramatically different making comparison testing very desirable for manifestation of different phenomena. Helium's thermal conductivity is notably several times greater than that of nitrogen, so considerably more energy and power was required to ignite the flames. The power level was slowly increased from test to test until ignition was finally achieved on the ninth test, where only seven flames were ignited of the 15 tests conducted on May 11 (20132), the first day for flames in Increment 63. The cautious approach was taken because of concerns that the igniter tip might fail (i.e., burn out) if too much power was applied. An igniter failure would have likely halted testing for weeks given the need to schedule the commander's time to replace the igniter tip.

After that first test day, the decision was made to have the igniter replaced with a coil rather than loop design which was believed to be more robust because of the thicker wire. Furthermore, the igniter tip had not been replaced in months, and the tips do degrade with time, especially when they are not quickly retracted from the ignited flames. The coil igniter was used successfully but still with caution, where flames were not always ignited successfully. Subsequently, some tests were conducted in atmosphere with ~30% oxygen (by volume) instead of the planned ~21%. As one would expect, that proved to be helpful for achieving ignition. But the end result was that some tests were conducted with 21% oxygen, some at 30%, and some at both nominal concentrations. Fortuitously, this mixture of conditions may provide deeper insight into the mechanisms of flame dynamics and response.

Tests are also unsuccessful when the flame fails to ignite, where that occurred, for example, during Increment 61 in tests 19304A4, 19304A6, 19308C1, and 20006A1. That can be corrected by adjusting various test parameters and other settings. In this case, adjustments were subsequently made to the ignition position (e.g., after 19304A4 and 19304A6), the relative timing of the fuel flow and igniter activation (e.g., after 19308C1), and the duration of the igniter activation (e.g., after 20006A1). As another example, failed ignition can result from false ignition detection and

deactivation resulting from the combination of settings for ignition and detection, e.g., the gain for the Photomultiplier Tube (PMT) module measuring the OH* chemiluminescence. The latter, in the ultraviolet at 310 nm, is a good indicator of an ignited flame rather than the glowing igniter, which emits primarily in the infrared and visible spectrum.

Testing on June 3 (20155) was cut short due to a combination of issues. The final test was aborted due to an error with an Image Processing and Storage Unit (IPSU). Consoles needed to be reset for the new increment, and there was an issue with the Fuel Oxidizer Management Assembly (FOMA) Control Unit, i.e. the FCU. Rebooting the FCU inadvertently set the valve safety timers to zero, which cannot be reset without venting and refilling the combustion chamber with the intended atmosphere. Those issues combined with an extended loss of commanding prevented further science operations on that day. Therefore, an additional test day (July 8, i.e., 20190) was added to make up for lost science, e.g., where only one test was successfully conducted at ~21% oxygen on June 3. The FCU was off nominal because of a high carrier gas pressure in the Gas Chromatograph Instrumentation Package (GCIP). Imaging of the hydrogen flames was often an issue, where they were not apparent or hardly visible to the operations camera and ACME DataCam, where the latter is the primary camera for ACME testing. The testing with the extremely dim flames was limited to that conducted on Dec. 18 (19352), Jan. 24 (20024), June 3 (20155), and July 8 (20190). This shortcoming unfortunately inhibits assessment of the three-dimensional flame symmetry as the two intensified cameras were opposite of each other.

The flow ranges of the selected Mass Flow Controllers (MFCs) was found to be a challenge, where it was discovered that the goal of studying both kinetic and radiative extinction was not always satisfactorily achieved because of flow rate constraints.

At high flow rates, the burner does not produce spherically symmetric flames. While nominally spherical at moderate flow rates, the flame's center becomes increasing offset from the burner center. More specifically, the flame is shifted toward the supply tube indicating that the sintered attachment of the porous sphere to the supply tube is more 'open' than the pores of the spherical burner. In other words, the flow from the burner is not spherically symmetric. At sufficiently high flow rates, the flame becomes obviously non-spherical, where it can appear to be ellipsoidal or even parabolic in shape. In the latter case, the flame locally quenches near the supply tube.

When at a small fraction of the maximum flow rate, Mass Flow Controller (MFC) performance has been seen to be poor during s-Flame testing, i.e., where actual flow rate differed from that desired. However, this was not a surprise to those familiar with the hardware.

Flow perturbations have been noticed in some s-Flame tests, where ACME's MFCs used to control the flow of fuel, nitrogen, and oxidizer have unfortunately not always

performed satisfactorily. In tests conducted using the 1-slpm fuel MFC (where the flow range is on a nitrogen basis), the flow has been seen to overshoot and undershoot when first opening. Small flow irregularities have been most apparent through their effect on the Photomultiplier Tube (PMT) measurements, e.g., of OH* and CH* chemiluminescence. A momentary increase in fuel flow, as has occurred in some s-Flame tests, naturally leads to increased light from the flame. The MFCs are tested - without flames - after they are first installed. Adjustments are typically made to (1) the setting that controls when the valves open and (2) the Proportional-Integral-Derivative (PID) settings. This ACME activity has involved significant testing, where there is an assortment of MFCs with different flow ranges for each of the three gas types. While the MFC adjustments have been beneficial, they have not eliminated all gas flow problems. It was seen in prior ACME testing that nitrogen oscillations (if any) tend to damp with time, so a nitrogen pre-flow can be used so that such oscillations might end before the flame is ignited. As another measure, the MFCs are re-zeroed prior to each test, where that was another measure adopted in earlier ACME testing.

As mentioned previously, s-Flame tests have sometimes automatically aborted (because of the controlling software script), where that is normally prior to gas flow and the activation of the igniter – which is typical for other ACME experiments. As a result, there was little lost other than the available experimental time in the failed test attempt. ACME aborts normally occur because of communication issues with an IPSU, which for example happened for tests 19304A5, 19352D3, 19352P1, 20024N1, and 20024J1. A few other tests aborted because flow rates were accidentally specified outside of the range of the installed mass flow controller(s), where that included for example 19352A1, 19352D2, and 20013C2. ACME aborts are infrequent, but there are sometimes days when they occur repeatedly as can be seen here.

C. Completeness/Quality of Data

During Increments 60 and 61, nine days of tests were conducted during which 145 flames were ignited for the first round of testing. In contrast, only 29 tests were specified as required or highly desired in the ACME Integrated Science Requirements Document (ISRD) for that round. Therefore, a wealth of data was obtained for this round of testing which featured gaseous fuels diluted by nitrogen.

However, there was a significant loss of data. Most importantly, the first fuel bottle appeared to leak on installation (in Increment 60) and later on-orbit testing (in Increment 61) with other hydrogen/methane blends suggests that its fuel mixture would have been ideal for achieving science success. Furthermore, there was significant image data loss on two different test days, specifically Sept. 13 (19256) and Dec. 18 (19352). As a result, those test days were nominally repeated on Jan. 29 (20029) and Jan. 24 (20024), respectively (i.e., in a reversed order). For these reasons, a simple comparison of the total number of flames ignited (145) with that specified as required or highly desired (29) for the first round of testing is somewhat misleading.

As discussed in the preceding section, the hydrogen and diluted-hydrogen flames were not apparent or extremely dim in imaging from the operations camera and ACME DataCam. Neglecting that factor, the cameras are sensitive to radiation where pixels gradually become 'hot' (i.e., dead) over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this; and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS).

D. Status of Data Analysis [provided by Prof. Stephen Tse, Rutgers U.]

Image files are currently being processed. Radial positions of maximum CH* chemiluminescence is being plotted against time. Different radii are being considered and plotted together, as some flames are not completely symmetric. MFC and PMT data are being compared with video images to assess and confirm experimental settings. Radiometer data is being reviewed. Analyses of the fiber pyrometry data are ongoing. MATLAB code is used to isolate fiber images, which are then color assessed for temperature.

Computational simulations with detailed chemistry and transport of the test cases are underway. Time-dependent simulations will be compared with the time-dependent experimental data, where tracking of the flamefront location is based on chemiluminescent emission. Steady-state simulations will produce the Isola curves mapping flame response versus system Damkhöler number for given fuel mixtures. The experimental cases will then be pin-pointed on the computational mappings to assess whether the experimental conditions are in quasi-steady state or near extinction limits, either kinetic or radiative. Oscillation behavior will be correlated to possible pulsating oscillations found in the simulations.

E. Preliminary Research Findings [provided by Prof. Tse and lightly edited by Stocker]

The s-Flame experiment utilizes spherically-symmetric diffusion flames and their dynamics and responses to investigate fundamental processes, such as extinction, instability, and soot formation.

The tests conducted on 19254 (near the end of Increment 60) revealed that a quasi-steady flame could be attained for extended duration at low flow rate, i.e., 3 cc/s for 20% C_2H_4/80% N_2 (on a volume basis). The flame was shut down by the automated system because the burner reached a cut-off temperature of 250°C. Although the flame oscillated near the end of the test, it was likely due to influence of the burner tube. Comparisons with computation simulations will help to clarify the involved processes. For higher flow rates, e.g., 5-10 cc/s, with the same mixture, radiative extinctions were observed. Generally, the flames showed unstable behavior near the burner tube prior to extinction, likely because of lack of fuel mixture and heat loss in that region. Other

mixtures, e.g., 25% C_2H_4/75% N_2 and 30% C_2H_4/70% N_2, also showed radiative extinctions at higher flow rates. For some cases, the pyrometry fiber interacted with the flame, allowing determination of temperature in on-going analyses. Soot was observed only during ignition, which went away as the flame established itself. To assess whether these nitrogen-diluted ethylene flames are non-sooty or sooty, cases were run where mixtures were ignited and established at low flow rates (e.g., 3 cc/s) with low ethylene contents (e.g., 25-30% C_2H_4) and then ramped to high flow rates (e.g., 5-10 cc/s) with higher ethylene contents (e.g., 40-50% C_2H_4). The flames all radiatively extinguished, but no sooting was observed. Observations of such radiative extinction behavior at high flow rates is an achievement because such phenomenon was not observed in our ground-based studies, given the limited microgravity durations. Radiative reabsorption extends flame size at extinction. In ISS testing, radiative extinction has been found to be a common event, which was predicted computationally, but not observed previously for our flame conditions.

The 19256 tests (in Increment 60) were aimed at examining kinetic extinction. Similar to the radiative extinction cases, sufficient microgravity duration is needed to observe kinetic extinction because a near steady flame needs to be established first, before ramp down in flow rate to find properly the chemical kinetic extinction limit. High flow rate cases (e.g., 13.5 cc/s, 15 cc/s) showed asymmetrical flames which radiatively extinguished. Flow rates were lowered so that quasi-steady flames could be established before ramp down in flow rate. The 15% C_2H_4 mixtures were not strong enough to sustain a quasi-steady flame at 3 cc/s before ramp down. However, for stronger mixtures, i.e., 20-30% C_2H_4, very nice flames were established at 3 cc/s; and kinetic extinctions were observed during ramp down before reaching 1 cc/s. Radiative extinctions for ramp up cases were also examined. Similarly, for stronger mixtures, i.e., 20-30% C_2H_4, very nice flames were established at 3 cc/s; and radiative extinctions were observed during ramp up before reaching 8 cc/s.

Testing on 19304 investigated 45% CH_4/55% N_2 mixtures issuing into an air atmosphere. The results indicated that the flow rates specified by the fuel MFC (0.5 slpm) were in the regime of kinetic extinction. Quasi-steady flames could not be established. Furthermore, larger flow rates could not be provided by the MFC, so observations of radiative extinction could not be achieved. Even with a larger capacity MFC, characteristically larger flames would be expected to be asymmetric. As such, nitrogen-diluted methane flames were planned to be reexamined in increased oxygen atmosphere, i.e., 30% O_2/70% N_2.

Consequently, the 19308 tests investigated nitrogen-diluted methane flames in 30% O_2/70% N_2 atmosphere. Various methane concentrations from 45% down to 15% were examined. For 45% CH_4/55% N_2, relatively long duration flames of ~15 s and ~10 s were established at flow rates of 2 cc/s and 3 cc/s, respectively. At larger flow rates of 5 and 9.4 cc/s, the flames expanded to large radii and then extinguished. These flames appeared to demonstrate radiative extinction. Reduction of flow rate from 2 cc/s to 1 cc/s resulted in extinction during ramp down, indicating kinetic extinction. For 30%

CH_4/70% N_2, flow rates of 3, 5, and 10 cc/s produced flames that expanded to extinction (assumed radiatively); while reduction of flow rate from 3 to 1 cc/s resulted in presumably kinetic extinction during ramp down. For 25% CH_4/75% N_2, flow rates of 5 and 10 cc/s yielded flames that expanded to extinction. For 20% CH_4/80% N_2, constant flow rate of 10 cc/s created a flame that expanded until extinction (assumed radiatively); while reduction of flow rate from 10 cc/s to 1 cc/s resulted in wavering extinction during ramp down, suggesting kinetic extinction. For 15% CH_4/85% N_2, flames were ignited, but extinguished very quickly. These tests of 19308 suggest that they could manifest both radiative and kinetic extinction for the conditions specified.

The 19352 tests examined various concentrations of H_2 (45%-70%) diluted with N_2 issuing into air. The installed fuel MFC had a maximum output of 0.1 slpm, which unfortunately provided a less-than-sufficient flow range where its use may have precluded the study of radiative extinction corresponding to larger flame sizes. Imaging the diluted hydrogen flames was a challenge, especially since the hobj image files from the ACME DataCam were mis-formatted. Furthermore, little could be seen in the analog ops imaging, e.g., where it, like the DataCam and Cool Flames Investigation (CFI) intensified camera, cannot image the ultraviolet OH* flame emission (at 310 nm). Analysis of images from the Low Light Level Ultraviolet (LLL-UV) camera and its OH* filter is ongoing. And as one might expect, little could be seen with the CFI intensified camera using its broadband CH_2O filter. Quasi-steady flames seem to be present at low flow rates (e.g., 2.4 cc/s for 70% H_2), as inferred from PMT, pyrometry fiber emission, and burner temperature observations. Kinetic extinction seems to occur as the flow rate is ramped down. There were various problems with the IPSU and some tests aborted. Using the 100% H_2 bottle with a higher flow rate MFC was later performed for tests of 20024 (see below), which was a make-up test day added because of the mis-formatting of the DataCam image files.

The tests conducted on 20006, using the 44.4% H_2/55.6% CH_4 bottle, examined 20% H_2/25% CH_4/55% N_2 mixtures issuing into air. Flames extinguished, some with wavy oscillating behavior prior to extinction. However, it seems that even for very small flow rates, e.g., 2 cc/s, extinction was by the radiative mechanism during flame expansion. There was too little parameter room in terms of flow rate and burner size to identify conclusively kinetic extinction during ramp down for these mixtures. In short, radiative extinction (with local flame extinction and re-ignition seemed to occur prior to ultimate extinguishment) during flame expansion after ignition was observed. However, it is unclear whether ramping down the flow rate resulted in kinetic extinction. Detailed analyses of the data are needed. The various cameras seemed to capture the tests well.

The 20013 tests, which used the 66.7% H_2/33.3% CH_4 bottle, mainly investigated 30% H_2/15% CH_4/55% N_2 mixtures issuing into air. Radiative extinction was difficult to identify conclusively because flames with larger flow rates, e.g. 15 cc/s, seemed to last, although with patchy extinction areas, until time out. Larger flow rates could not be examined either because of limited MFC range or distorted flame shape cause by burner limitations at high flow rates. Also, kinetic extinction during ramp down could not be identified

conclusively, as some flames seemed to extinguish because they were "going into the burner."

The 20024 tests using the 100% H_2 bottle repeated and augmented conditions for those of 19352 where the 0.1-slpm fuel MFC range was too low to examine radiative extinction and the analog and intensified cameras did not pick up the flame images at all or well. Here, the 0.5-slpm fuel MFC was used and only a fused silica filter was used for the Cool Flames Investigation (CFI) intensified camera. 45%-15% H_2/N_2 balance mixtures were investigated. The flames at high flow rates, e.g., 10 cc/s, did not extinguish, appearing quasi-steady until time out. The pyrometry fibers glowed, perhaps making good flame-structure measurements, which will be evaluated subsequently. The flames seemed to extinguish during ramp down, but further analysis is needed to assess kinetic extinction.

The tests conducted on 20029 used 50% C_2H_4/50% N_2 and repeated and augmented conditions of 19256, when the images were not good. Again, the results indicated both kinetic and radiative extinction phenomena under corresponding conditions. However, no sooting flames were established, necessitating 100% C_2H_4 and an enriched oxygen atmosphere to facilitate sooting behavior in possible future tests.

The 20132 tests kicked off a round of tests with helium-diluted fuel and oxidizer. The ambient was 21% O_2 and 79% He. The fuel was composed of 25% H_2, 20% CH_4, and 55% He. Using He as inert made igniting the mixture much more difficult compared to that for the N_2 inert cases of Round 1, using the same ignition parameters. The reason is believed to be due to the 6X higher value in thermal conductivity of He versus that of N_2. As a result, ignition parameters were varied, e.g., with a longer preheat time and higher power, to achieve ignition for the He-diluted mixtures and ambient. The tests produced flames at 3, 5, 7, and 10 cc/s that expanded outwardly quickly and extinguished. For 5, 7, and 10 cc/s, the flames seemed to disappear and 're-ignite' with a "flash" before final extinction. We are currently investigating the flash phenomenon. Moreover, the radiation mechanism is not conclusive, as it appears different than for the N_2-diluted cases. We did not have a chance to examine lowering of flow rates to assess kinetic extinction.

The 20136 tests examined 20% H_2/25% CH_4/55% He issuing into 21% O_2/He. For 5 cc/s and 7 cc/s, the mixture ignites but immediately extinguishes, with almost no observable flame-front expansion. For 10, 13, 15, and 17.5 cc/s, the mixtures ignite, expand quickly, and extinguish. We believe that the extinction mechanism is radiative and will compare with computational simulations. For the cases with a ramp down in flow rate, the flames seemed to extinguish before ramp down. Thus, we were not able to assess purely kinetic extinction. Ignition parameters were varied to investigate thresholds for igniting the mixture for different flow rates. Only flow rates above 7 cc/s showed flame-front expansion leading to extinction, with 17.5 cc/s showing a flash or wavering of the flame before final extinction.

The tests conducted on 20142 investigated 30% H_2/15% CH_4/55% He issuing into 30% O_2/He. Elevated oxygen was approved for study, since ignition and extinction limits in previous testing were affected by the high thermal conductivity of He. For 5, 7, 10, 15, 13, and 16.5 cc/s flow rates, nice flames were established that expanded to wavy/oscillating extinction at the end, indicative of radiative extinction. For the cases with a ramp down in flow rate, the flames seemed to extinguish during the ignition flow rate stage before ramp down. Thus, we were not able to assess kinetic extinction.

The 20155 tests examined 45% H_2/55% He issuing into 30% O_2/He and 21% O_2/He. Previous tests using He as inert had shown that much larger energy and power are needed to ignite the mixtures. For the 30% O_2/H_2 atmosphere, the tests produced flames at 10 and 14 cc/s (MFC limit of 1 SLMP) that expanded after ignition, but were extinguished by the automated protocol because the burner reached the 250°C temperature limit before any self-extinction could occur. The flames were not too large at the end. Presumably, the thermally-conductive He diluent facilitates heat transfer to the burner. For the cases with a ramp down in flow rate, i.e., starting at 10 and 14 cc/s and ramping down, the flames seemed to self-extinguish during ramp down. Flow rate at 5 cc/s results in flames that only existed briefly. Extinction at this lower flow rate is probably due to heat loss to the burner, rather than pure kinetic extinction. Ramp down from 14 cc/s towards 5 cc/s resulted in the burner reaching the 250°C limit during ramp down. More analyses are needed. Tests at 21% O_2/He atmosphere were limited. Fuel flow of 5 cc/s showed flash ignition, but no sustained flame. Meanwhile, 10 cc/s resulted in a flame that seemed to expand to self-extinction.

Testing on 20160 investigated 45% CH_4/55% He issuing into 30% O_2/He and 21% O_2/He. Without H_2 in the mixture, even larger energy and power were needed to ignite the mixtures. For the 30% O_2/H_2 atmosphere, the 5, 10, and 15 cc/s flow rates were sooty only at ignition; and the subsequent non-sooty flames expanded to self-extinction. At 3 and 4 cc/s, the mixtures flashed at ignition, and extinguished almost immediately. It was not clear whether extinction was due to kinetic extinction or heat loss to the burner. At 6 and 7 cc/s, flames ignited with less soot, expanding with no soot to self-extinction. The case with a ramp down in flow rate could not conclusively identify kinetic extinction. More analysis is needed. Tests at 21% O_2/He atmosphere needed even more energy and power for ignition. At 5, 7, 10, and 15 cc/s, flames ignited with soot, and expanded quickly to extinction.

The 20164 tests began study of ethylene-based flames, where 20% C_2H_4/80% He issued into 30% O_2/He and 21% O_2/He atmospheres. For 5 cc/s, the flame expanded and remained quasi-steady until the burner limit temperature (250°C) was reached. At 6, 7, 10, and 15 cc/s, the flames expanded to extinction, with the higher flow rates (i.e., 10 and 15 cc/s) exhibiting sooty ignition. Fuel flow of 5.5 cc/s seemed to correspond to a condition where the onset of oscillating extinction was simultaneous with the burner reaching the 250°C temperature limit. For the cases with a ramp down in flow rate, the flames seemed to extinguish during ramp down, but heat loss to the burner may be the

mechanism of extinction. Although 13 tests were conducted with the 21% O2/He atmosphere, ignition was not achieved.

The tests performed on20170 investigated 25% C_2H_4/75% He issuing into 30% O_2/He and 21% O_2/He. At 5 and 5.5 cc/s, nice blue chemiluminescent flames were ignited, but eventually were cut off because of the burner temperature limit being reached. At 6 cc/s, a nice blue flame was ignited which expanded slowly to extinction. At 7, 10, and 15 cc/s, sooty ignition occurred, but as the flames expanded, the soot went away, eventually resulting in radiative extinction. By ramping the flow rates up, starting at 5 cc/s, soot at ignition was minimized, and the flames could expand to extinction. Ramp down of the flow rate resulted in almost immediate extinction. For tests at 21% O_2/He atmosphere, cases at 5, 10, and 15 cc/s resulted in flames that expanded quickly to extinction. Less soot was present at ignition compared with that for the 30% O_2/He atmosphere cases. Consistent ignition was not always achieved for the same conditions, which is puzzling.

Testing on 20176 explored 30% C_2H_4/70% He issuing into 30% O_2/He and 21% O_2/He. At 5 cc/s, a long duration flame was established that eventually reached the burner temperature limit. At 6, 7, 10, and 15 cc/s, the flames were sooty at ignition, but the soot dissipated; and they expanded to extinction. Ramping up the flow rates, starting at 5 cc/s, minimized soot at ignition and resulted in flames that slowly expanded to extinction. For tests at 21% O_2/He atmosphere, cases at 5, 10, and 15 cc/s resulted in flames that expanded to extinction. At 4 cc/s, the flame lasted a while before eventual extinction. The 3 cc/s case did not ignite. Ramp down of the flow rate seemed to result in flames that extinguished quickly before the protocol could ramp down the flow rate.

The last testing of Round 2, which was a makeup of 20155, occurred on 20190, with 45% H_2/55% He issuing into 30% O_2/He and 21% O_2/He. For both oxygen concentration environments, flow rates resulting in expanding flames lasted for a while, but were cut off by automated protocol before self-extinction when the burner temperature limit was reached. As for cases where the flow rates were ramped down, extinction detection was observed for both oxygen concentration environments. We are currently analyzing the data to determine if extinction was kinetic or due to heat loss to the burner. A last set of tests were conducted with a reduced oxygen concentration atmosphere, i.e., 17.5% O_2/82.5% He. Some cases indicated that radiative and kinetic extinction were possibly observed.

In summary, selected preliminary findings from this testing include:

- Cases run for nitrogen-diluted methane and ethylene flames and in air seemed to indicate kinetic extinction at low flow rates and radiative extinction at high flow rates. As previously mentioned above (and in our report for the preceding increment pair), such phenomena were not observed in our ground-based research, given the limited microgravity durations. Nevertheless, comparisons with computational simulations are needed to assess better the extinction behavior.

- Cases run for nitrogen-diluted hydrogen flames seemed to show kinetic extinction at ramp down. However, for the mass flow rates that could be accommodated, radiative extinction was not observed.
- Cases run for nitrogen-diluted hydrogen/methane mixture flames manifested extinction, which are presumed to be radiative. However, kinetic extinction at low flow rates could not be identified. More analyses of the data are needed.
- Cases for helium-diluted cases shifted the extinction limits with respect to the nitrogen-diluted cases. In general, kinetic extinction was difficult to observe for the conditions examined. More analyses are needed to assess instability near the flame limits, which was predicted to occur computationally.

Early ISS results were presented in the following presentations:

1. Chung K. Law and Stephen Tse, "ACME - s-Flame," ISS Increments 61 & 62 Science Symposium, 17-19 Sept. 2019. (oral presentation)
2. Stephen D. Tse, Jonathan Shi, Delin Zhu, and Chung K. Law, "Dynamics of Burner-Generated Spherical Diffusion Flames In Microgravity," 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, 20-23 Nov. 2019. (poster presentation)
3. Jonathan Shi, Stephen D. Tse, Dennis Stocker, and Chung K. Law, "ISS Experiments of Gaseous Spherical Diffusion Flames," 36th Annual Meeting of the American Society for Gravitational and Space Research, 5-6 Nov. 2020. (pre-recorded oral presentation)

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Video(s) describing the ACME project can be found at https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/.

Videos related to s-Flame and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

Product references are included for clarification and do not indicate an endorsement by NASA or the federal government.

Appendix E. Operational Timeline

This summary timeline of s-Flame operations was prepared by Dennis Stocker, where the red text indicates the key days when flames were ignited. The dates for crew activities were based on plans, where it is possible that the dates for some simple tasks, e.g., hardware gather or stowage, may have shifted slightly.

Date	GMT	Activity
s-Flame round 1, featuring nitrogen dilution		
2019		
July 30	211	Bottle swaps by Koch
July 31	212	Bottle trouble shooting by Hague
Aug. 7	219	Bottle swap by Parmitano
Aug. 13	225	Hardware gather, chamber insert reconfigure, and window cover removal for s-Flame by Koch
Aug. 14	226	Hardware stow by Koch
Aug. 15	227	Hardware check-out, etc. in preparation for s-Flame
Sept. 9	252	Bottle swap by Koch
Sept. 11	**254**	**s-Flame tests using 50% C_2H_4 with N_2**
Sept. 13	**256**	**s-Flame tests using 50% C_2H_4 with N_2**
Sept. 23	266	Swap of bottles and fuel Mass Flow Controller by Parmitano
Sept. 25	268	MFC testing and tuning, launch of Skripochka, Meir, and Mansouri
Sept. 27	270	IPSU-G diagnostic testing
Oct. 2-3	275-276	IPSU-G diagnostic testing
Oct. 24	297	IPSU-G replacement by Meir
Oct. 28	301	IPSU-G checkout
Oct. 31	**304**	**s-Flame tests using 45% CH_4 with N_2**
Nov. 4	**308**	**s-Flame tests using 45% CH_4 with N_2**
Nov. 18	322	GC sampling with check gas, file maintenance
Nov. 19	323	Bottle swaps by Morgan
Dec. 13	347	Swap of fuel and nitrogen Mass Flow Controllers by Morgan
Dec. 18	**352**	**s-Flame tests using 100% H_2**
Dec. 24	358	Swap of fuel and nitrogen Mass Flow Controllers by Parmitano
2020		
Jan. 2	002	Bottle swap by Skripochka (with Parmitano's assistance)
Jan. 6	**006**	**s-Flame tests using 44.4% H_2 with CH_4**
Jan. 8	008	Bottle swap by Skripochka
Jan. 13	**013**	**s-Flame tests using 66.7% H_2 with CH_4**
Jan. 17	017	Bottle swaps by Skripochka
Jan. 24	**024**	**s-Flame tests using 100% H_2**
Jan. 28	028	Bottle swap by Skripochka
Jan. 29	**029**	**s-Flame tests using 50% C_2H_4 with N_2**
s-Flame round 2, featuring helium dilution		
April 9	100	Launch of Cassidy, Ivanishin, and Vagner
April 23	114	Swap of FOMA calibration unit by Cassidy
April 24	115	FOMA calibration by Cassidy
April 28	119	Swap of fuel Mass Flow Controller and setting of valve timers by Cassidy
April 29	120	MFC testing & tuning

May 1	122	Bottle swap by Ivanishin (with Cassidy's assistance)
May 11	**132**	**s-Flame tests using 25% H_2 / 20% CH_4, with He**
May 13	134	Bottle swaps by Ivanishin
May 15	**136**	**s-Flame tests using 20% H_2 / 25% CH_4, with He**
May 19	140	Bottle swaps by Ivanishin
May 21	**142**	**s-Flame tests using 30% H_2 / 15% CH_4, with He**
May 26	147	Bottle swaps by Ivanishin
June 2	154	Swap of igniter tip (to coil type) and fuel Mass Flow Controller by Cassidy, hardware checkout
June 3	**155**	**s-Flame tests using 45% H_2 with He**
June 4	156	Bottle swap by Ivanishin
June 8	**160**	**s-Flame tests using 45% CH_4 with He**
June 10	162	Bottle swap by Ivanishin
June 12	**164**	**s-Flame tests using 20% C_2H_4 with He**
June 16	168	Bottle swaps by Ivanishin
June 18	**170**	**s-Flame tests using 25% C_2H_4 with He**
June 19	171	Bottle swap by Ivanishin
June 24	**176**	**s-Flame tests using 30% C_2H_4 with He**
July 6	188	Bottle swaps by Ivanishin
July 8	**190**	**s-Flame tests using 45% H_2 with He**

Appendix F. Astronauts and Cosmonauts who Conducted ISS Operations for s-Flame

This list was prepared by Dennis Stocker from his log of ACME operations.

Eight cosmonauts and astronauts participated in ISS operations for this investigation, where they are listed with the dates of their activity (at least as planned). They are listed by the number of days in which they conducted supporting on-orbit activities and alphabetically. The crew activity on each of those dates is very briefly summarized in the Appendix D. Additional information about the crew activity can be found in some of the subsequent appendices.

Anatoly Ivanishin (RSA)	2020	122, 134, 140, 147, 156, 162, 168, 171, 188	9 days
Chris Cassidy (NASA)	2020	114-115, 119, 122 (with Ivanishin), 154	5 days
Christina Koch (NASA)	2019	211, 225-226, 252	4 days
Oleg Skripochka (RSA)	2020	002, 008, 017, 028	4 days
Luca Parmitano (ESA)	2019	219, 266, 358	3 days
	2020	002 (with Skripochka)	1 day
Drew Morgan (NASA)	2019	323, 347	2 days
Nick Hague (NASA)	2019	212	1 day
Jessica Meir (NASA)	2020	297	1 day

Together, the 6 men and 2 women represented three space agencies and three nations, namely Italy, Russia, and the United States.

Appendix G. ISS Operations per the Payload Weekly Science Summary

This section is from Dennis Stocker's log of ACME operations.

s-Flame round 1, featuring nitrogen dilution

2019

Week 21: July 26 – August 1

On GMT 206, we performed the final test points for the first round of testing for ACME Flame Design. Test points were performed using ethylene fuel (sometimes diluted) in a chamber environment of air (21% oxygen, 79% nitrogen at 1 atm). The flames for these tests were short-lived and non-sooty—unlike the flames from the tests at higher oxygen concentrations. The goal of Flame Design is to improve our understanding of soot processes so that flames can be "designed" to be soot free.

To prepare for the start of ACME s-Flame (Structure and Response of Spherical Diffusion Flames), Christina Koch replaced the bottles on CIR Manifolds #2 and #4 on GMT 211. While monitoring the bottle pressures, it was noted that the pressure of the bottle at Manifold #4 was slowly decreasing. The rack was powered down, and Nick Hague and Luca Parmitano both helped with troubleshooting operations. Currently, a forward plan is being developed.

Week 22: August 2 - 8

On GMT 219, Luca Parmitano replaced the CIR Manifold #4 bottle, which was losing pressure while installed and the valve opened, with a known good bottle. While monitoring, the known good bottle maintained its pressure! Next week, we plan to continue transitioning to the next ACME experiment, s-Flame (Structure and Response of Spherical Diffusion Flames).

Week 23: August 9 - 15

On GMT 225, Christina Koch configured the ACME Chamber Insert and the CIR Combustion Chamber to support the next ACME experiment, s-Flame (Structure and Response of Spherical Diffusion Flames).

Week 27: September 6 - 12

On GMT 254, the first test points were performed for ACME s-Flame (Structure and Response of Spherical Diffusion Flames) with 13 spherical flames ignited. This investigation is the fifth to be conducted as part of the ACME project, which features studies of burner-stabilized flames of gaseous fuels. s-Flame's goal is to advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames—where today's testing was focused on soot-free flames. Without the buoyant flows created by Earth's gravity, flames in microgravity are typically spherical, which allows the analysis to focus on the combustion chemistry and details of the transport phenomena.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

In this test point video, a non-sooty (blue) flame is ignited with a retracting hot-wire igniter, where the gaseous fuel is flowing from a porous spherical burner. After ignition, the flame grows slightly as a result of the diminishing oxygen gradient. An array of 14-micron ceramic fibers is then moved toward the flame, where the temperature of the flame can be determined from the color of the fiber's glow—where yellow is hotter than orange, and orange is hotter than red. While only a single fiber can be seen at first, a second becomes visible when the flame suddenly gets bigger

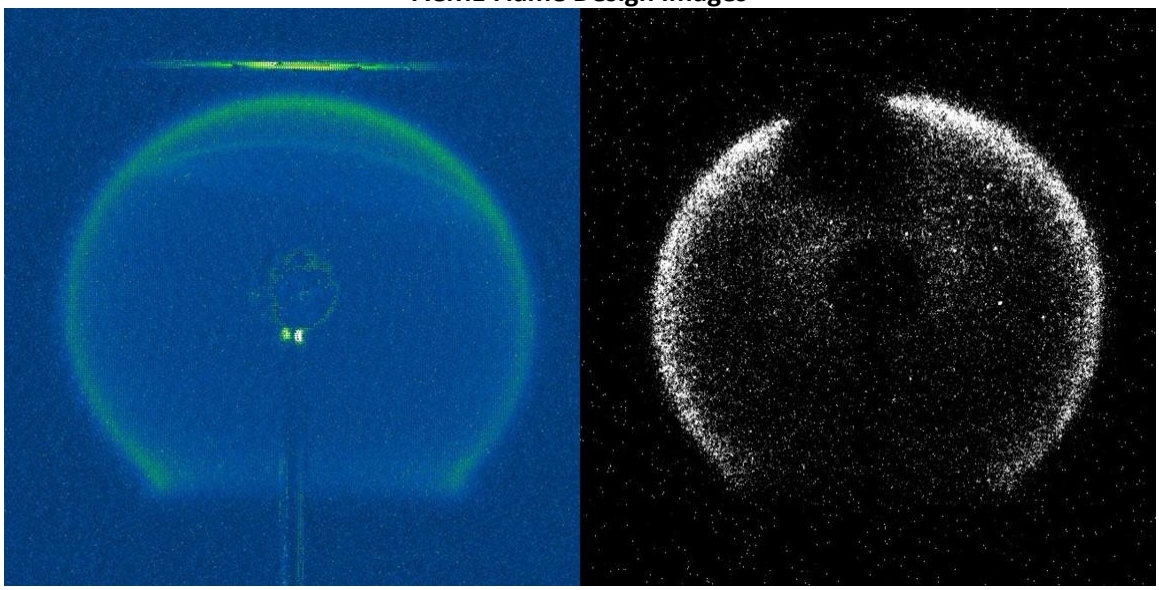
and engulfs the first fiber. The diluted ethylene flow to the burner was abruptly increased and made to be more fuel rich. As a result, the flame needed to grow to reach enough oxygen to burn all the fuel.

Throughout the test, the spherical flame is open near the burner's tube because of heat loss. As the flame grows, the opening around the tube increases (while the flame weakens and dims). The flame edge then pulls away from the burner tube and leaves just a small portion of the spherical shell. This large hole in the flame shell allows for fuel and air mixing and the flame momentarily reestablishes and brightens before it extinguishes.

Week 28: September 13 - 19

On GMT 256, 16 spherical flames were successfully ignited for ACME s-Flame (Structure and Response of Spherical Diffusion Flames). Even though ethylene's double bond tends to make it a sooty fuel, these nitrogen-diluted ethylene flames were all non-sooty (except for during ignition). For each test point, the harp-like array of 14-micron ceramic fibers was used to determine temperatures in or near the flame, while an array of thermocouples was used to measure the spread of the thermal field from the flame.

ACME Flame Design Images



These two images are of the same flame at nearly the same time. The image on the left is a false-colored image. The porous spherical burner can be seen in the middle of the flame. The flame shell surrounds the burner except at the fuel supply tube, which causes the flame to locally quench because of heat loss. A glowing ceramic fiber can be seen above the flame, where the fiber's glow is used to determine the local temperature by comparison of the red, green, and blue color planes in a measurement technique known as pyrometry. The faint lines at the top of the flame suggest that the fiber may have locally quenched the flame even though it was above the flame.

The image on the right is from an intensified camera looking at the emission from the excited molecule CH*, which emits a photon of blue 431-nm light when it drops to a lower energy state in a process referred to as chemiluminescence. The location of the burner sphere is noticeable due to the diminished light in the middle of the flame. The opening in the flame shell due to the burner's tube is also noticeable. The

apparent opening in the top of the flame may be caused by the fiber array arm, which, in this camera's view, blocks the flame.
In microgravity, flames aren't supplied oxygen through buoyant convention. As a result, low-momentum flames like these in a quiescent (still) atmosphere tend to be quite weak and self-extinguish. Microgravity is an extremely useful for environment for this experiment since a key objective is to study flame dynamics and extinction.

Week 29: September 20 - 26

On GMT 266, Luca Parmitano replaced the ACME Fuel Controller and the bottles at CIR Manifold #2 and CIR Manifold #4. These changes support the next ACME s-Flame test points, which will be in a chamber environment of air (21% oxygen, 79% nitrogen at 1 atm) and will use diluted methane fuel at faster flow rates.

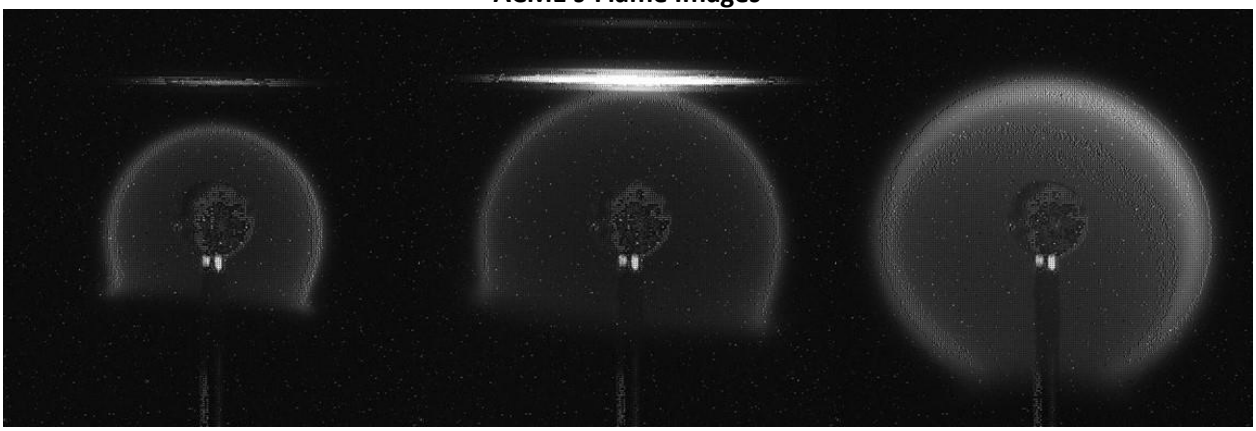
Week 5: October 25 - 31

On GMT 297 on the back of the CIR Optics Bench, Jessica Meir removed a failed hard drive from the FCF Image Processor and then replaced the FCF Image Processor. On GMT 301, the CIR was powered up and a checkout of the new FCF Image Processor was successful. We returned to performing test points later in the week!

Week 6: November 1 - 7

On GMT 304 and 308, 30 spherical flames were successfully ignited for ACME s-Flame (Structure and Response of Spherical Diffusion Flames). These tests burned methane fuel diluted with nitrogen in atmospheres of 21% and 30% oxygen (with a balance of nitrogen). Methane is the primary component of natural gas. These test points focused on determining where the radiative heat loss or velocities are too high for the flame to be sustained, which are known as the radiative and convective extinction limits. Radiative heat transport is much more important under microgravity conditions because the hot gases don't buoyantly rise from the flame.

ACME s-Flame Images



This figure shows three flames of 45% methane burning in 30% oxygen. The gaseous methane fuel flows from the porous sphere of the burner, which is visible in the flame's center. The flow of fuel increases from left to right, which increases the size of the flame. The burner's fuel supply tube is visible below the sphere. The flame loses heat to this tube causing an opening, which typically grows with time as the flame weakens. In the flame on the right, the opening is comparatively small, which signifies a stronger flame.

In the left and center images, the glow of a 14-micron ceramic fiber is visible above the flame. Where the fiber is visibly glowing, the flame's temperature at that point can be determined via color-ratio pyrometry, which uses a color camera characterized for that purpose. The faint glow of a second fiber further from the flame can be seen at the top of the center image. When the image on the right was captured, the fiber array was not yet inserted into the flame.

Week 8: November 15 - 21

On GMT 323, Drew Morgan replaced the bottles at CIR Manifold #2 and CIR Manifold #4 to support ACME s-Flame test points using pure hydrogen fuel in air (21% oxygen, 79% nitrogen at 1 atm).

Week 12: December 13 - 19

On GMT 347, Drew Morgan replaced the ACME Bypass and Fuel Controllers to support the ACME s-Flame (Structure and Response of Spherical Diffusion Flames) test points on GMT 352, which flowed nitrogen and fuel at slower flow rates.

Ten flames were successfully ignited on GMT 352. These tests were conducted with nitrogen-diluted hydrogen fuel making the flames very faint. The CIR is equipped with an intensified camera, which images these very faint flames by visualizing the ultraviolet emission of the OH molecule (at 310 nm) as it drops to a lower energy level. This process is known as chemiluminescence. Thus far, there has only been a preliminary review of some image data. These tests used hydrogen fuel in differing concentrations with both steady and ramped flow rates. The tests with ramped flow rates investigated extinction characteristics of the flame.

Week 14: December 27 - January 2

On GMT 358, Luca Parmitano replaced the ACME Bypass and Fuel Controllers to support ACME s-Flame (Structure and Response of Spherical Diffusion Flames) test points with faster nitrogen and fuel flow rates.

On GMT 002, Oleg Skripochka and Luca Parmitano replaced the bottle at CIR Manifold #4 to support ACME s-Flame test points using a hydrogen/methane fuel mixture in air (21% oxygen, 79% nitrogen at 1 atm).

2020

Week 15: January 3 - 9

On GMT 006, 17 flames were successfully ignited for ACME s-Flame (Structure and Response of Spherical Diffusion Flames), which uses spherically-symmetric diffusion flames to investigate fundamental processes, such as flame extinction. These tests were conducted with a mixture of hydrogen (20%), methane (25%), and nitrogen (55%) ejected from the porous spherical burner into a chamber atmosphere of air (21% oxygen, 79% nitrogen at 1 atm). The dim, soot-free flames exhibited local quenching due to radiative heat loss and reignition, which sometimes occurred repeatedly. In these cases, the dynamic parts of the broken flame sometimes appear to dance around the burner.

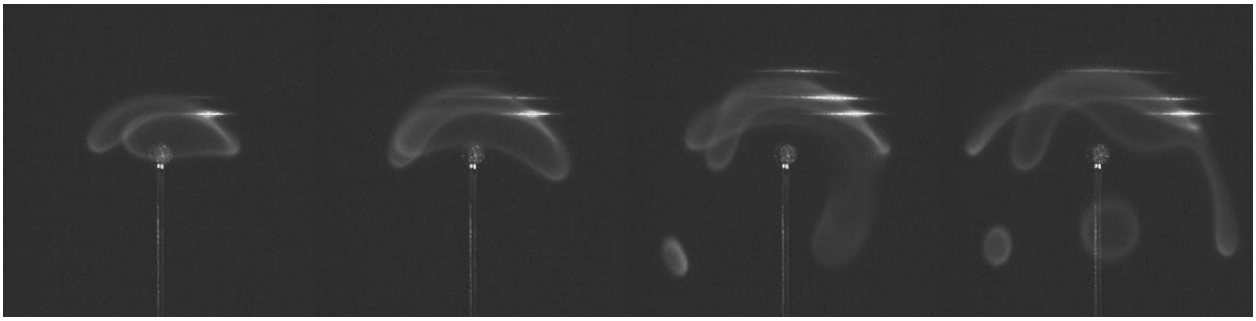
CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

In this test point video, a non-sooty (blue) flame is ignited with a retracting hot-wire igniter. The harp-like array of 14-micron ceramic fibers moves in from the left to a fixed position. The glow from the fibers allows for a good estimation of the local temperature. The flame starts "dancing," where it partially quenches due to radiative heat loss and the remaining flame stretches and circles around the burner through a repeated combination of quenching and reignition. The fuel

and oxygen, which are normally separated by the flame, mix where the flame quenches, making the local gas more flammable.

Week 16: January 10 - 16

On GMT 013, 22 flames were successfully ignited for ACME s-Flame (Structure and Response of Spherical Diffusion Flames), which is a record number of sustained flames in a single day of testing for all the ACME experiments! Like the tests performed on GMT 006, these tests burned a mixture of hydrogen, methane, and nitrogen but with a greater hydrogen concentration relative to the methane concentration. Even though these flames were less dynamic than the "dancing" flames from GMT 006, they did break apart and create flamelets. The flamelets were about 1 cm across and were shaped like curved bands or circles rather than spheres or ellipsoids.



This figure shows four different tests from GMT 013, where the flow rate of the hydrogen/methane fuel increases from left to right while the nitrogen flow rate remains constant. The burner, with its supply tube and porous sphere, is opposite the flame. The fuel and nitrogen both flow from the burner sphere into a chamber atmosphere of ~19% oxygen and nitrogen at a pressure of ~1 atm.

The bright horizontal lines in each flame are the glowing 14-micron ceramic fibers, where that glow can be used to determine the flame's temperature. As the fuel flow rate increases, the flames become larger and separate flamelets form. For these tests, the flow rate was increased to explore flame stability and extinction limits.

Week 17: January 17 - 23

On GMT 017, Oleg Skripochka replaced the bottles at CIR Manifold #2 and CIR Manifold #4 to support ACME s-Flame test points using pure hydrogen fuel in air (21% oxygen, 79% nitrogen at 1 atm).

Week 18: January 24 - 30

On GMT 024, 16 flames were successfully ignited for ACME s-Flame (Structure and Response of Spherical Diffusion Flames). These tests were conducted with nitrogen-diluted hydrogen fuel making the flames very faint. Even though the flames were barely visible, their presence was obvious due to the measurements of various instrumentation, including thermocouples, radiometers, and photomultiplier tubes. The glowing ceramic fibers were also a positive indication of a flame. The harp-like array of 14-micron ceramic fibers was translated into the flame to make optical measurements of the flame temperature from the hot fibers' visible glow.

Week 19: January 31 - February 6

On GMT 029, 21 flames were successfully ignited for ACME s-Flame (Structure and Response of Spherical Diffusion Flames). These were the final tests for the first round of testing for s-Flame. In this round, a total

of 145 flames were ignited using nitrogen-diluted gaseous fuels. In the upcoming second round, tests will be conducted with the same fuels but diluted with helium, which has much different transport properties than nitrogen.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

In this test point video, a sooty (yellow) flame is ignited with a retracting hot-wire igniter. This flame is burning ethylene fuel diluted by nitrogen. Ethylene's double bond tends to make flames sooty. However, the soot (which is black, but thermally glows yellow) quickly disappears, leaving a blue soot-free spherical flame.

Following ignition, the flame grows to reach enough oxygen to consume the fuel. The harp-like array of 14-micron ceramic fibers moves in from the left to a fixed position. The glow from the fibers allows for a good estimation of the local temperature. These flames are a bit like bubbles, where the fuel is within the flame, the oxygen is outside of it, and the light, heat, and chemical products are emitted at the surface where they come together and react.

The base of the flame begins to retreat from the burner's fuel supply tube due to heat loss. Both the flame and the glowing fibers can be seen to dim as the flame weakens. Eventually the base of the flame begins to oscillate. The flame locally quenches—creating a hole in the flame. This hole allows for increased mixing of the fuel and oxygen, which in turn allows the flame to temporarily recover. As the oscillations grow in strength, the flame goes out.

s-Flame round 2, featuring helium dilution

Week 2: April 17 - 23

On GMT 114, Chris Cassidy replaced the CIR FOMA Cal Unit on the back of the CIR Optics Bench. This activity allowed us to perform the calibration of the CIR's FOMA (Fuel/Oxidizer Management Assembly) system. This calibration needs to be completed yearly to ensure that all the pressure transducers in the CIR are accurate.

On GMT 115, in support of the CIR FOMA Calibration, Chris Cassidy closed and opened valves on the gas bottles installed in CIR.

Week 3: April 24 - 30

On GMT 119, Chris Cassidy replaced the ACME Fuel Controller and set the CIR Valve Timers to support the upcoming second round of testing for ACME s-Flame (Structure and Response of Spherical Diffusion Flames) starting in a couple weeks.

s-Flame's goal is to advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames. Without the buoyant flows created by Earth's gravity, flames in microgravity are typically spherical, which allows the analysis to focus on the combustion chemistry and details of the transport phenomena. Results may lead to more efficient and less polluting combustion technology for use on Earth.

Week 4: May 1 - 7

On GMT 122, Anatoly Ivanishin and Chris Cassidy replaced three CIR Manifold Bottles to support the upcoming second round of testing for ACME s-Flame (Structure and Response of Spherical Diffusion Flames) starting next week.

s-Flame's goal is to advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames. Without the buoyant flows created by Earth's gravity, flames in microgravity are typically spherical, which allows the analysis to focus on the combustion chemistry and details of the transport phenomena. Results may lead to more efficient and less polluting combustion technology for use on Earth.

Week 5: May 8 - 14

On GMT 132, we began the second round of testing for ACME s-Flame (Structure and Response of Spherical Diffusion Flames) with seven flames ignited using a hydrogen/methane fuel mixture diluted with helium. While the first round of testing used nitrogen to dilute the atmosphere and fuels, this round of testing uses helium. Helium is more diffusive and has a greater thermal conductivity than nitrogen. These factors can strongly influence combustion, which when coupled with the microgravity environment, can lead to new insight and understanding.

On GMT 134, Anatoly Ivanishin replaced the bottles on the CIR Manifold #4 and the CIR Check Gas Manifold to continue supporting s-Flame test points.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

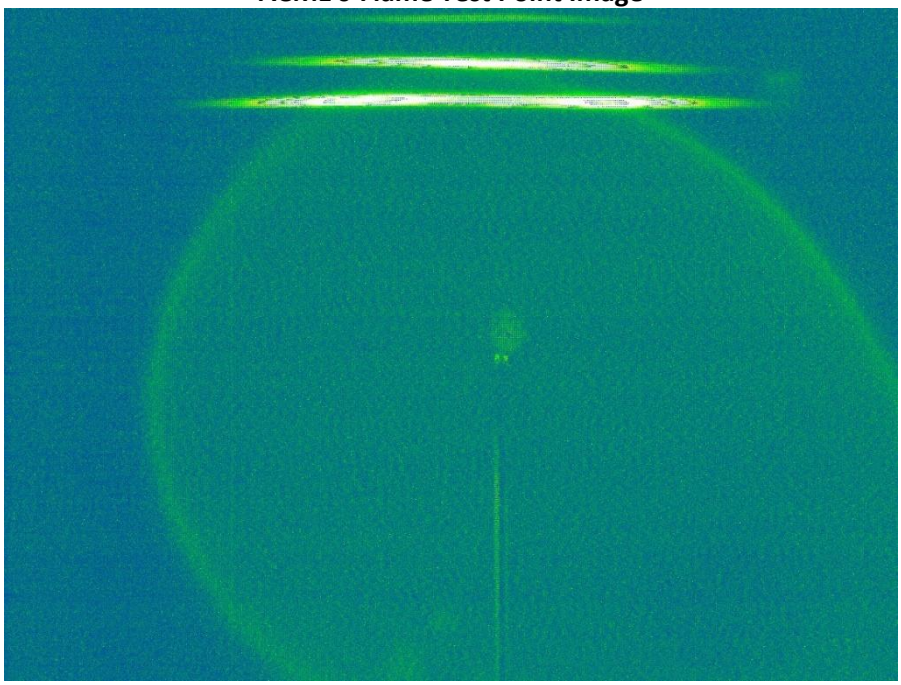
In this test point video, a porous sphere at the end of the burner's fuel supply tube is used to create spherical flames. The gaseous fuel is ejected from this sphere, and the flame is ignited with a retracting hot-wire igniter. The flame "retreats" from the burner's tube due to heat loss leaving just a flame "cap." As the igniter retracts, a second ignition occurs while the flame "cap" is still burning. This second flame, which is large and spherical, is initially bright but quickly fades and extinguishes. We're not sure what caused the second ignition, but regardless, the resulting data provides an opportunity to learn about the inner workings of flames.

s-Flame's goal is to advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames. Without the buoyant flows created by Earth's gravity, flames in microgravity are typically spherical, which allows the analysis to focus on the combustion chemistry and details of the transport phenomena. Results may lead to more efficient and less polluting combustion technology for use on Earth.

Week 6: May 15 - 21

On GMT 136, 11 flames were successfully ignited for ACME s-Flame (Structure and Response of Spherical Diffusion Flames). These tests burned a hydrogen/methane fuel mixture diluted with helium but with a different ratio of flammable gases than investigated in the first test day. The results from both days will be compared with those from a third ratio, which will be tested on our next test day. On GMT 140, Anatoly Ivanishin replaced the bottles at CIR Manifold #1 and CIR Manifold #4 to support ACME s-Flame test points using the third hydrogen/methane fuel mixture diluted with helium.

ACME s-Flame Test Point Image



In this false-colored preliminary image, the flame encompasses the porous spherical burner and its fuel supply tube. The bright lines at the top are 14-micron ceramic fibers that are glowing because of the flame's heat. Where the fibers are visibly glowing, the flame's temperature at that point can be determined by color-ratio pyrometry, which uses a color camera characterized for that purpose. Qualitatively, a yellow glow is hotter than orange, which is in turn hotter than red—just like metal in a forge. Given the extremely fine fibers, the measurement effectively provides the flame temperature, which can be compared with the results of computational modeling.

s-Flame's goal is to advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames. Without the buoyant flows created by Earth's gravity, flames in microgravity are typically spherical, which allows the analysis to focus on the combustion chemistry and details of the transport phenomena. Results may lead to more efficient and less polluting combustion technology for use on Earth.

Week 7: May 22 - 28

On GMT 142, 17 flames were successfully ignited for ACME s-Flame (Structure and Response of Spherical Diffusion Flames). These tests burned a hydrogen/methane fuel mixture diluted with helium but with a different ratio of flammable gases than investigated in the first two test days. These tests were performed in a chamber environment of 30% oxygen (the remainder was helium); whereas, the first two test days were performed in 21% oxygen. Flames are strongly dependent on the oxygen concentration, so these flames lasted longer before extinguishing than the earlier tests. On GMT 147, Anatoly Ivanishin replaced the bottles at CIR Manifold #2 and CIR Manifold #4 to support ACME s-Flame test points using 45% hydrogen fuel diluted with helium.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

Following ignition in this test point video, the spherical flame expands to reach enough oxygen to consume the fuel flowing from the burner. The flame retreats from the burner's fuel supply tube due to heat loss leaving just a flame "cap." The harp-like array of 14-micron ceramic fibers moves

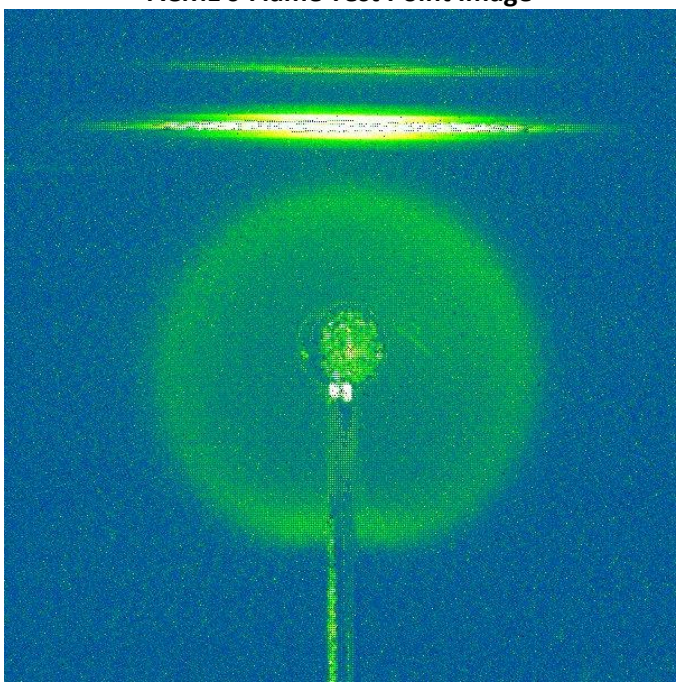
in from the left to a fixed position. The glow from the fibers allows for a good estimation of the local temperature. As the flame opens, the fuel and oxygen, which are normally separated by the flame, mix together. This mixing allows the flame to temporarily recover. The flame flashes back three times before it fully extinguishes.

s-Flame's goal is to advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames. Without the buoyant flows created by Earth's gravity, flames in microgravity are typically spherical, which allows the analysis to focus on the combustion chemistry and details of the transport phenomena. Results may lead to more efficient and less polluting combustion technology for use on Earth.

Week 8: May 29 - June 4

In support of ACME s-Flame (Structure and Response of Spherical Diffusion Flames) test points, Chris Cassidy replaced the ACME Igniter Tip and the ACME Fuel Controller on GMT 154. The next day, nine helium-diluted hydrogen flames were successfully ignited for s-Flame, where most of these tests were performed in a chamber environment of ~30% oxygen with a balance of helium. Previously, similar tests were performed with nitrogen dilution, but helium is dramatically different than nitrogen. Helium is more diffusive and has a greater thermal conductivity than nitrogen. Studying these differences can enable new learning, particularly when combined with the microgravity environment. On GMT 156, Anatoly Ivanishin replaced the bottle at CIR Manifold #4 to support ACME s-Flame test points using methane fuel diluted with helium.

ACME s-Flame Test Point Image



One benefit of performing combustion research in microgravity is spherical flames. This image (preliminary and false-colored) shows a spherical flame of diluted hydrogen ~25 mm in diameter. Reducing the flame geometry to a single dimension dramatically simplifies analysis and modeling. It also enables computation focused on chemical kinetics and transport phenomena rather than fluid dynamics and flame structure.

The two bright lines above the flame are 14-micron ceramic fibers, which are visibly glowing because of the flame's heat. The temperature distribution along the glowing fiber can be quantified through Thin Filament Pyrometry, where the hue is indicative of the temperature. Flame measurements are also made with intensified cameras, thermopile detectors, photomultiplier tubes, and thermocouples. All this data is used to improve and validate flame modeling and through that enable improvements in practical terrestrial combustion.

Week 9: June 5 - 11

On GMT 160, 12 helium-diluted methane flames were ignited in oxygen/helium atmospheres for ACME s-Flame (Structure and Response of Spherical Diffusion Flames). Compared to other gaseous hydrocarbon fuels, methane (CH4) is more difficult to ignite because of its lack of carbon-carbon bonds. Methane ignition hadn't been an issue in earlier testing with nitrogen dilution, but it was challenging in this week's tests because of helium's much greater thermal conductivity. Nonetheless, the recently installed coiled igniter tip enabled us to ignite these flames.

The results from the current series of s-Flame tests will be compared with the results from the previous tests using nitrogen dilution. In that earlier testing, it was found that radiative extinction (i.e., extinction caused by radiative heat loss) is a common event, which had been predicted computationally. However, we did not observe this in preliminary ground-based testing. It was also seen in the on-orbit tests that radiative reabsorption extends the flame size at extinction. Another type of extinction was sometimes observed, where the gas velocities caused nitrogen-diluted ethylene flames (under certain conditions) to blow out. It will be interesting to see what the future comparisons will reveal about the nature of flames.

On GMT 162, Anatoly Ivanishin replaced the bottle at CIR Manifold #4 to support ACME s-Flame test points using 20% ethylene fuel diluted with helium.

Week 10: June 12 - 18

On GMT 164, 11 helium-diluted ethylene flames were ignited for ACME s-Flame (Structure and Response of Spherical Diffusion Flames). This is a shift in testing for the s-Flame experiment from non-sooty fuels (i.e., hydrogen, methane, or hydrogen/methane blends) to a sooty fuel. Ethylene (C2H4) tends to be a sooty fuel because it has a carbon double-bond. Soot forms when hot fuel combines with itself in the absence of oxygen, where the union of fuel molecules is facilitated by the double bond. The flames in these tests were sooty at ignition, but the soot quickly disappeared resulting in blue flames, where blue is the natural color of flames with hydrocarbon fuels. On GMT 168, Anatoly Ivanishin replaced the bottles at CIR Manifold #1, CIR Manifold #2 and CIR Manifold #4 to support ACME s-Flame test points using 25% ethylene fuel diluted with helium.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

This test point video runs at half speed to better show what is happening with the flame. A flame of 20% ethylene fuel ignites in a chamber atmosphere of about 30% oxygen (both diluted with helium). The flame is initially very bright. Although the soot particles are black, their glow depends on their temperature, where yellow is hotter than orange which is in turn hotter than red. While some soot clusters escape the flame, much of the soot becomes trapped within the flame by the temperature gradient.

After some time, the array of ceramic fibers is translated into the flame to determine the radial temperature distribution where the fibers are glowing. Despite their 14-micron diameter, the

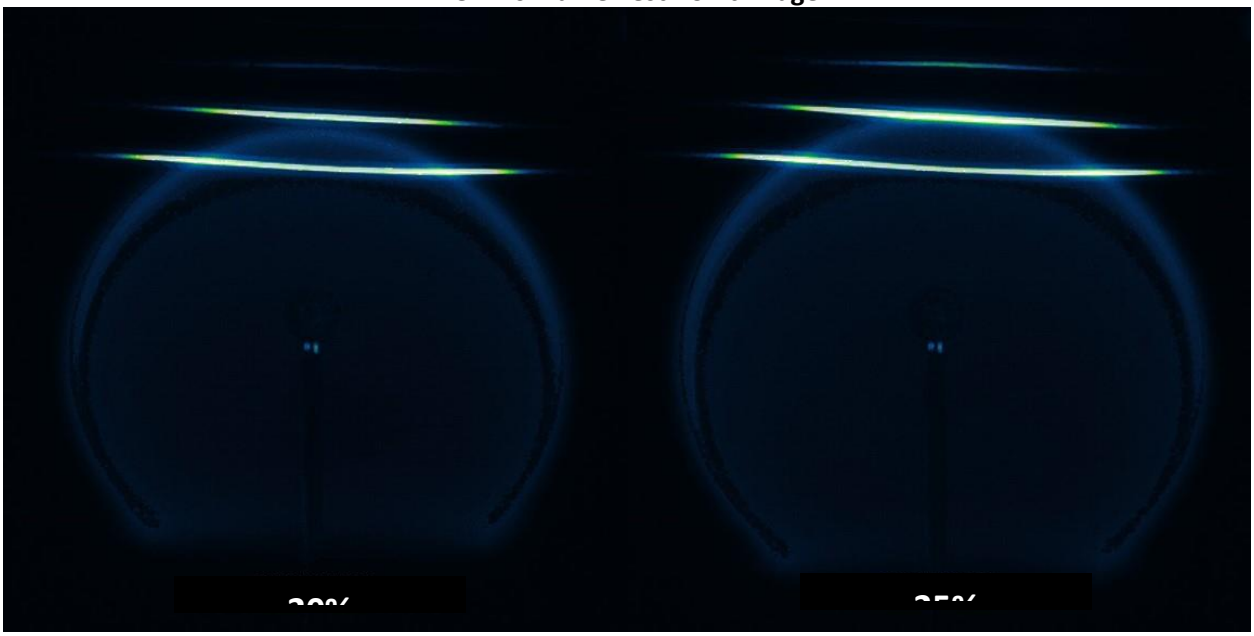
fibers appear to cool or push away the soot because that side of the flame becomes increasingly blue.

The flame becomes all blue and retreats from the fuel supply tube because of heat loss. The opening in the flame allows fuel and oxygen to directly mix, where they are normally separated by the presence of the flame shell. This mixing enhances flammability and the reinvigorated and locally brighter flame flashes back to the supply tube reestablishing the nominally spherical flame. But it is short-lived, and the flame quickly retreats again until it is just a "cap" on the left. Mixing again leads to another flame flash, but it quickly extinguishes due to radiative heat loss, heat loss to the supply tube, and conductive heat loss enhanced by the helium's high thermal conductivity.

Week 11: June 19 - 25

On GMT 170, 20 flames were ignited for ACME s-Flame (Structure and Response of Spherical Diffusion Flames). These flames burned a 25% ethylene/75% helium blend; whereas, the flames on GMT 164 burned a 20% ethylene/80% helium blend. Comparing these data will enable analysis of the effects of this fuel/inert blend on flame dynamics, such as extinction behavior and structure. On GMT 171, Anatoly Ivanishin replaced the bottle at CIR Manifold #4 to support ACME s-Flame test points using a 30% ethylene/70% helium blend.

ACME s-Flame Test Point Image



In this image (preliminary and false-colored), the flame on the right has a higher ethylene concentration than the flame on the left. As expected, the flame on the right is larger where the spherical flame expands until the fuel reaches enough oxygen to be consumed.

These flames, which are non-premixed, are like bubbles in that the gaseous fuel is inside the bubble and the oxygen is outside the bubble. The flame shell, like the film of a bubble, is a thin sheet where the fuel and oxygen come together and react, producing light, heat, and chemical products, such as carbon dioxide and water vapor. Meanwhile, the helium—which readily diffuses—is free to cross the flame shell.

However, diffusion was minimal because the helium concentrations within the fuel and the surrounding atmosphere were similar.

Both flames were bright and sooty at ignition, but these images are from late in each test when the flames became dim. The bright lines at the top are the 14-micron ceramic fibers used for optical determination of the flame temperature using a camera characterized for that purpose.

s-Flame's goal is to advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames. Without the buoyant flows created by Earth's gravity, flames in microgravity are typically spherical, which allows the analysis to focus on the combustion chemistry and details of the transport phenomena. Results may lead to more efficient and less polluting combustion technology for use on Earth.

Week 12: June 26 - July 2

GMT 176 was a record-breaking day for ACME with 23 flames ignited—the most we've ignited in one day! We've now ignited over 250 flames for s-Flame (Structure and Response of Spherical Diffusion Flames) and over 1000 flames for ACME since it started in Nov. 2017. The tests on GMT 176 were performed with 30% ethylene in helium for comparison with the results from the two preceding test days, which burned 20% and 25% ethylene/helium blends.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

In this test point video, helium-diluted ethylene fuel is ignited with a low fuel flow rate to limit the soot produced with ignition. After ignition, the flow rate is tripled in a smooth ramp, so that the flame response can be observed. As expected, the flame quickly grows, where it must expand to reach enough oxygen to consume all the fuel.

Before the fuel flow rate has completely tripled, the flame retreats from the burner's fuel supply tube because of heat loss—creating a hole in the flame near the tube. This hole allows for increased mixing of the ethylene and oxygen, which are normally separated by the flame shell. This mixing causes the flame to momentarily reestablish around the burner tube with a bright flash of light. Mixing again leads to another flame flash, after which the flame fully extinguishes. The fuel flow rate did completely triple before the flame extinguished.

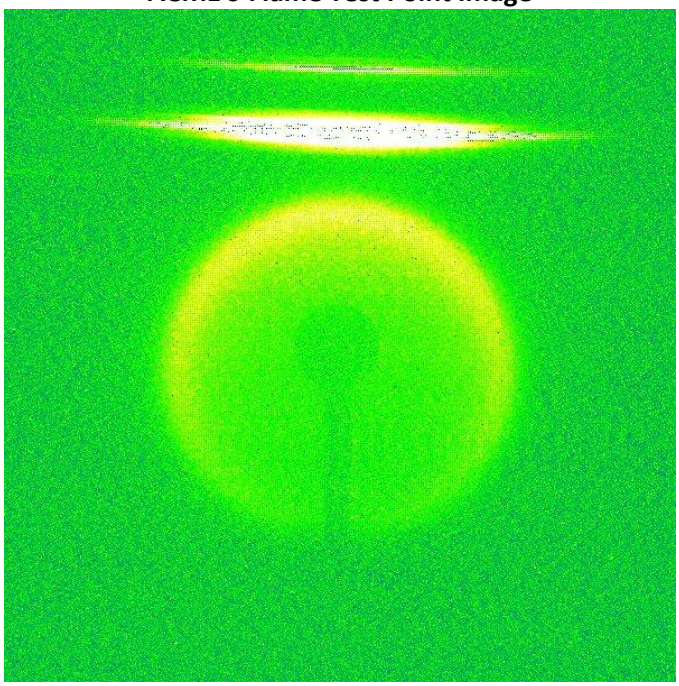
Although not visible in the video, the measured OH* and CH* chemiluminescence was relatively constant throughout this test with the important exceptions of ignition and the bright flashes of light prior to extinction. In contrast, the measured chemiluminescence declined with time in similar tests with constant fuel flow rates. OH* and CH* are molecules at an elevated energy state which emit a photon of a specific wavelength as their energy level drops a level. The CH* emission is the source of most of the blue in a hydrocarbon flame, while the OH* emission is in the ultraviolet. ACME uses the latter measurement as an indicator of ignition and extinction because it is insensitive to the hot-wire igniter and hot burned gases.

Using this unique microgravity data, the s-Flame investigation seeks to test and improve computational models, which may enable improvements to practical combustion on Earth.

Week 13: July 3 - 9

On GMT 188, Anatoly Ivanishin replaced the bottles at CIR Manifold #1, CIR Manifold #2 and CIR Manifold #4 to support ACME s-Flame (Structure and Response of Spherical Diffusion Flames) test points using 45% hydrogen fuel diluted with helium. On GMT 190, 21 sustained flames using this fuel were ignited for ACME s-Flame. These are the final test points for the second round of ACME s-Flame testing. Helium's characteristics are dramatically different from that of nitrogen, which was used for dilution in the first round of s-Flame testing.

ACME s-Flame Test Point Image



In this image (preliminary and false-colored), the burner tube and porous sphere are visible in the middle of the flame. In this test, a hydrogen/helium blend was ejected into an oxygen/helium atmosphere from the porous sphere at the end of the burner. The two bright lines above the flame are 14-micron ceramic fibers, which are glowing because of the flame's heat and are used for optical temperature measurements.

s-Flame's goal is to advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames. Without the buoyant flows created by Earth's gravity, flames in microgravity are typically spherical, which allows the analysis to focus on the combustion chemistry and details of the transport phenomena. Results may lead to more efficient and less polluting combustion technology for use on Earth.

Appendix H. Crew Activity per Space Station Science Highlights

The subsequent text and images are from the weekly highlights prepared by the Lead Increment Scientist and made available via e-mail or online via www.nasa.gov/mission_pages/station/research/.

Note that hyperlinks were generally not included in the excerpts below, but they typically go to web pages accessible via https://www.nasa.gov/mission_pages/station/research/experiments/explorer/. Those pages are secondary sources of information for the ACME project, for which the primary web site is https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/.

s-Flame round 1, featuring nitrogen dilution

Week of July 29, 2019

Other investigations on which the crew performed work:

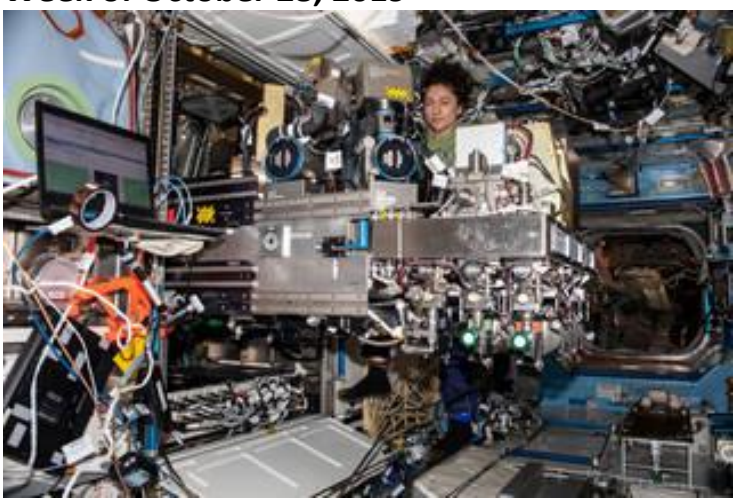
- **ACME** Flame Design, which studies the production and control of soot to optimize oxygen-enriched combustion and the design of robust, soot-free flames, is part of a series of independent ACME experiments using the orbiting laboratory's Combustion Integrated Rack (**CIR**).
  The note above is in error, where the first round of Flame Design operations ended with the data transfer and downlink on July 26. Crew activity during this week was in preparation for the s-Flame experiment.

Week of August 5, 2019

Other investigations on which the crew performed work:

- **ACME** Flame Design, which studies the production and control of soot to optimize oxygen-enriched combustion and the design of robust, soot-free flames, is part of a series of independent ACME experiments using the orbiting laboratory's Combustion Integrated Rack (**CIR**).
  The note above is in error, where the crew activity during this week was in preparation for the first round of the s-Flame experiment.

Week of October 28, 2019



NASA astronaut Jessica Meir works on components of the Combustion Integrated Rack (CIR), a platform for the Advanced Combustion via Microgravity Experiments (**ACME**) project. This set of six independent studies includes ACME Flame Design, which studies the production and control of soot to optimize oxygen-enriched combustion and the design of robust, soot-free flames. ***Credits: NASA***

Week of January 13, 2020

Other investigations on which the crew performed work:

- Advanced Combustion via Microgravity Experiments (**ACME**) is a set of six independent studies of gaseous flames conducted in the Combustion Integrated Rack (CIR). ACME supports spacecraft fire prevention as well as improved fuel efficiency and reduced pollutant production in practical combustion on Earth.
- The Structure and Response of Spherical Diffusion Flames (**s-Flame**), part of the ACME project, advances prediction of the structure and dynamics of soot-free and sooty flames. Results could contribute to engines with improved efficiency and reduced emissions on Earth.

Week of January 27, 2020

Other investigations on which the crew performed work:

- The Structure and Response of Spherical Diffusion Flames (**s-Flame**), part of the **ACME** project, advances prediction of the structure and dynamics of soot-free and sooty flames. Results could contribute to engines with improved efficiency and reduced emissions on Earth.

s-Flame round 2, featuring helium dilution

Week of April 27, 2020

Predicting flame structure

During the week, the crew prepared for continuation of runs for Structure and Response of Spherical Diffusion Flames (**s-Flame**). This investigation advances prediction of the structure and dynamics of soot-free and sooty flames, which could contribute to development of engines with improved efficiency and reduced emissions on Earth. S-Flame is part of the Advanced Combustion via Microgravity Experiments (**ACME)** project, a series of independent studies of gaseous flames performed in the station's Combustion Integrated Rack (**CIR**). The project's primary goals are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. A secondary goal is improving spacecraft fire prevention through innovative research focused on materials flammability.

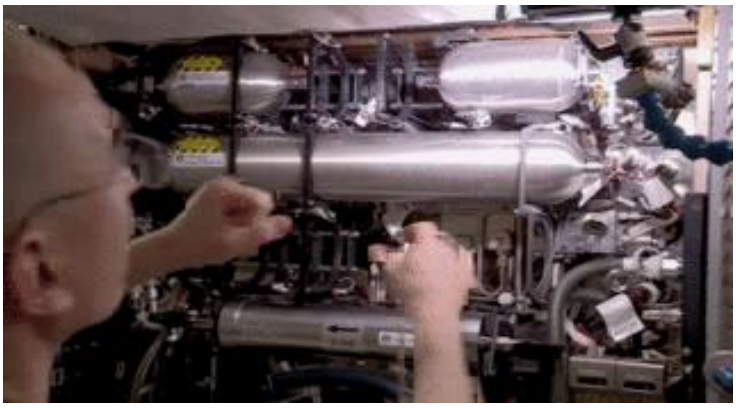


NASA astronaut Chris Cassidy works on the Combustion Integrated Rack, a device that enables safe fuel, flame and soot studies in microgravity. ***Credits: NASA***

Week of May 18, 2020

Other investigations on which the crew performed work:

- Structure and Response of Spherical Diffusion Flames (**s-Flame**) studies the structure and dynamics of soot-free and sooty flames. Findings could contribute to development of engines with improved efficiency and reduced emissions on Earth. S-Flame is part of the Advanced Combustion via Microgravity Experiments (**ACME**) project, a series of independent studies of gaseous flames performed in the station's Combustion Integrated Rack (**CIR**).



ROSCOSMOS cosmonaut Anatoly Ivanishin replaces a gas bottle in the Combustion Integration Rack (CIR) in preparation for flame studies controlled by the ground. ***Credits: NASA***

Week of May 25, 2020

Other investigations on which the crew performed work:

- Structure and Response of Spherical Diffusion Flames (**s-Flame**) studies the structure and dynamics of soot-free and sooty flames. Findings could contribute to development of engines with improved efficiency and reduced emissions on Earth. S-Flame is part of the Advanced Combustion via Microgravity Experiments (**ACME**) project, a series of independent studies of gaseous flames performed in the station's Combustion Integrated Rack (**CIR**).

Week of June 1, 2020

Cleaner, more efficient flames

During the week, the crew made changes for different runs of the Structure and Response of Spherical Diffusion Flames (**s-Flame**) investigation. This experiment is part of the Advanced Combustion via Microgravity Experiments (**ACME)** project, a series of independent studies of gaseous flames performed in the station's Combustion Integrated Rack (**CIR**). In particular, s-Flame advances prediction of the structure and dynamics of soot-free and sooty flames with the goal of improving fuel efficiency and reducing pollutants in routine fuel combustion activities on Earth. It also could contribute to better spacecraft fire prevention through innovative research focused on materials flammability.

Week of Jun 15, 2020

Other investigations on which the crew performed work:

- Structure and Response of Spherical Diffusion Flames (**s-Flame**) studies the structure and dynamics of soot-free and sooty flames. Findings could contribute to development of engines with improved efficiency and reduced emissions on Earth. S-Flame is part of the Advanced

Combustion via Microgravity Experiments (**ACME**) project, a series of independent studies of gaseous flames performed in the station's Combustion Integrated Rack (**CIR**).

Week of July 6, 2020

Other investigations on which the crew performed work:

- Structure and Response of Spherical Diffusion Flames (**s-Flame**) studies the structure and dynamics of soot-free and sooty flames. Findings could contribute to development of engines with improved efficiency and reduced emissions on Earth. S-Flame is part of the Advanced Combustion via Microgravity Experiments (**ACME**) project, a series of independent studies of gaseous flames performed in the station's Combustion Integrated Rack (**CIR**).

Appendix I. Crew Activity per the ISS Daily Summary Report

*The subsequent text was prepared by Dennis Stocker, where the source information is available from https://blogs.nasa.gov/stationreport. Text was added identifying the investigation's two rounds, but s-Flame operations listed under the heading "*All activities are complete unless otherwise noted*" were not included in this report. Some days of crew activity were only listed in that way, so this section does not comprehensively documenting crew activity in support of the s-Flame experiment.*

s-Flame round 1, featuring nitrogen dilution

2019 July 30: GMT 211

Combustion Integration Rack (CIR)/ Advanced Combustion via Microgravity Experiments (ACME) Manifold Bottle Exchange: The crew replaced the 40% O_2 gas bottle with a 30% O_2 bottle. They also replaced the ethylene fuel bottle with a hydrogen/methane mixture fuel bottle. Telemetry showed a slow leak in the hydrogen/methane bottle and teams are discussing a recovery plan. Flame Design part 1 investigation is complete and preparations are in work for transition to the s-Flame investigation. The purpose of s-Flame is to advance our ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth.

2019 July 31: GMT 212

Combustion Integration Rack (CIR) Manifold Bottle Inspection: Following yesterday's discovery of a slow leak, the fuel bottle connection was demated, inspected, cleaned, and remated to the CIR manifold. The pressure was subsequently monitored and the leak was reduced, but still present. The crew then looked for the leak source (suspected to be either the quick-disconnect or pressure relief valve) with the ultrasonic leak detector but were unable to isolate the source of the leak. The fuel bottle valve was closed and teams are discussing a forward plan. This fuel bottle will be used for the upcoming s-Flame investigation.

2019 August 13: GMT 225

Advanced Combustion via Microgravity Experiments (ACME): The crew configured the ACME Chamber for s-Flame by partially reinstalling the chamber insert to replace the igniter tip, then reinstalled the chamber insert, replaced the fuel controller, and closed the Front End Cap and rack doors. The ACME experiment series includes five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2019 September 9: GMT 252

Combustion Integrated Rack (CIR): The crew replaced the Manifold 4 fuel and Nitrogen bottles in preparation for upcoming s-Flames experiment runs. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2019 September 23: GMT 266

Combustion Integration Rack (CIR)/ Advanced Combustion via Microgravity Experiments (ACME): In support of the ongoing s-Flame experiment, the crew replaced the fuel and oxidizer manifold bottles as

well as the mass flow controller. The purpose of s-Flame is to advance the ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. Results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth. S-Flame is one of five experiments hosted by CIR/ACME, which is designed to study gaseous flames.

2019 October 24: GMT 297

CIR (Combustion Integrated Rack): The crew did a full replacement of the FCF CIR image processor (IPSUG) due to a previous failure of the hard drive. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2019 November 19: GMT 323

CIR/ACME/s-Flames manifold bottle exchange: As a continuation of the s-Flame experiment, the crew performed a routine manifold bottle exchange. In this activity, the crew replaced a used 30% O_2/70% N_2 bottle with a new one of the same composition, and replaced the used 45% CH_4/55% N_2 fuel bottle with a 100% H_2 fuel bottle. The purpose of s-Flame is to advance our ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth. S-Flame is one of five experiments hosted by CIR/ACME which is designed to study gaseous flames.

2019 December 13: GMT 347

Advanced Combustion via Microgravity Experiments (ACME): The crew replaced the ACME Bypass Controller and the ACME Fuel Controller, in order to continue s-flame operations. The ACME experiment series being performed in the CIR includes five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2019 December 24: GMT 358

CIR/ACME (Combustion Integration Rack/Advanced Combustion via Microgravity Experiments): In support of the continuing s-Flame investigation, the crew exchanged the ACME Bypass Controller and the ACME Fuel Controller. These controllers allow different flow regimes to be used for the experiment. The purpose of s-Flame is to advance our ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth. S-Flame is one of five experiments hosted by CIR/ACME which is designed to study gaseous flames.

2020

2020 January 2: GMT 002

Combustion Integrated Rack (CIR): As a continuation of the s-Flame experiment, the crew performed manifold H2 and CH4 bottle exchanges. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2020 January 8: GMT 008

Combustion Integrated Rack (CIR): As a continuation of the s-Flame experiment, the crew performed a manifold H2 and CH4 bottle exchanges. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2020 January 17: GMT 017

CIR/ACME (Combustion Integration Rack/Advanced Combustion via Microgravity Experiments): In support of the continuing s-Fames investigation, the crew replaced a used 3.8L Nitrogen/Oxygen manifold bottle with a new one. They also replaced the 1.0L hydrogen/methane manifold fuel bottle with a 100% hydrogen fuel bottle. The purpose of s-Flame is to advance our ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth. S-Flame is one of five experiments hosted by CIR/ACME, which is designed to study gaseous flames.

2020 January 28: GMT 028

CIR (Combustion Integrated Rack): The crew replaced the CIR manifold #4 with new H2 and C2H4/N2 gas bottles in preparation of continued s-Flame test runs. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

s-Flame round 2, featuring helium dilution

2020 April 23: GMT 114

CIR (Combustion Integrated Rack): The crew performed a replacement of the CIR FOMA (Fuel Oxidizer Management Assembly) Calibration Unit in preparation for the next set of s-flame runs. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2020 April 24: GMT 115

CIR (Combustion Integrated Rack): The crew prepared the CIR to allow ground commanding of the FOMA (Fuel Oxidizer Management Assembly) Calibration. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2020 April 28: GMT 119

Advanced Combustion via Microgravity Experiments (ACME): The crew replaced an ACME controller In preparation for the continuation of s-flame runs. The ACME experiment series being performed in the CIR includes five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2020 May 1: GMT 122

Combustion Integrated Rack (CIR): The crew replaced Nitrogen/Oxygen manifold bottles in support of the continuing s-Flames investigation. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2020 May 13: GMT 134

Combustion Integration Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME) : As a continuation of the ACME/s-Flame part 2 investigation, the crew removed a nitrogen check gas bottle and replaced it with a helium check gas bottle. They also removed a H_2/CH_4/He fuel bottle and replaced it with one of the same gas mixture, but different percent composition. Following the installation of the gas bottles, it was noted the actual pressure in the helium check gas bottle was significantly lower than expected and the team is assessing the impacts. The purpose of s-Flame is to advance our ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth. S-Flame is one of five experiments hosted by CIR/ACME which is designed to study gaseous flames.

2020 May 19: GMT 140

Advanced Combustion via Microgravity Experiments (ACME): The crew replaced Nitrogen/Oxygen manifold bottles in support of the continuing s-Flames investigation. The ACME experiment series being performed in the CIR includes five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2020 May 26: GMT 147

Advanced Combustion via Microgravity Experiments (ACME): The crew replaced Nitrogen/Oxygen manifold bottles in support of the continuing s-Flames investigation. The ACME experiment series being performed in the Combustion Integrated Rack (CIR) includes five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2020 June 2: GMT 154

Combustion Integration Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): In a continuation of the s-Flame part 2 investigation, the crew performed the necessary activities to change out a mass flow controller and igniter. The mass flow controller allows the ACME experimenters to study combustion under different gas flow regimes. The igniter was changed from a well-used hoop style igniter to a coil style igniter which can deliver more energy to ignite a gas mixture. A minor leak was seen in one of the coolant Quick Disconnect (QD) fittings when it was removed, but this is not expected to be an issue once installed again. The purpose of s-Flame is to advance our ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth. S-Flame is one of five experiments hosted by CIR/ACME which is designed to study gaseous flames.

2020 June 4: GMT 156

Combustion Integration Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): In support of the on-going s-Flame part 2 investigation, the crew removed a 45% H_2/ 55% He fuel bottle and replaced it with a 45% CH_4/ 55% He fuel bottle. The purpose of s-Flame is to advance our ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth. S-Flame is one of five experiments hosted by CIR/ACME which is designed to study gaseous flames.

2020 June 10: GMT 162

Combustion Integration Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): In support of the on-going s-Flame part 2 investigation, the crew removed a 45% CH_4/ 55% He fuel bottle and replaced it with a 20% CH_2=CH_2/ 80% He fuel bottle. The purpose of s-Flame is to advance our ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth. S-Flame is one of five experiments hosted by CIR/ACME which is designed to study gaseous flames.

2020 June 16: GMT 168

Advanced Combustion via Microgravity Experiments (ACME): In support of the on-going s-Flame part 2 investigation, the crew replaced C2H4, He, and O2 gas bottles in preparation for upcoming test point runs. The ACME experiment series being performed in the CIR includes five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2020 June 19: GMT 171

Advanced Combustion via Microgravity Experiments (ACME): In support of the on-going s-Flame part 2 investigation, the crew replaced the C2H4 fuel bottle in preparation for upcoming test point runs. The ACME experiment series being performed in the CIR includes five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

Appendix J. Crew Activity per the NASA Space Station Blog

This section consists of excerpts from https://blogs.nasa.gov/spacestation/. Hyperlinks were generally not included, but clarifications have sometimes been added in red.

s-Flame round 1, featuring nitrogen dilution

2019 August 13 – Crew Gets Ready for Next Spacewalk and New Spaceships

Christina Koch of NASA serviced and replaced hardware that fuels experiments inside the Combustion Integrated Rack.

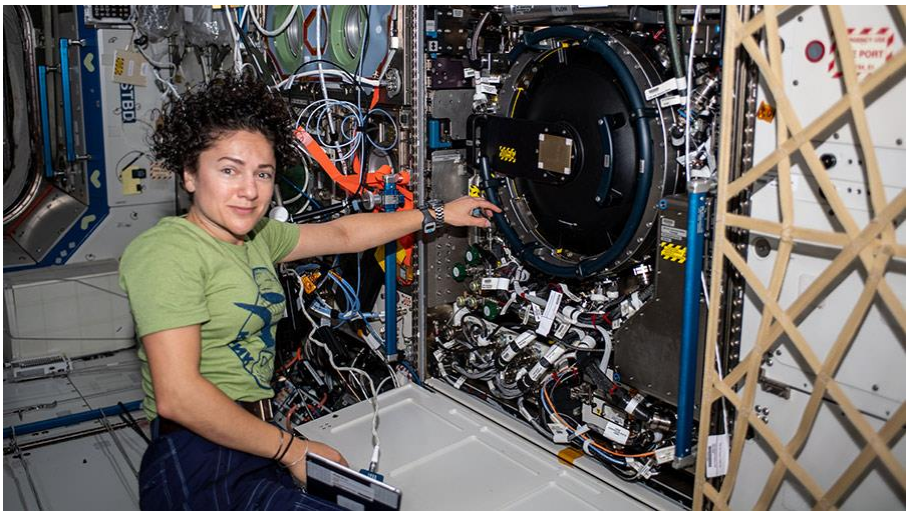
2019 September 23 - Japanese, Russian Rockets Prepare to Launch Cargo and Crew This Week

Luca Parmitano of ESA (European Space Agency) replaced fuel bottles in the Combustion Integrated Rack ….

2019 October 24 - Crew Explores Space Biology, Reviews Particle Detector Repairs

Meir later swapped out a failed computer hard drive that supports combustion experiments.

2019 October 29 - Spacesuits, Human Research, Robotics Training Ahead of Cargo Missions



NASA astronaut Jessica Meir works to swap out a failed computer hard drive that supports experiments inside the Combustion Integrated Rack aboard the International Space Station's U.S. Destiny laboratory module.

2020 January 8 - Spacewalk Preps Underway Amidst Space Research

Cosmonauts Alexander Skvortsov and Oleg Skripochka split their time on Russian science and maintenance tasks. … Skripochka replaced fuel bottles for combustion research.

2020 January 17 – Nutrition Studies, Spacewalk Preps Ahead of SpaceX Crew Escape Test

Fellow cosmonaut Oleg Skripochka serviced combustion research gear ….

s-Flame round 2, featuring helium dilution

2020 April 21 - Advanced Space Research Ahead of Weekend Cargo Mission

In the afternoon, Cassidy moved over to the U.S. Destiny laboratory module and swapped computer parts inside the Fluids and Combustion Facility (FCF). The FCF consists of two refrigerator-sized research racks enabling safe observations of the behavior of fluids and flames in microgravity. Scientists use the data to design advanced fuel transfer systems and increase fire safety.

While this may be true, Cassidy did not make such a change on the Combustion Integrated Rack (CIR) during April 2020.

2020 April 28 - Cygnus Prepped for Departure During Station Science and Upkeep

Cassidy also opened up the Combustion Integrated Rack for maintenance replacing components in the research device that enables safe fuel, flame and soot studies in microgravity.

2020 April 29 - Spacesuit Work and Heart Research Fill Crew Day



Expedition 63 Commander Chris Cassidy works in the Combustion Integrated Rack, a research device that enables safe fuel, flame and soot studies in microgravity.

While Cassidy had worked on the Combustion Integrated Rack (CIR) on April 28, he is not doing so in this photo (in which the CIR facility does not appear) and as such the caption is in error. In this photo, he appears to be working at a Materials Science Research Rack (MSRR).

2020 May 13 - Flying Robots, Ultrasound Eye Scans Top Science Schedule

Ivanishin started his workday swapping fuel bottles inside the Combustion Integrated Rack which enables safe studies of fuels, flames and soot in microgravity.

2020 May 19 - Japan Ready for Cargo Launch, Crew Works Science and Emergency Procedures

Veteran cosmonaut Anatoly Ivanishin continued servicing the Combustion Integrated Rack as he swapped out fuel bottles inside the device that enables fuel, flame and soot research in microgravity.

2020 May 26 - Expedition 63 Awaits SpaceX Crew, Unpacks Japanese Cargo

Veteran cosmonaut Anatoly Ivanishin swapped out more fuel bottles inside the Combustion Integrated Rack to maintain research operations inside the U.S. Destiny laboratory module. The refrigerator-sized research rack enables fuel, flame and soot research in microgravity.

2020 June 4 - Advanced Station Science Benefiting Humans

Ivanishin was inside the U.S. Destiny laboratory swapping fuel bottles inside the Combustion Integrated Rack to enable safe fuel and flame science.

2020 June 10 - Station Buzzing With Science in Middle of Spacewalk Preps

Today, the three NASA astronauts and two Roscosmos cosmonauts juggled a variety of experiments encompassing fluids, combustion, genetics and vision among others. … Russian Flight Engineer Anatoly Ivanishin … swapped fuel bottles inside the Combustion Integrated Rack to continue safe fuel and flame research.

2020 June 16 - Biology and Physics as Crew Enters BEAM, Preps for Spacewalk

Tuesday's science aboard the International Space Station encompassed life science, fluids and flames to help humans on Earth and in space. … Veteran cosmonaut Anatoly Ivanishin continued to service the Combustion Integrated Rack replacing fuel bottles to maintain safe fuel and flame research in the device. Insights could improve fire safety as well as combustion processes for Earth and space industries.

2020 June 19 - Crew Readies Satellite and Organizes Station

Cosmonaut Anatoly Ivanishin changed out fuel bottles in the Combustion Integrated Rack this morning to continue the safe research of flames in microgravity.

2020 June 22 – Astronauts Check Spacewalk Tools and Suits, Study Space Physics

Hurley spent the first half of his Monday working on fluid and combustion physics. He first explored how microfluidics can cause biochemical reactions in blood revealing mechanisms hidden on Earth. Next, he researched fabricating composite materials to learn how to repair and build structures on future space missions.

While Hurley worked on fluid physics, the reference to combustion is presumably in error. At the least, he did not work on the ACME project on that day, if at all.

2020 July 6 - Cargo and Science Operations Start Work Week

Cosmonaut Anatoly Ivanishin replaced fuel bottles inside the Combustion Integrated Rack to continue safe fuel and flame research.

Appendix K. ACME Operations per @ISS_Research on Twitter

These tweets (from https://twitter.com/ISS_Research/) were from the period of s-Flame operations, i.e., from July 2019 to July 2020. However, please note that some are in regard to other ACME investigations.

2019 Sept. 10: GMT 253

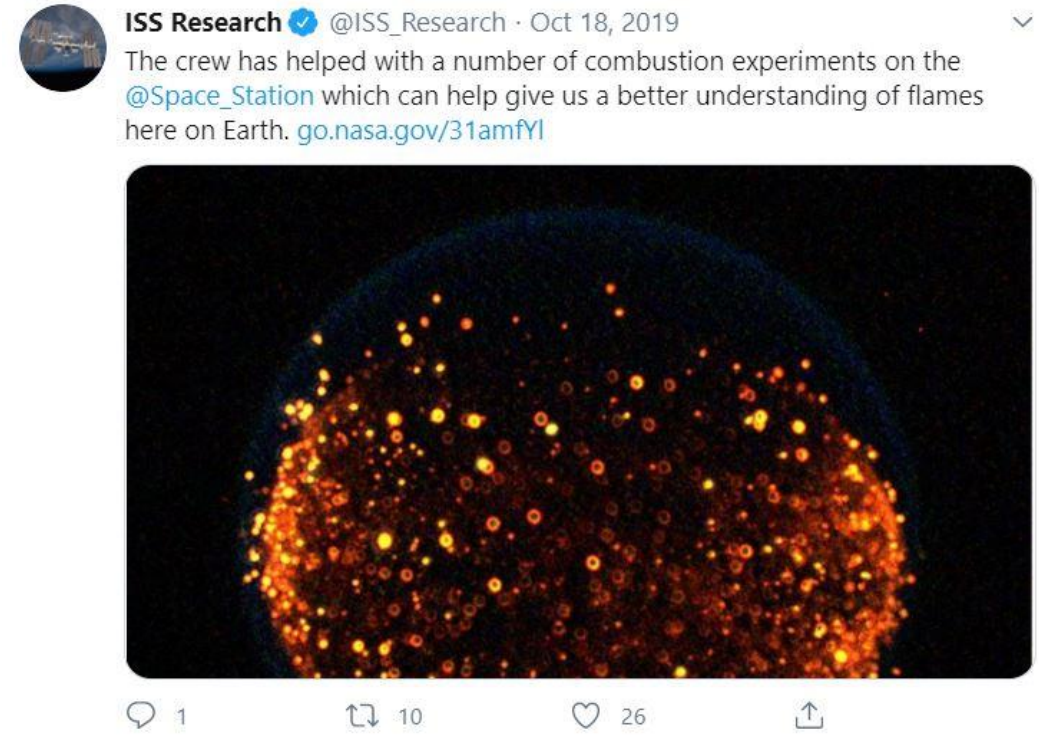


2019 Sept. 12: GMT 255



This Flame Design video clip can be viewed at, and downloaded from, https://images.nasa.gov/details-jsc2019m000710_Designing-Flames-Aboard-ISS-MP4.html.

2019 Oct. 18: GMT 291



2019 Dec. 28: GMT 362



2020 Jan. 21: GMT 021

ISS Research @ISS_Research · Jan 21, 2020

From combustion to microbiology, research aboard the @Space_Station spans many different scientific disciplines.

Learn the basics of some of these areas of study in our Station Science 101s! go.nasa.gov/2NJCxTq

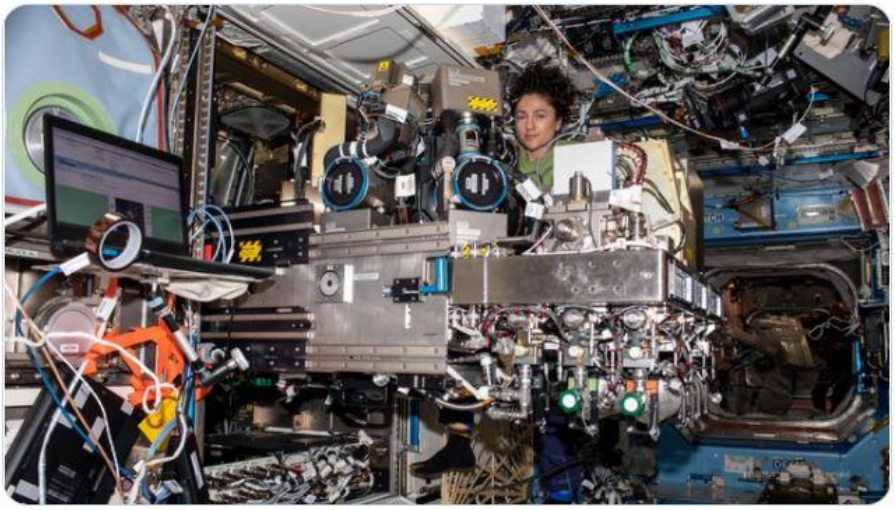


2 52 253

2020 April 14: GMT 105

ISS Research @ISS_Research · Apr 14

One of the jobs of the space station crew is to run and maintain the science hardware. Meir worked on the Combustion Integrated Rack (CIR), a facility in which combustion experiments can be performed in microgravity. go.nasa.gov/2V5QyPw



1 14 87

2020 June 25: GMT 177



Appendix L. ACME Operations per Astronauts on Twitter

These tweets are from ISS crew members who conducted operations in support of the s-Flame experiment.

Jessica Meir (@Astro_Jessica) on Twitter, https://twitter.com/astro_jessica

2019 Nov. 06: GMT 310



Chris Cassidy (@Astro_SEAL) on Twitter, https://twitter.com/astro_seal
2020 Sept. 1: GMT 245



Appendix M. Summary of the ISS data collection

The s-Flame data is all digital and includes both image and non-image data, where this is true for the other ACME experiments. To be clear, there was no collection of soot or other physical samples that were subsequently transported to the ground.

The s-Flame image data includes images from the following cameras:

1. ACME DataCam, used with a clear fused-silica 'filter' and various zoom and focus settings (but those settings were consistent within each individual test)
2. CFI intensified camera, used with CH*, formaldehyde, and UV fused silica filters, but where only a single filter was used in each test
3. Low Light Level Ultraviolet (LLL-UV) camera, equipped with an OH* filter
4. High-Bitdepth Multispectral (HiBMs) camera, equipped with a red filter corresponding to the opposing CIR illumination package, which was sometimes left off

The raw DataCam images are in a Halcon Iconic Object (hobj) format, while the images from the CFI, HiBMs, and LLL-UV cameras are provided in a Tagged Image File (tif) format. In contrast, the analog video was recorded on the ground in a Transport Stream (ts) format.

The key non-imaging data from a test is a comma-delimited file that is described in detail in the Software Interface Design Description (ACME-DOC-010 Rev. D).

The raw data is currently archived at NASA and work is underway to make it available publicly through NASA's Physical Science Informatics System (PSI).

Appendix N. Publications, presentations, awards, and outreach

Papers, etc. listed in italics are general to ACME rather than specific to the s-Flame experiment.

N.1 Publications and Presentations with ISS Results

Conference Proceedings, Presentations, and Posters

1. *"Investigations at ISS on Combustion of Gas Mixtures. Sample Results and Technical Experience," Sergey Pichugin, Oleg Artemiev, Sergey Prokopiev, Oleg Kononenko, Oleg Novitsky, and Sergey Frolov, XIII International Scientific and Practical Conference 'Manned Flights in Space,' Gagarin Cosmonaut Training Centre, Star City, Russia, Nov. 13-15, 2019.*
2. "Dynamics of Burner-Generated Spherical Diffusion Flames in Microgravity," Stephen Tse, Jonathan Shi, Delin Zhu, and Chung K. Law, 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, Colorado, November 20-23, 2019.
3. "ISS Experiments of Gaseous Spherical Diffusion Flames," Jonathan Shi, Stephen D. Tse, Dennis Stocker, and Chung K. Law, 36th Annual Meeting of the American Society for Gravitational and Space Research, Nov. 5-6, 2020.
4. *"Flames Investigations at ISS," Cosmonautics of XXI Century Conference, TSNIIMash, Moscow, Russia, S. Pichugin, D. Samsonov, S. Frolov, O. Korobeinichev, and A. Shmakov, May 12-14, 2021.*
5. *"Advanced Combustion via Microgravity Experiments on the International Space Station," Dennis P. Stocker, Work in Progress Poster, 12th U.S. National Combustion Meeting, May 24-26, 2021.*

N.2 Publications and Presentations (comprehensive list)

Papers, etc. in this section which include ISS results are enclosed in borders like this.

Peer-Reviewed Journal Papers

1. "Opportunities and challenges of combustion in microgravity," by C. K. Law and G. M. Faeth, *Progress in Energy and Combustion Science,* Vol. 20, pp. 65-113 (1994). https://doi.org/10.1016/0360-1285(94)90006-X
2. "On burner-stabilized cylindrical premixed flames in microgravity," by J. A. Eng, C. K. Law, and D. L. Zhu, *Twenty-Fifth Symposium (International) on Combustion*, The Combustion Institute, Pittsburgh, PA, pp. 1711-1718 (1994). https://doi.org/10.1016/S0082-0784(06)80819-2
3. "On the structure, stabilization, and dual response of flat-burner flames," by J. A. Eng, D. L. Zhu, and C. K. Law, *Combustion and Flame*, Vol. 100, pp. 645-652 (1995). https://doi.org/10.1016/0010-2180(94)00102-X
4. "Extinction mechanisms of near-limit premixed flames and extended limits of flammability," by C. J. Sung and C. K. Law, *Twenty-Sixth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, PA, pp. 865-873 (1996). https://doi.org/10.1016/S0082-0784(96)80296-7
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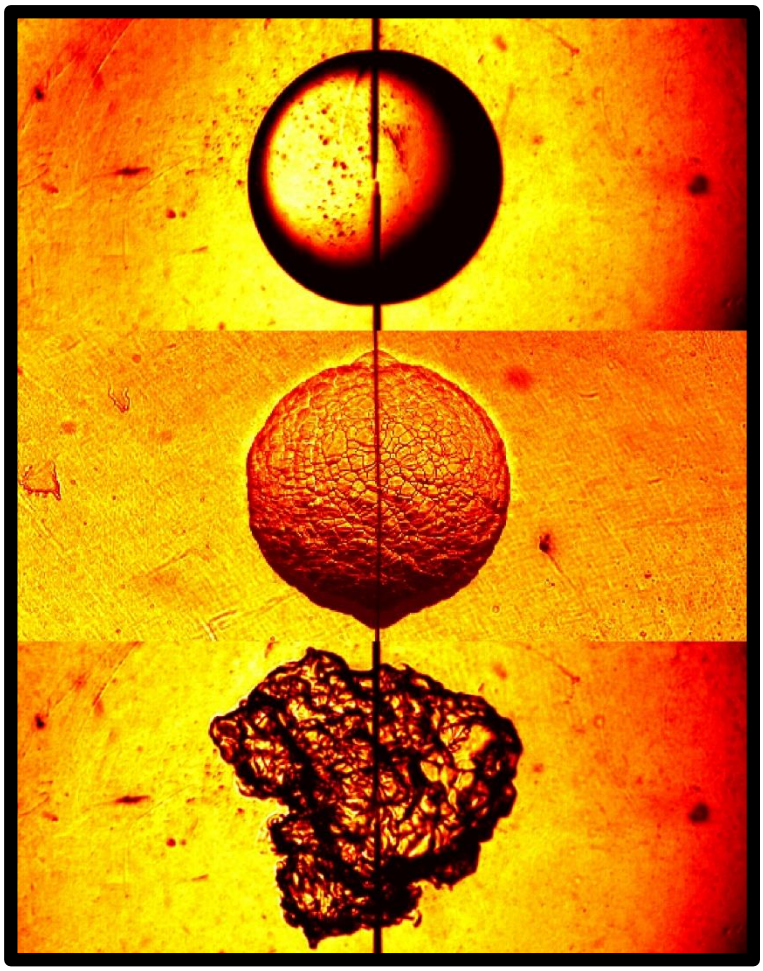
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2. Laurel Paxton, BSE '14, Undergraduate Thesis, "Weakly Buoyant Spherical Diffusion Flames: Properties of Hydrogen-CO/Ethylene Flames," Princeton University, Princeton, NJ. http://arks.princeton.edu/ark:/88435/dsp014f16c301w
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N.3 Awards

1. First Place, Combustion Art Competition, "Three Faces of An Expanding Flame," Chung K. Law, Swetaprovo Chaudhuri, and Fujia Wu, 8th U.S. National Combustion Meeting, Park City, Utah, May 19-22, 2013.

   The three images are snap shots of a spark-ignited expanding flame in different environments of the same hydrogen-air mixture. The top flame shows the ideal, reference case of a stable, smooth flame surface in a quiescent environment at atmospheric pressure. The middle flame is taken under elevated pressure simulating that within an internal combustion engine. The flame surface is hydrodynamically unstable and develops cells, causing the flame to burn faster due to the increased flame surface area. The bottom flame is taken in a highly turbulent environment which simulates another aspect of the engine interior. Now the flame surface is distorted by the multi-scale turbulence, which again leads to an increase of the burning rate.



2. Morgan McKinzie Senior Thesis Prize (*for excellence in undergraduate research*), "Weakly buoyant spherical diffusion flames: properties of hydrogen-CO/ethylene flames," Laurel Paxton, Dept. of Mechanical and Aerospace Engineering, Princeton University, June 2014.

N.4 Outreach

NASA Project Scientists Dennis Stocker created the Space Flames Facebook page (www.facebook.com/space.flames) and Flickr site (https://www.flickr.com/photos/space-flames), where some entries are specific to s-Flame. He continues to administer these sites, where most content has been about the ACME project of which s-Flame is a part. Meanwhile, Stocker has made educational and general information is available via https://tinyurl.com/space-flames. As an educational example, a simple board game related to the ACME's spherical flames experiments, such as s-Flame, which can be downloaded. As another example, an ACME information sheet providing an overview and recent status can also be downloaded through that site. Such flyers have been distributed at meetings of The Combustion Institute and the American Society for Gravitational and

Space Research (ASGSR). Other downloads, such as the s-Flame Science Requirements Document, are available through https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/downloads/.

Both Princeton and Rutgers have promoted awareness of the s-Flame investigation online, for example, through Rutgers' 2019 article, "From the Ground Up: Conducting International Space Station Experiments from a Rutgers Office."

Meanwhile, NASA promoted awareness of ISS combustion research with the video 'The Power of Studying Combustion on the ISS' which was posted on March 13, 2018.

Awareness of ISS combustion research, including the ACME project, was promoted on Oct. 15, 2018 by The Combustion Institute in their online news article, "International Space Station (ISS) Combustion Experiments."

Stocker is involved in a broad range of NASA educational outreach to students and teachers, Scouts of both genders, and the general public. His primary outreach focus is the fundamentals of microgravity (for which resources are available via http://tinyurl.com/SEECmicrogravity) and human spaceflight. But he periodically addresses other topics, e.g., where he sometimes speaks on 'NASA and the Science of Superman' – for example at the Superman Celebration in Metropolis, IL. In this regard, his NASA outreach is carried out locally, regionally, nationally, and online. For example, he regularly presents on microgravity at Space Center Houston's Space Exploration Educators Conference (SEEC). He has exhibited the last five National Scout Jamborees, four national Boy Scout conferences, the last three national Girl Scout conventions, and the 2019 World Scout Jamboree (which was held in the U.S.). Stocker has been a key lead for the agency in NASA's participation in these Scouting events, where as an example the latter outreach is visually described at www.nasa.gov/feature/stem/world-scout-jamboree.html.